

AN ARCHIVAL ALMA SEARCH FOR SPECTRAL TECHNOSIGNATURES TOWARD 82 STELLAR SYSTEMS WITHIN 40 PARSECS

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ABSTRACT

We have searched the public ALMA archive for artificial narrowband radio emission from nearby stars. “Narrowband” means narrow compared with an ALMA channel, 15.3–1953 kHz here, not the hertz-scale resolution of centimetre-wave work. The primary experiment seeks an unresolved drifting carrier in 403 fine-channel windows toward 60 systems over 47.7 GHz of unique sky frequency; a secondary search of 1252 coarse windows adds 70.4 GHz but cannot discriminate drift.

Sensitivity is measured, by injecting carriers and recovering them through the complete automated selection. On each system’s best window that selection recovers a carrier nine times in ten above an equivalent isotropic radiated power of 1.5×10^{15} W at the median system, over 8.7×10^{13} – 4.4×10^{16} W. This is an upper bound on reach, not a central estimate: the calibration’s window-to-window transfer is uncertain by $\times 0.48$ – $\times 1.35$, which dominates the budget.

13 windows were flagged by a screen comparing the stellar position with 512 controls in the same field. The screen generates candidates and is not a significance test: the stellar rank is not exchangeable with the controls, and a radial correction does not repair that on independent data. Flagged windows were therefore judged on physical evidence. 9 are circumstellar or foreground carbon monoxide, including the β Pictoris disc, recovered blind in two transitions as a positive control. The other 4 have no identification, but none is detected as a point source at the stellar position in the visibilities and none reappears in a repeat observation. **We identify no independently confirmed narrowband carrier.**

This applies to the ALMA archive-selected sample within 40 pc, not the solar neighbourhood: the targets come from disc and planet-formation programmes, and with only 21 of 5908 catalogued M dwarfs covered the sample cannot constrain the local habitable-zone planet population.

Subject headings: extraterrestrial intelligence, radio lines: stars, submillimetre: stars, surveys, astrobiology

1. INTRODUCTION

A technological civilisation, signalling deliberately or going about its affairs, may imprint the electromagnetic spectrum detectably, a *technosignature* (Sheikh 2020; Vidal et al. 2026). Six decades of radio searches cover many thousands of stars yet sample only 6×10^{-18} of the haystack of transmitter frequency, bandwidth, power and duty cycle (Wright et al. 2018). Three gaps persist. The millimetre and submillimetre regime is very sparsely searched, and everything above that axis’s ~ 115 GHz upper limit more sparsely still. Most surveys are aimed at interest-selected classes of star, leaving the local stellar population unsampled as a population. Few reprocess *existing* interferometric archives.

This paper addresses all three, asking how strongly existing ALMA observations constrain unresolved artificial spectral emission from nearby stellar systems. To be unambiguous about the signal searched: we look for an intrinsically narrow coherent carrier whose intrinsic linewidth is unresolved by the native ALMA spectral channel, and we do not claim hertz-scale spectral resolution. The fine channels used here are 15.3–1953 kHz

wide, orders of magnitude broader than a centimetre-wave narrowband backend, so we write “unresolved narrowband carrier” below wherever ambiguity could arise. We mine the archive, correcting for proper motion and Doppler drift, for unresolved spectral carriers (channel-confined excess at native ALMA resolution) and anomalous millimetre continuum. Throughout, “narrowband” describes the assumed intrinsically narrow transmitter and never the resolving power of the data (§4). Channelisation sets which of two search classes a window belongs to, and the completeness of each class is measured in §4.4: 76 per cent of the searched windows carry no drift discrimination at all, and only 403 support the fine-channel search that “narrowband SETI” normally means (§4). Three contributions follow. The first is the systematic exploitation of archival ALMA stellar fields, 113 target/band datasets mined from public data with no new observing time. The second is spatial-control screen calibration for interferometric SETI, in which the same statistic is evaluated at the star and at 512 control positions so that the false-alarm rate is measured and not assumed. The third is a quantitative accounting of the frequency, power and drift domain actually searched, which gives the non-detection explicit boundaries. No ancillary analysis carries a technosignature disposition. One worked case runs through the paper, a crossing towards β Pictoris, which is the largest star-versus-control contrast here and is attributed to the disc’s known circum-

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stellar CO across two transitions and 8 execution blocks (§5.2). Figure 3 is the roadmap: it gives the selection from catalogue to searched sample on the left, and the decision chain from window to disposition on the right.

Two properties of the design bear on how every number below should be read, and both are developed where they belong rather than argued here. The statistic and the molecular-line mask were defined on these data, so the rank probabilities are exploratory rather than pre-registered in any conventional sense; what we claim is narrower and is set out with its repository timestamps and its out-of-sample test in §4.1 and §C.0.2. And the statistic compares a single position with single positions, which rules out the obvious region-maximum alternative for reasons of exchangeability; both forms are scored for every window and released, and the comparison is audited in Appendix L.2.

In plain terms: we looked for a radio transmitter’s carrier signal in every millimetre-wave observation ALMA has made public of a star within 40 parsecs, about 130 light years. We found no signal that repeats or that can be confirmed independently. Thirteen features passed the first automated cut; nine are carbon monoxide gas around or in front of the stars, and the remaining four are no more numerous than chance predicts and show no sign, in the raw interferometer data, of coming from the star itself. The main limitations are that these observations were taken for other purposes, so the stars are not a representative sample, and that most of them were observed only once, so a signal that came and went could not have been confirmed.

Which sensitivity to quote. Four power scales appear below and only the last is a sensitivity: the trigger level P_{trig} , the power P_{eff} a transmitter must radiate to reach it, the trigger’s own 90 per cent recovery P_{90} , and P_{90}^{sel} , the power the *complete* automated selection recovers nine times in ten. P_{90}^{sel} is the survey sensitivity and the only one of the four quoted in the abstract, the discussion and the conclusions; Appendix E defines all of them and every other term used here in a particular sense.

What the sample is, and is not. The targets were not chosen for a technosignature search. They are the stars that archival disc and planet-formation programmes happened to observe, so the sample is enriched in F and G stars and depleted in M dwarfs relative to the 40 pc census, and only 21 of 5908 catalogued M dwarfs within 40 pc appear in it at all. Every statistical statement below therefore concerns *the ALMA archive-selected stellar sample within 40 pc* and not the solar neighbourhood; in particular the survey cannot constrain the local habitable-zone planet population, which M dwarfs dominate (§6.3).

2. BACKGROUND

Radio searches, from Cocconi & Morrison (1959) and Drake (1960) through the programmes reviewed by Tarter (2001) and Project Phoenix (Backus & Project Phoenix Team 2002), divide into wide-field commensal and targeted surveys (Zhang et al. 2020; Czech et al. 2021; Morrison et al. 2023; Li et al. 2026; Johnson et al. 2023), with wide-field low-frequency coverage from the MWA (Tremblay et al. 2024) and a single-dish C- and K-band programme at 6 and 18 GHz with the Sardinia Radio Telescope (Manunza et al. 2025), with machine-

learning interference rejection now central (Ma et al. 2023; Choza et al. 2024; Jacobson-Bell et al. 2025; Poznanski et al. 2025; Garrett et al. 2026), a capability our classical threshold search lacks. Essentially all of that work lies below ~ 10 GHz, and the millimetre regime above it has been searched only once before the present work, with ALMA (Mason et al. 2024). There are physical reasons to go higher: interstellar scattering and dispersion fall steeply with frequency, a given aperture delivers a tighter beam per watt of EIRP, and directional communication, radar and beamed power are already engineered as coherent mm-wave emission. Atmospheric transmission, receiver noise and phase stability raise the cost without forbidding it.

Mason et al. (2024) searched archival Band 3 windows toward 28 “stellar bycatch” stars, observed for other science and usable for free, placing $\text{EIRP}_{\text{min}} > 7 \times 10^{17}$ W for the nearest and identifying the challenges of high-frequency work: drift rates scale with frequency, smearing erodes sensitivity without a drift search, and the dense forest of molecular lines raises the confusion background. A pilot study by one of us (White 2026; submitted, and not yet available to the reader) extended that methodology to TRAPPIST-1 across Bands 3, 6 and 7 and returned no detection. Nothing in the present paper rests on it: the ranked target list it produced is superseded by the frozen work list of §3, which is released in full, and every window here was re-reduced and re-searched from the raw archive data with the pipeline of §4. Where a star appears in both, TRAPPIST-1 included, the numbers to cite are this paper’s frozen products.

Figure 1 locates this survey in the *parameter space* of published searches, and it is not a sensitivity comparison. The vertical coordinates of the different programmes are not comparable completeness limits: each is its own threshold statistic, evaluated over channel widths that differ by seven orders of magnitude, and we know of no way to convert them to a common completeness. What the figure shows is that ALMA occupies a different combination of frequency and spectral resolution from cm-wave SETI, not that it is more or less sensitive.

Where a numerical comparison is possible it should be stated plainly rather than read off the figure. Essentially all previous narrowband technosignature work lies below ~ 10 GHz, the upper bound of the Breakthrough Listen L- and C-band surveys and of the Wright et al. (2018) haystack axis, so a search spanning 89.6–873.1 GHz moves the searched frequency range up by about two orders of magnitude relative to that boundary. Any EIRP advantage toward the nearest systems is a d^2 effect and not an instrumental one: §6.2 decomposes it. This survey samples far fewer stars, far fewer epochs and a channelisation seven orders of magnitude coarser than a dedicated narrowband search, and it is complementary to those surveys rather than a deeper version of them.

A non-detection must become a quantitative exclusion statement: Wright et al. (2018) formalised the “Cosmic Haystack” search volume, Margot et al. (2023) its conversion into an upper bound on the transmitter-hosting fraction, and Sheikh et al. (2025) asked where Earth’s technosignatures would be detectable. That bound is valid only if the sample represents the underlying population, which interest-selected samples do not. Most sur-

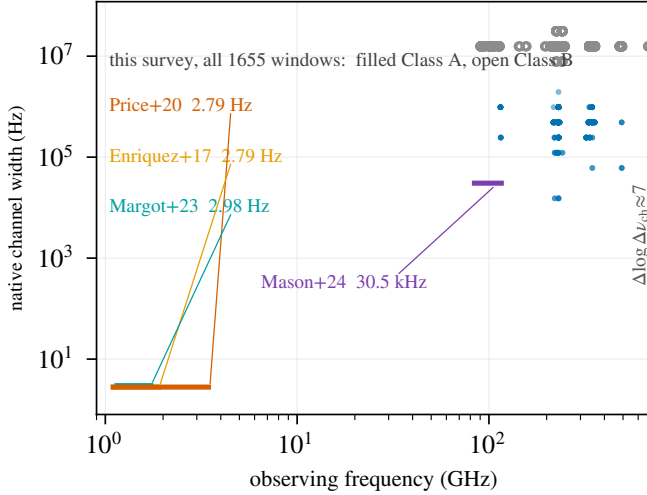


FIG. 1.— Where this survey sits in spectral resolution. Native channel width against observing frequency for every searched window (filled: Class A, fine-channel; open: Class B, coarse-channel), with the comparison programmes as horizontal segments: Enriquez et al. (2017), Margot et al. (2023), Price et al. (2020) and Mason et al. (2024). The distinctive point of this survey is its frequency regime, not its depth: the channel widths plotted span about seven orders of magnitude — which is what the $\Delta \log \Delta \nu_{\text{ch}} \approx 7$ annotation marks — so sensitivities are not comparable between programmes. The corresponding EIRP-versus-distance comparison, which requires that warning to be read correctly, is Fig. 11 in Appendix E.

veys select by proximity, spectral type, planets or Earth Transit Zone membership; the ALMA-specific form, disc selection, skews the sample young and dynamically active. We instead take every star within 40 pc lying in the usable field of at least one public ALMA observation (§3). That sample is *work-list complete but epoch-incomplete*, and this paper uses no other completeness formulation: every star on the frozen work list that proves genuinely covered is searched, but not every execution block those stars hold is in this release. Of 168 catalogue cross-matches, 115 correspond to genuine ALMA coverage and 90 were searched; of the 656 public execution blocks they hold, 479 have been searched and 404 form the frozen science sample reported here. It makes no completeness claim for the local population itself. That choice removes technosignature-specific selection and leaves the archival astrophysical selection in place; the 168 sample entries are ~ 1 per cent of the catalogued population within 40 pc (the 90 actually searched here are ~ 0.5 per cent), and the residual coverage selection is characterised as a completeness function (§3, Table 2).

3. SAMPLE SELECTION

The sample is *work-list complete but epoch-incomplete* within 40 pc (Table 12), which is a much weaker condition than volume completeness in the first half and an explicit limitation in the second. The execution-block accounting closes as follows, and every number in this paper refers to one of these populations.

1. 656 public execution blocks lie in the parent member observing-unit sets of the sample stars. This is the population in scope, and it splits in two.
2. 479 of them have been downloaded, calibrated and searched; 177 have not. Every one of the 177 is repeat coverage: each sits in an observing-unit set from

which another block was already taken, and the de-duplication rule below keeps one block per star and tuning. 20 of them the archive offers as raw visibilities with no pipeline calibration at all. That is the first identity, $479 + 177 = 656$.

3. 484 blocks were processed in total, 4.1 TB of raw data through one frozen pipeline. This is a *different* population from the 479: it is those plus 5 whose member-OUTS progenitor link the archive does not expose, so they cannot be placed in the ledger above. That is the second identity, $479 + 5 = 484$. The two counts must not be added to each other.
4. Of the 484, 404 form the *frozen science sample* reported here, 77 are the reserved calibration hold-out of §C.0.2, and 3 left no surviving window after the quality cut.

We therefore make no claim to have searched the whole ALMA archive, only every star on the work list. Table 24 gives the provenance of every dataset used anywhere in this paper.

The procedure admits every star within 40 pc of the Sun (Gaia DR3 parallaxes, distances direct inversions $d = 1/\pi$; every entry $\pi \geq 25$ mas with a parallax measured to better than 20 per cent, which is the condition the archive query applied) whose proper-motion-corrected position at the observation epoch falls in the quoted field of view of at least one public ALMA execution block. The star need not sit at the pointing centre, and §A.3 shows the quoted field of view admitted 53 stars no observation in fact contains, so “covered” is a property of our selection and not a guarantee. Disc, planet and spectral type enter as descriptive metadata only. Units are defined in §A.2. This gives 168 unique target stars, processed nearest-first. Because trigger thresholds scale with d^2 at fixed noise, that order is sensitivity-ranked, so the prevalence statements of §6 are conditional on it.

Composition follows from the archival selection and is tabulated against the reference census in Table 13. The sample is F- and G-enriched and deficient in M dwarfs, and 285 of the 1655 windows come from 3 systems; §6.3 gives the figures and their cost. One catalogue entry needs flagging here rather than only in the appendix, because the released catalogue and the text would otherwise appear to disagree: the entry “ γ Lupi” resolves by parallax to HD 139664 at 17.40 pc, not to the naked-eye star, and is relabelled accordingly. No measured value changes. Appendix L lists the other cross-match and exclusion edge cases in the same way, and any reader comparing the released catalogue against this text should read it. The spectral-class proxy is `teff_gspphot`, available for 53 of 90 sample stars and missing preferentially for the coolest objects, so the M fraction is a lower limit and the earlier-type ratios upper limits. The debris-disc hosts contribute dusty continua, kept from masquerading as anomalies by the continuum lane’s screen (§4).

“Usable public ALMA coverage” is a five-criterion decision tree, frozen absent an erratum.

No star or window was excluded by an unlisted criterion, and none for short integration.

Because the rest of the paper is organised around that figure, its right-hand panel is worth reading now rather than at §4.5, where the same eleven steps are given with every number attached. In brief, and in order: the 656

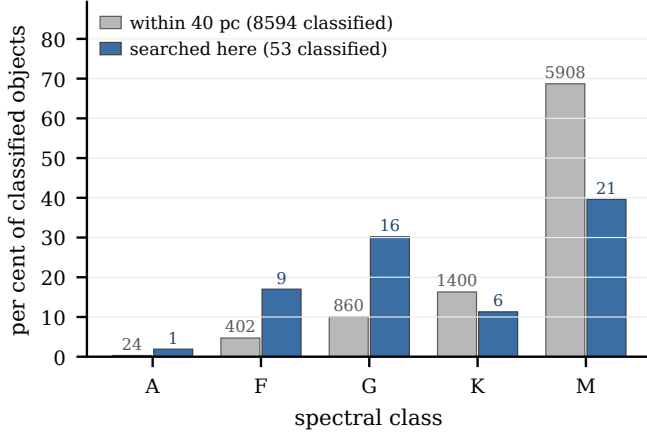


FIG. 2.— The archival selection, which no limit in this paper escapes. Grey: the spectral-class composition of the Gaia DR3 objects within 40 pc that carry a temperature. Blue: the same for the searched sample. Counts are printed above each bar. The sample is enriched in F and G stars by factors of four and three and depleted in M dwarfs by a factor two, so 21 of the 5908 classified M dwarfs within 40 pc carry a searched window. Because M dwarfs host most of the habitable-zone planets in the solar neighbourhood, this is the single most important limitation on what the null result means (§6.3).

TABLE 1

THE ARCHIVAL SELECTION BY SPECTRAL CLASS, INCLUDING THE COVERAGE THAT MATTERS. $N_{40\text{ pc}}$ COUNTS GAIA DR3 OBJECTS WITHIN 40 PC CARRYING A TEMPERATURE; N_{searched} THOSE WITH AT LEAST ONE SEARCHED WINDOW; AND THE LAST COLUMN THOSE WITH AT LEAST ONE *Class A* WINDOW, WHICH IS THE ONLY COVERAGE THAT CAN RESOLVE A DRIFTING CARRIER. 13 M DWARFS HAVE *Class A* COVERAGE, OF 5 908 CLASSIFIED WITHIN 40 PC.

Class	$N_{40\text{ pc}}$	N_{searched}	f_{searched}	with <i>Class A</i>
A	24	1	0.0417	1
F	402	9	0.0224	4
G	860	16	0.0186	15
K	1 400	6	0.0043	6
M	5 908	21	0.0036	13

public execution blocks in scope are reduced to 484 with a usable calibration and 404 in the science sample once the pre-registered hold-out is withheld; those yield 1655 spectral windows, of which 403 are fine-channel enough to resolve drift; 75 windows carry a threshold crossing at the stellar position; 13 of those also outrank all 512 spatial controls and 10 still do after the radial correction of §C.0.3; 6 survive the visibility-domain test; 4 have no astrophysical attribution; and 0 are independently confirmed. The first five steps assemble the data and its inventory and the remaining six test it; the boundary is the threshold crossing, and everything from it onward is the candidate sequence of §4.1.

4. DATA AND METHODOLOGY

Abbreviations used below. QA2 is ALMA’s second-stage quality assurance, the stage that delivers calibrated data; a member observing unit set (MOUS) is the scheduling container holding one or more execution blocks; TDM and FDM are the correlator’s time- and frequency-division modes, giving coarse and fine channelisation respectively. Symbols, including T_* , T_{trig} and the power scales, are defined in Appendix E.

The design is one shared archive-ingestion stage feed-

TABLE 2

THE SURVEY AS SEARCHED, AND THE PAPER’S HEADLINE NUMBERS IN ONE PLACE. EVERY COUNT QUOTED IN THIS PAPER RECONCILES WITH IT. THE TWO COMPLETENESS EXPERIMENTS MEASURE DIFFERENT QUANTITIES AND ARE KEPT IN SEPARATE TABLES. EVERY FREQUENCY UNION QUOTED HERE IS AN AGGREGATE OVER A CLUSTERED, REPEATEDLY REVISITED COVERAGE, WHOSE ACTUAL DISTRIBUTION IS FIG. 4; THE UNION SHOULD NOT BE READ AS CONTINUOUS COVERAGE OF ITS ENDPOINTS.

Quantity	Value
Stars within 40 pc with qualifying public ALMA coverage	168
genuinely pointed at; never observed (§A.3)	115; 53
Stars searched; independent systems	90; 82
Star/band datasets	113
Spectral windows searched	1655
12 m array; ACA 7 m	601; 1054
extracted; repeats, defective, withheld	1725; 45, 13, 12
Class A fine-channel / Class B coarse-channel	403 / 1252 (76% Class B)
Execution blocks	404
Distance range	1.30–39.63 pc
$\Delta\nu_A$, unique frequency union, Class A (the drifting-carrier experiment)	47.7 GHz
$\Delta\nu_B$, full Class B union (overlaps Class A by 20.1 GHz)	90.5 GHz
$\Delta\nu_{B\text{-only}} = \Delta\nu_{A \cup B} - \Delta\nu_A$, coverage Class B adds	70.4 GHz
$\Delta\nu_{A \cup B}$, total archival coverage	118.1 GHz (46 intervals, 89.6–873.1 GHz; Fig. 4)
usable after the molecular-line mask	113.3 GHz
Windows by ALMA band (3/4/5/6/7/8/9/10)	112/4/4/1072/427/24/4/8
Effective after line-exclusion mask	≈113.3 GHz
Native channel widths	15.3 kHz–31.25 MHz; median 15.625 MHz
Drift-rate ceilings	1.1–10.5 kHz s ^{−1} (12.0–13.3 GHz s ^{−1} GHz ^{−1})
On-source time per window	21–5274 s (median 2812 s)
Nominal S_{min}	0.41 mJy–1.30 Jy (median 9.8 mJy)
90 per cent selection P_{90}^{sel} , Class A, per window	8.7×10^{13}–1.4×10^{17} W (median 2.8×10^{15}) ^a
per system, best window	8.7×10^{13} – 4.4×10^{16} W (median 1.5×10^{15})
90 per cent trigger P_{90} , Class A	6.3×10^{13} – 1.1×10^{17} W (median 2.4×10^{15})
Effective $P_{\text{eff}} = P_{\text{trig}} C_{\text{resp}} C_{\text{smear}}$, per window, Class A	5.1×10^{13} – 9.1×10^{16} W (median 1.9×10^{15})
Class B	3.7×10^{13} – 6.4×10^{17} W (median 1.4×10^{15})
Nominal trigger P_{trig} , Class A (15.3–1953 kHz)	2.2×10^{13} – 6.8×10^{16} W (median 9.4×10^{14})
Class B (7.812–31.25 MHz)	1.6×10^{13} – 2.8×10^{17} W (median 6.0×10^{14})
Completeness, by class and experiment	Table 4 (amplitude); Table 4 (dwell)
Threshold crossings ($T_* \geq T_{\text{trig}}$ at the stellar position)	75 windows (71 Class A, 4 Class B; Table 1)
Star-exceeds-ring windows ($T_* \geq T_{\text{trig}}$)	13 windows (13 Class A, so all of them)
circumstellar or foreground CO	9
unattributed, absent in the repeat block	4
Persistent, independently confirmed narrowband spectral technosignatures, within the domain searched	0

^a Power and frequency quantities are given to two significant figures throughout, except where a value is a channel or sky frequency that must be matched to the catalogue, which is given to the precision the grid supports. The internal consistency checks quoted elsewhere to a per cent (§L.1.2) bound *relative* quantities within one dataset; the absolute flux scale rests on ALMA’s delivered calibration, whose accuracy across the 2013–2025 span is several per cent and is not improved by anything done here.

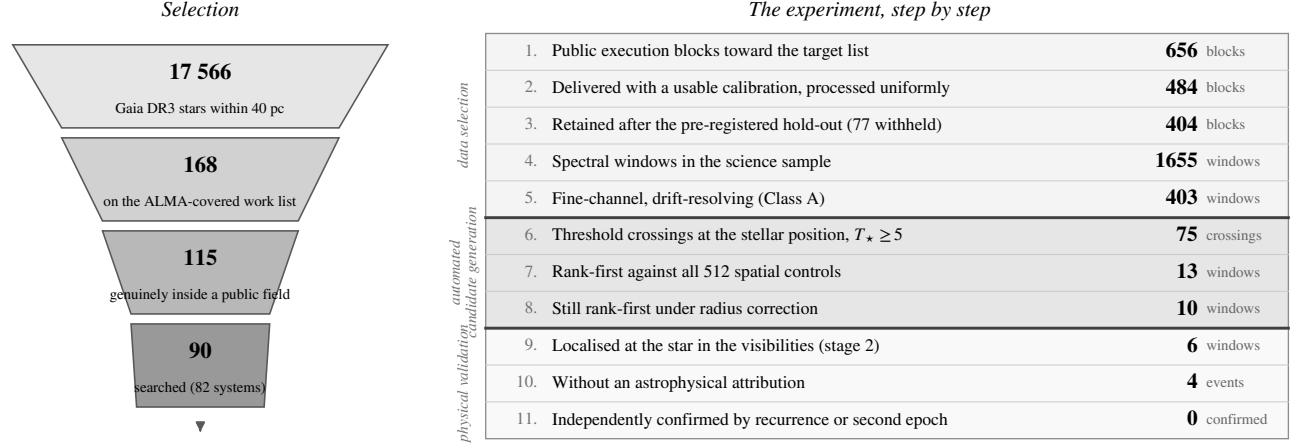


FIG. 3.— Left: the selection funnel. The three denominators the text keeps apart are the catalogued population, the 168-entry ALMA-covered work list, the ~ 115 stars a public field genuinely contains, and the 90 searched. Right: the experiment in eleven steps with the number surviving each, matching §4.5 line for line — from the public execution blocks in scope, through the pre-registered hold-out and the window inventory, to the candidate tests. The 512-control comparison at step 7 is a candidate-generation screen and not a test of the null; step 8 is the same screen with the radial gradient removed. Every count is read from the generated macros, so the figure, the list and the released catalogue cannot disagree.

ing two independent search lanes, and Table 3 gives the data path of the primary lane in order. For each target we mine, within a bounded budget, the calibrated archival execution blocks covering the star’s Gaia-propagated position. Their visibilities feed (i) the *primary* lane, a Doppler-drift-corrected spectral-carrier search following Mason et al. (2024) and Enriquez et al. (2017), which converts each non-detection into an equivalent isotropic radiated power limit $\text{EIRP}_{\text{trig}} = 4\pi d^2 S_{\text{min}} \Delta\nu$. The product $S_{\text{min}} \Delta\nu$ is an integrated power only if the channel’s effective noise bandwidth equals its separation, which for a smoothed correlator it does not; that is precisely what the response factor C_{response} corrects, and we calibrate the correction against injected signals. Unresolved tones of known amplitude were injected into real visibilities and recovered through the unmodified pipeline. The measured factor is $\times 2.31$ at the median, spanning $\times 2.02$ – $\times 2.69$ across correlator modes, against the $\times 2.29$ predicted analytically for the Hanning response (Appendix G): the two medians agree to one per cent. The spread is window-to-window variation in the correlator response and not an error on the estimator, so $P_{\text{eff}} = S_{\text{min}} \Delta\nu C_{\text{response}} C_{\text{smear}}$ recovers the injected power. EIRPs are quoted to two significant figures, which is the precision the flux scale supports. Here S_{min} is the primary-beam-corrected trigger-level flux density defined below, $\Delta\nu$ the native channel width and d the distance (worked numerically in Appendix F). Five terms enter the systematic budget, each multiplying S_{min} and hence $\text{EIRP}_{\text{trig}}$ directly.

- (i) The ALMA absolute flux scale: 7 per cent band-weighted, up to 20 per cent at Bands 9 and 10, and epoch-dependent.
- (ii) Residual atmospheric phase after water-vapour correction and phase referencing: a coherent point-source loss $\exp(-\sigma_\phi^2/2)$, 5–20 per cent at 230–345 GHz. It is common to star and annulus within the isoplanatic patch and so no threat to exchangeability, but it is *not* common to the injected tones that calibrate P_{90} , which is why §5.3 carries it as a one-sided bias.

- (iii) The primary-beam model at the stellar offset: negligible on axis, 1.9 per cent at the worst pointing error.
- (iv) Visibility calibration: bounded empirically at 4.5 per cent. If the delivered amplitude scale were inconsistent between execution blocks, the ratio of each window’s measured per-integration rms to its radiometric prediction would scatter from block to block beyond what thermal noise explains. Over 2271 windows in 559 blocks the per-block median of that ratio is 0.956 with a robust scatter of 4.5 per cent, against 0.8 per cent within a block — so the term is real, is the larger of the two, and is smaller than the flux-scale term. Earlier versions of this paper left it unquantified while quoting the combined budget as authoritative; including it raises that combination from 8 to 9 per cent, and 9 per cent is what we now quote.
- (v) The instrumental spectral response: median $\times 2.29$, the largest of the five by an order of magnitude, carried explicitly as P_{eff} by the reading rule below.

Gaia distance errors are negligible, 0.2 per cent typically. The same visibilities feed (ii) an *ancillary* line-excluded continuum lane (Appendix J). In most windows an Hz-wide carrier is unresolved within a 15.625 MHz channel, so the data cannot show a one-channel excess to be intrinsically narrow. Two survey classes are reported, distinguished by the dimensionless drift parameter

$$\eta_{\text{drift}} = |\dot{\nu}_{\text{max}}| \tau_{\text{track}} / \Delta\nu_{\text{ch}}, \quad (1)$$

the channel-width fraction the *maximum* searched drift $|\dot{\nu}_{\text{max}}|$ traverses over the full on-source track τ_{track} at native channel width $\Delta\nu_{\text{ch}}$. The track duration is written τ_{track} throughout, leaving T for the search statistic alone. The track is the right time-scale, the de-drift step summing along it; η_{smear} (Appendix F) is the same ratio over one integration. A window is *drift-resolving* when $\eta_{\text{drift}} \gtrsim 1$, carrying a carrier across enough channels for its motion to be measured. The drift ceiling inside η_{drift} is a search-grid boundary, the widest trial drift the stack evaluates, not a prior on transmitter drift rates; carriers

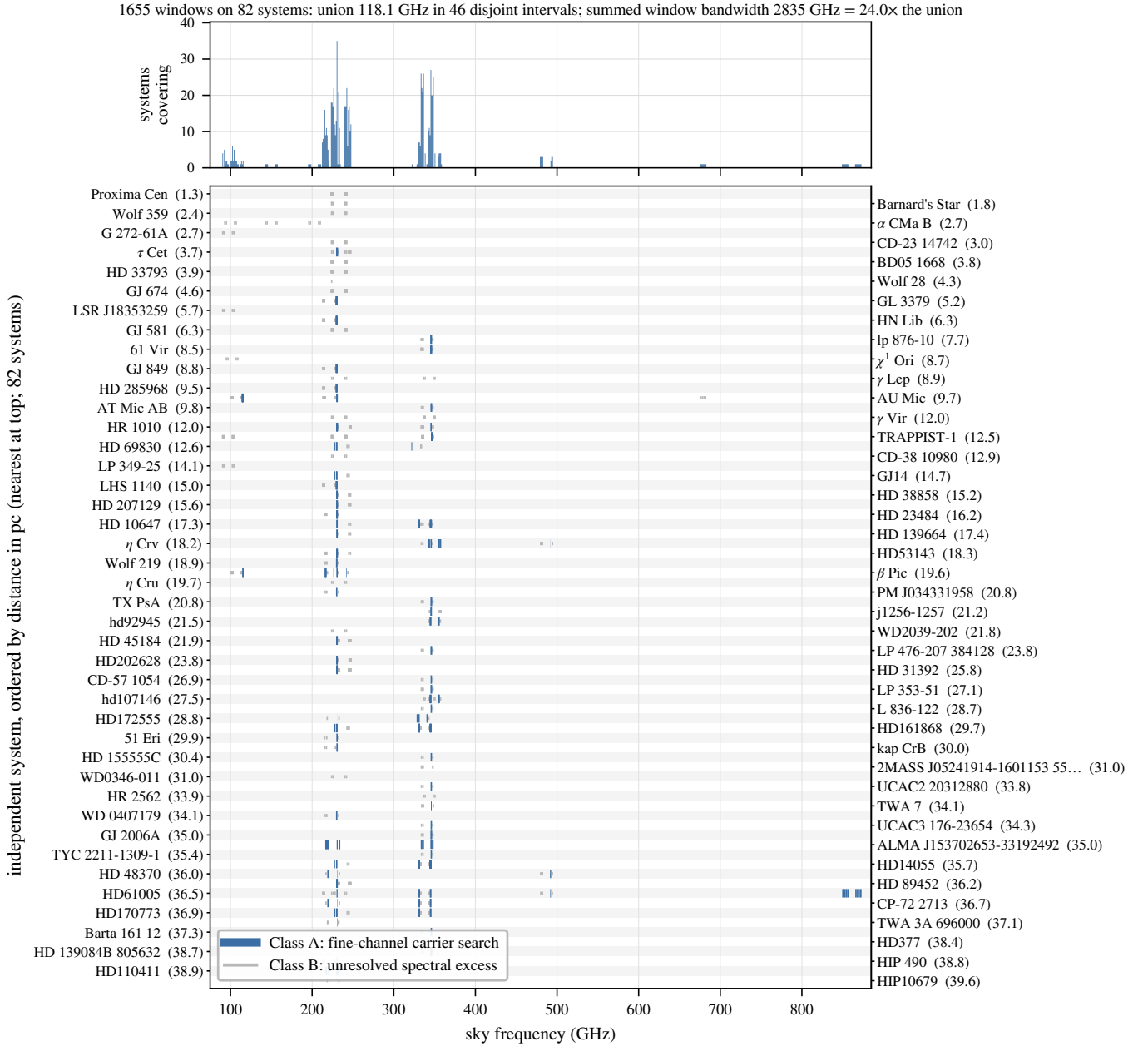


FIG. 4.— Where this survey searched. *Lower panel*: each searched window is a horizontal segment on its host system's row, rows ordered by distance and labelled alternately on the left and right axes; heavy blue marks the fine-channel carrier search (Class A) and light grey the coarse-channel spectral-excess search (Class B). *Upper panel*: the number of systems covering each gigahertz of sky frequency. The panel shows coverage and search class only; the threshold distribution is Fig. 1. The coverage is narrow, clustered and repeatedly revisited: 46 islands concentrated near 230 and 345 GHz, with summed window bandwidth 24.0× the union. The union is the frequency coverage; the summed bandwidth is the integrated search effort, and the two answer different questions (Scope of the constraints, §4).

drifting faster are missed, and §6.1 gives the resulting selection function (Fig. 6).

Class A is the fine-channel spectral-carrier search: the 403 windows of fine channelisation, native widths 15.259–1953.125 kHz (median 488.3 kHz; 60 per cent at the 488.3-kHz mode), spanning $\eta_{\text{drift}} = 0.17\text{--}422$ (median 14.7). It is the only class with measured end-to-end recovery of a drifting carrier. *Class B* is the coarse-channel spectral-excess search: the 1252 coarse windows, 76 per cent of the sample, native widths 7.812–31.25 MHz (1212 windows at the 15.625-MHz mode), spanning $\eta_{\text{drift}} = 0.004\text{--}1.35$ (median 0.51). It is sensitive to persistent near-stationary excess power, carries no drift discrimi-

nation, and is never described as a narrowband search anywhere in this paper.

η_{drift} is the physical criterion and the classes are labelled by the instrument. Channelisation is bimodal, with no window between 1953.125 kHz and 7.812 MHz, an empty gap of factor 4.0, so any boundary inside the gap assigns identical membership and figures label the classes by name. The two criteria do not coincide exactly, and the overlap is quantified: 38 long-integration Class B windows reach $\eta_{\text{drift}} \geq 1$, and only marginally (maximum 1.35), where a one-to-two channel traverse supports no discriminative drift fit after baseline removal; 13 short Class A windows fall below unity and are con-

servatively kept in Class A. On the physical criterion alone 428 windows of 1655 reach $\eta_{\text{drift}} \geq 1$, against 403 in Class A, so the labels are channelisation labels and “drift-resolved” is used of the class only where the two agree. Each window’s η_{drift} , η_{smear} , trial drift count and maximum searched acceleration $a_{\text{max}} = 3.60\text{--}4.00 \text{ m s}^{-2}$ are columns of the released catalogue, so any reader may reclassify on the physical criterion alone. Results are given per class first and combined only where statistically appropriate (§5.2).

Both lanes exclude catalogued molecular transitions at tolerances suited to their statistics. The continuum lane masks the common species at native resolution before integrating; the carrier lane applies a $\pm 50 \text{ km s}^{-1}$ stellar-frame velocity exclusion mask (§5.2.6) to any threshold crossing. That half-width was widened after the β Pictoris crossings had been seen, so the safeguard belongs here rather than in a later section: reclassifying all 75 crossings at ± 20 , ± 30 , ± 50 and $\pm 100 \text{ km s}^{-1}$ admits no new unattributed outlier — a one-sided test, see §5.2.6 — and at every width CP–72 2713 remains the only unmasked one. The mask defines frequency space *excluded from the experiment*: the unmasked survey covers 118.1 GHz of unique sky frequency in 46 clustered islands (Fig. 4), the effective default quoted throughout is the $\approx 113.3 \text{ GHz}$ surviving it, and no constraint is placed on carriers inside the excluded velocity regions. Masking reduces the frequency space a stage-1 event can occupy while leaving the per-channel noise untouched, so no threshold moves. All flux densities and EIRP thresholds are total intensity (Stokes I), summed over the two parallel-hand cross-correlations with QA2 weights; no cross-hand extraction or circularity test was performed, so the construction is polarisation-blind in amplitude and a strongly polarised artificial signal responds like an unpolarised one at fixed total power. The estimator is safe in this respect, though the data have not been audited for it. Both parallel hands are delivered for 198 of the 200 searched execution blocks whose polarisation states the archive lets us resolve (50 per cent of the 404), so the discrimination described next is available from the existing data for most of the sample and needs no new observation. What the retained products do not record is per-correlation flagging inside QA2, so we cannot say what fraction kept both hands unflagged all the way through; one that lost a hand would be $\sqrt{2}$ less sensitive than its threshold implies, with nothing here to catch it. What Stokes I costs is discrimination, not sensitivity: an artificial coherent carrier is more likely to be strongly polarised than the unpolarised line and dust emission that fills this archive, so a polarisation measurement would separate the two for free.

The three sensitivity quantities of Table 12 are defined here in full.

The trigger power P_{trig} is five times the window’s per-channel rms at its native channel width, for a transmitter at the star’s position matching the searched morphology. It is the level at which the search fires and what every reference to the trigger below means; it is an internal pipeline quantity, not a sensitivity to compare with a transmitter, and not a significance. The two numbers to quote are P_{eff} and P_{90} .

The effective unresolved-carrier threshold P_{eff} is P_{trig}

corrected for the instrument’s spectral response. ALMA applies online Hanning smoothing, which leaves only 0.38 to 0.50 of a sub-channel tone’s power in the peak channel depending on where in the channel it falls, median 0.44 (Fig. 5, Appendix F). Nominal thresholds are therefore optimistic by $\times 2.00$ to $\times 2.67$, median $\times 2.29$, and $P_{\text{eff}} = 2.29 P_{\text{trig}}$ is the number to compare with any transmitter power. It follows from the 0.25/0.5/0.25 kernel and is window-specific ($\times 1.33\text{--}2.00$ for the 146 windows with online channel averaging), and injection through the response kernel reproduces it to 1.1 per cent (median $\times 2.31$; Appendix G). It is a sensitivity correction and not a completeness, it is not small, and we treat it as the principal physical threshold.

The completeness limit P_x , the power at which measured recovery reaches x per cent, is a different quantity again: $P_{50} = 1.10 P_{\text{trig}}$ and $P_{90} > 2 P_{\text{trig}}$ on the calibration window’s drifting class (Appendix G). Transfer to other bands, channelisations and integration times is measured on 28 further configurations, where P_{50} spans $0.64\text{--}1.13\times$ its median (Appendix G); P_{95} remains *not determined*. What this survey has is a measured reference recovery function whose transferability is now bounded by measurement, and not a measured Class A completeness function.

Every headline EIRP is quoted beside its native channel width, and the catalogue carries P_{trig} and both response-corrected values per window.

Three signal morphologies are detectable here, and §6.1 quantifies each against real windows: a continuous frequency-stationary carrier, a continuously drifting one, and an intermittent one.

4.1. Final candidate definition and statistical interpretation

Which statistic is primary, and why. The spatial screen was frozen before the candidate list was examined, in the form that ranks the stellar position against 512 single-position controls under one global noise normalisation. Validation showed that form to be non-exchangeable: the stellar rank is displaced from the uniform expectation, and the mechanism is understood. Images are corrected for the primary-beam response, so the true noise rises with distance from the pointing centre, while the search divides every position by a single scale estimated over the whole window. The control probes therefore sit at systematically different effective noise from the star, which is at the phase centre.

A radius-corrected normalisation removes that radial component by construction, standardising each probe within its own radius bin with no free parameter (Appendix L). An earlier version of this paper adopted it as the primary statistic. We have since tested it prospectively, on 778 windows that played no part in building it, and it does not restore exchangeability (§5.2.3). The corrected stellar rank remains inconsistent with uniformity out of sample, and its median moves further from the exchangeable value rather than toward it. A correction that does not deliver a calibrated statistic cannot be preferred merely because its mechanism is understood.

The originally frozen screen therefore remains the primary candidate-generation statistic — the statistic that decides which windows are examined — and neither form

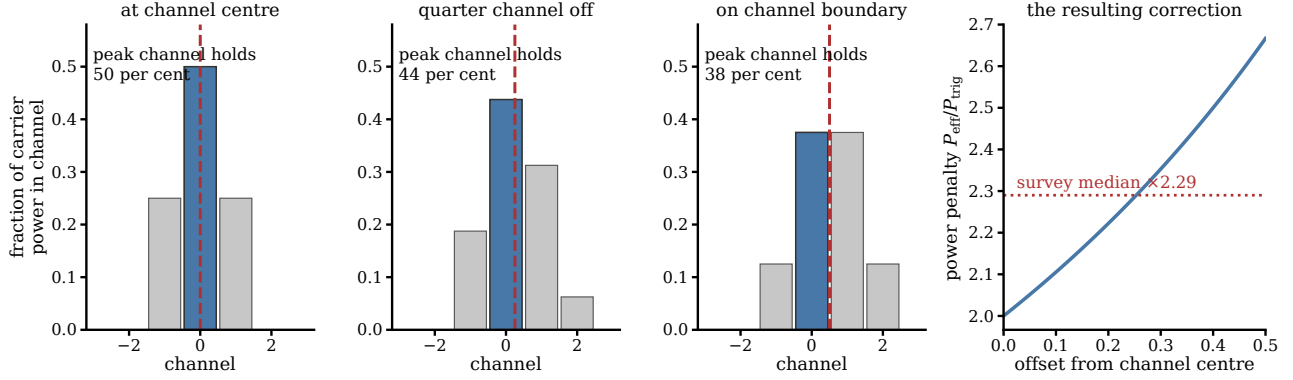


FIG. 5.— Why the nominal channel threshold is not the transmitter threshold. ALMA applies online Hanning smoothing by default, so an intrinsically unresolved carrier is spread over neighbouring channels and the peak channel holds only part of its power: half at a channel centre, 0.38 on a boundary (first three panels; blue is the peak channel, the dashed line the carrier). The search threshold is set in units of the channel noise, so the power a transmitter must radiate to reach it is larger than the nominal figure by the reciprocal of that fraction (right-hand panel), from $\times 2.00$ to $\times 2.67$ with a survey median of $\times 2.29$. This is the whole of the difference between P_{trig} and P_{eff} in Table 12.

is treated as a significance test. The word “primary” is used in that sense throughout. It is not to be confused with the separate question of which normalisation gives the preferred comparison for the false-alarm *rate*, which §5.2 treats on its own and where both forms are reported side by side. Because that distinction is easy to lose while skimming, we fix a notation for it here and keep it for the rest of the paper: a tail quantity computed against the screen’s own exchangeability-violating null is written $\mathcal{P}_{\text{scr}}$, never p . It is the probability of the observed count under a stated and imperfect null, not a calibrated significance. The symbol p is reserved for genuine significance levels, such as the Kolmogorov–Smirnov tests on rank distributions. Both are reported for every window in the released catalogue, the corrected one as a robustness comparison, and the consequences for the candidate list are given in full in Appendix L. What disposes of an event is not its rank under either normalisation but the physical evidence: whether the emission is localised at the stellar position in the visibilities (§5.2.1), whether it coincides with a catalogued transition in the stellar frame, and whether it recurs in an independent epoch.

Neither form is a significance test, and we say so before developing either. Ranking the star against 512 control positions drawn from the same field is a convenient and powerful way to decide which windows deserve examination. It is not a p -value: $1/(512 + 1) = 1.95 \times 10^{-3}$ is the *resolution* of the ensemble, the smallest rank it can express, and not the probability that noise produced the window.

Two consequences follow immediately and both are structural. First, no single window in this design can carry survey-level significance by rank alone: against a survey-wide Bonferroni scale

$$\alpha_{\text{survey}} = \frac{0.05}{1655} = 3.0 \times 10^{-5}, \quad \frac{1.95 \times 10^{-3}}{\alpha_{\text{survey}}} = 65, \quad (2)$$

the ensemble’s floor is 65 times too coarse (Appendix K). Second, a technosignature claim here therefore requires independent recurrence or independent astrophysical confirmation, and never rank alone.

The decisive test is interferometric, not statistical. A carrier at the stellar position must appear coherently in

the real part of the visibilities after phase rotation to that position, with an imaginary part consistent with zero. That is a physical statement about where the emission is, it does not rest on the exchangeability assumption the spatial screen violates, and it is therefore the test on which every disposition in this paper turns. The screen that ranks the star against 512 control positions decides only which windows are examined.

The selection logic is five steps, in order. The first two generate candidates; the third is the primary physical test.

- (i) Threshold crossing. A channel $\times$ drift cell at the stellar position reaches $T_{\star} \geq T_{\text{trig}}$. A fixed trigger power, not a calibrated p -value.
- (ii) Spatial prioritisation. The stellar statistic exceeds all 512 control positions in the same window. A candidate-generation mechanism: it decides which windows are examined and carries no significance of its own, for the reasons above.
- (iii) Visibility-domain localisation. The emission is at the stellar position in the visibilities. Each event’s own channel is followed along its recovered drift rate, the local continuum is removed per integration, and the visibilities are phase-rotated to the star: emission centred there gives a positive real part with an imaginary part consistent with zero (§5.2.1). This is the first test in the chain to ask a physical question about the sky, where the two before it ask a comparative question about an ensemble; it is also the first a window can fail on evidence, not on rank.
- (iv) Astrophysical identification. The event is checked against catalogued molecular transitions in the stellar frame and against instrumental signatures.
- (v) Independent recurrence. It repeats at the same tuning in an *independent epoch*, or acquires independent confirmation.

The visibility test was applied to the 13 windows the screen flagged, not to all 75 threshold crossings, because the recalibrated visibilities were retained only for windows that reached that stage. Extending it to the remaining 60 crossing windows requires re-downloading and recalibrating 50 execution blocks, 0.31 TB, which the measured cost model of this pipeline puts at about

15 h of processing; that extension is under way and will be reported separately. We note the asymmetry plainly: the windows tested in the visibility domain are those the image-plane screen selected, so the test as applied here vets the screen’s output rather than replacing the screen. Stage (iii) does not use the control ensemble at all, which is why it is worth its cost. It runs on recalibrated raw data, it can fail a window the image-plane statistic promotes, and it carries an internal positive control: the same fit recovers the β Pictoris CO emission at the stellar position in 6 of its 8 windows at 6.1–9.5 σ . Every stage-1 event is carried through all five stages in Table 7. Nothing short of (v) is called a candidate.

All delivered channels of each spectral window were searched: no edge-channel or bandpass-roll-off trimming was applied before extraction, so residual bandpass structure near a window edge is carried entirely in the calibration-residual term of the systematic budget rather than removed by flagging. The consequence is checked directly — the distance of every threshold crossing from its window edges is a released catalogue quantity, and §5.2 reports it for the flagged windows.

The primary statistic is formed on extracted spectra, never on images or visibilities, from the QA2-calibrated measurement set as delivered, per integration and per native channel, with the archive’s flagging carried through untouched and no regridding (Table 3, step 3). What the extraction returns at a propagated sky position $\mathbf{x}$ is the real part of the weighted vector average of the phase-shifted visibilities, which is the natural-weighted dirty-map value there. There is no deconvolution, so a quoted S_{peak} in a field with extended emission is a dirty-map amplitude. Baseline removal follows extraction (step 4). The primary-beam correction enters the quoted thresholds and fluxes (§4.2) and leaves untouched the extracted amplitudes the statistic ranks. For a sky position $\mathbf{x}$, each per-integration spectrum $I(\mathbf{x}, t, \nu)$ is baseline-subtracted by a block median along the *frequency* axis. No temporal median is applied at any stage (§4.3). The residual dynamic spectrum is stacked along each trial linear drift rate $\dot{\nu}$ with inverse-variance weights, and the statistic is the maximum of that stack over the channel $\times$ drift grid,

$$T(\mathbf{x}) = \max_{\dot{\nu}, \nu} \frac{\sum_t I(\mathbf{x}, t, \nu + \dot{\nu}t) / \sigma^2(t, \nu)}{[\sum_t 1/\sigma^2(t, \nu)]^{1/2}}, \quad (3)$$

a matched filter whose denominator is the noise of the stacked spectrum. $T(\mathbf{x})$ is neither assumed nor shown to be Gaussian or χ distributed under the null, which is why T_{trig} here denotes a trigger level and not a tail probability; Appendix K quantifies the departure. Only when σ is the same for every integration and channel does it reduce to a single-channel amplitude divided by one noise scale; otherwise the weights vary along the track. The scale $\sigma(t, \nu) = 1.4826 \text{ MAD}_{i=1\dots 512}[I_i(t, \nu)]$, that is 1.4826 $\times$ the median absolute deviation taken *across the control positions* at each integration t and each channel ν , which makes it global in position, local in frequency and in time, and shared by the star and by every control.

The across-position median enters only in forming that deviation and is never subtracted from the numerator, so T is an absolute stacked signal-to-noise carrying a cross-position noise scale, and it measures no local contrast.

Why this form and not a region maximum. T is eval-

uated at one position, identically at the star and at each control, and that is what makes the rank interpretable. If instead the stellar side takes a maximum over n_{src} positions while the controls remain single positions, the two sides are no longer exchangeable: under the null the region maximum exceeds all N_{ctrl} controls with probability $n_{\text{src}}/(n_{\text{src}} + N_{\text{ctrl}})$, not $1/(N_{\text{ctrl}} + 1)$. With $n_{\text{src}} = 135$ and $N_{\text{ctrl}} = 512$ that is 21 per cent rather than 0.2, a floor under which most crossings flag whatever the data contain. The argument is combinatorial and refers to no window: it holds for any statistic, any noise distribution and any field. A region-maximum form of the statistic fails for exactly this reason; §L.2 gives the comparison and Table 19 audits every window either form flags. A line filling the primary beam therefore drives star and annulus up together instead of cancelling.

The searched drift-rate range is the larger of (a) a generic frequency-scaled default (12–13 Hz s $^{-1}$ GHz $^{-1}$), the Mason et al. (2024) fiducial with a threefold safety margin, and (b) for confirmed exoplanet hosts, the maximum line-of-sight orbital acceleration of the innermost known planet, so a transmitter on a short-period planet cannot drift out of the window. Drift rate maps onto line-of-sight acceleration through the first-order Doppler relation

$$a_{\text{los}} = c \frac{\dot{\nu}}{\nu}, \quad (4)$$

so the ceiling is quoted in cross-band-comparable form $\dot{\nu}/\nu$: 12–13 Hz s $^{-1}$ GHz $^{-1}$ is $(1.2\text{--}1.3) \times 10^{-8}$ s $^{-1}$, that is $|a_{\text{los}}| = 3.6\text{--}3.9$ m s $^{-2}$ independent of observing frequency, the acceleration at ~ 0.04 au around a solar-type star. Earth-motion contributions lie two orders of magnitude below and are subsumed by the grid. The grid is uniform in $\dot{\nu}$ over n_{drift} steps (2–1944 across windows, recorded per window), so a carrier midway between trial rates leaves a residual drift traversing at most 0.26 of a native channel over the longest fine-class track, and the resulting amplitude loss is an order below the recovery slope measured between $T_{\star} = 5$ and 10. Two limits of the model matter more. The rates searched are *apparent* topocentric ones, so the survey does not separate a deliberately chirped carrier from an accelerating platform and constrains only linear apparent drift inside the ceiling; and an emitter need not be tied to a known planet at all, a free-flying platform or rotating structure accelerating as it chooses and falling outside these constraints without being excluded by them. For circular orbits the ceiling covers a transmitter co-orbiting an Earth analogue by three orders of magnitude and Proxima b comfortably, while TRAPPIST-1 b ($a_{\text{los}} = 4.00$ m s $^{-2}$) sits at it and c (2.14 m s $^{-2}$) reaches it at modest eccentricity, which raises the periastron value by up to $(1+e)/(1-e)^2$ (Fig. 6). That value is the maximum over orbital phase, not a typical one. TRAPPIST-1 transits, so its inclination is measured and effectively edge-on, and the relevant figure is the edge-on one: TRAPPIST-1 b exceeds the ceiling over 5–29 per cent of its orbital phase, depending on which end of the 12.0–13.3 Hz s $^{-1}$ GHz $^{-1}$ ceiling applies, and is searchable through the rest of it.

Across the 1655 windows maximum searched drift rates span 1.1–10.5 kHz s $^{-1}$ and trial drift rates per window 2–1944 (median 4), the largest being the narrow, finely channelised β Pictoris Band 6 window (§D.0.3). 11 win-

TABLE 3

THE STATISTICAL DATA PATH, IN ORDER. STEPS 1–8 RUN IDENTICALLY AT THE STELLAR POSITION AND ALL 512 CONTROLS; ONLY STEP 9 ONWARD DISTINGUISHES THEM.

1	calibrated measurement set from the ALMA archive
2	stellar position propagated to the observation epoch (Gaia DR3)
3	spectral extraction at that position, native channelisation
4	per-integration baseline removal: block median along frequency ($W = 65$)
5	stack along each trial linear drift rate
6	robust noise map $\sigma(t, \nu)$: MAD across the 512 controls
7	$T(\mathbf{x})$ from Eq. 3, an inverse-variance-weighted stack
8	repeat 3–7 at 512 control positions on a ring
9	threshold: $T_* \geq 5$ (a crossing)
10	rank test: $T_* > \max_i T_i$ (stage-1 flagged window)
11	molecular-line mask, $\pm 50 \text{ km s}^{-1}$ stellar frame
12	vetting: recurrence and astrophysical attribution

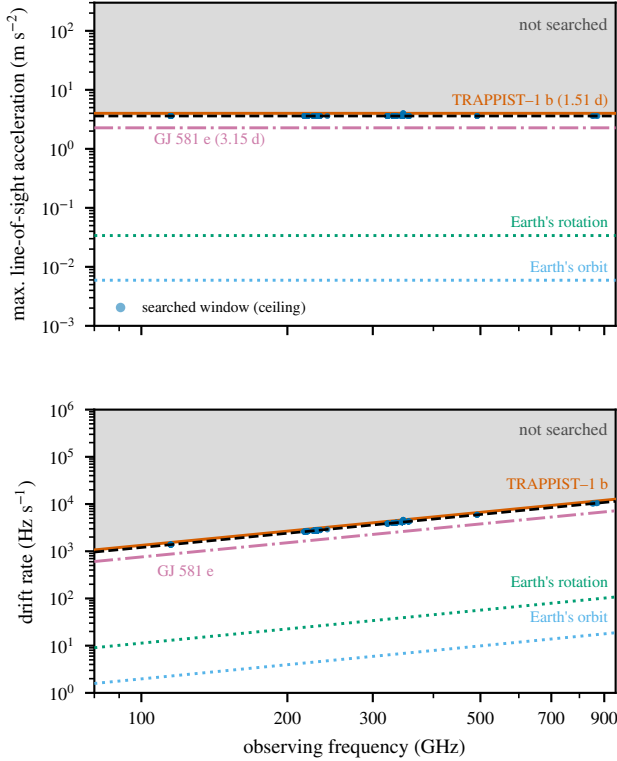


FIG. 6.— What the drift search does and does not reach, as a physical selection function. *Upper*: the drift ceiling of each of the 403 Class A windows (points), converted to the line-of-sight acceleration it corresponds to, $|a| = c\dot{\nu}/\nu$, against observing frequency; shaded above is unsearched. The boundary is flat because the grid was built at constant acceleration, 3.60 m s^{-2} , and not at constant drift rate. *Lower*: the same boundary in the units the search works in, where it is not flat: the same 3.60 m s^{-2} is 1.4 kHz s^{-1} at 115 GHz and 10.5 kHz s^{-1} at 872 GHz . Reference accelerations, all frequency-independent and so horizontal in the upper panel: Earth’s rotation (0.034 m s^{-2} at the equator) and orbit (0.0059 m s^{-2}), which a terrestrial transmitter carries whether or not it intends to; the most demanding known planet of any target, TRAPPIST-1 b (4.00 m s^{-2}); and the next most demanding, GJ 581 e (2.27 m s^{-2}). TRAPPIST-1 b is the one case the grid does not contain, and it misses by 0.01 per cent (§6).

dows are narrower than 100 MHz . The threefold margin matters. At $1\times$ the search would have been blind to two of its threshold-crossing windows, including the AU Mic crossing, while at $5\times$ no disposition changes; transmitters drifting faster than the $12.0\text{--}13.3 \text{ Hz s}^{-1} \text{ GHz}^{-1}$ ceiling remain unconstrained. Intra-integration smearing is negligible for 446 of the 455 windows that carry a value: at most 1.10 channels, median 0.001. 9 windows have

$\eta_{\text{smear}} < 0.99$ (Appendix F); 5 of them, all 15.3 kHz Band 6 windows and so among the most drift-capable here, carry effective thresholds $1.65\text{--}1.74\times$ nominal, and 2 are stage-1 flagged windows. Channel widths are effective resolutions from the measurement sets’ spectral window tables (Appendix F).

4.2. Frequency and velocity reference frames

Every frequency axis here, searched spectra, tabulated ranges and crossing frequencies alike, is *topocentric* as delivered by the ALMA correlator, with no velocity-frame shift applied at any stage. Velocities enter only in the vetting stage’s line mask, whose conversion chain is in Appendix F. The stacking is drift-aligned and inverse-variance weighted: coherent in frequency trajectory, and not in electric-field phase.

The native channel width is also the linewidth convention for every EIRP threshold, and the commonest confusion when limits are compared across surveys. Channel dilution spreads a sub-channel carrier’s total flux across the channel, so $\text{EIRP}_{\text{trig}}$ is the minimum *total* power of a channel-confined carrier, independent of the transmitter’s own width below the channel, and never a spectral power density. The Hz -resolution surveys’ limits are total powers in the same sense, each the best case of its own channelisation, which is what makes the axes of Fig. 1 commensurable. We do not project an Hz -resolution sensitivity from ours, but it is worth one number so that the reader has a sense of scale rather than only a disclaimer. *Illustratively and not achievably*: the median Class A channel is 488 kHz , so rechannelising the same data to the 2.79 Hz of a dedicated narrowband search would, on $P_{\text{min}} \propto \sqrt{\Delta\nu}$, divide the threshold by 418 and put the median Class A selection power at $\sim 6.8 \times 10^{12} \text{ W}$. ALMA cannot do this — the correlator delivered the channelisation the archive holds, and no reprocessing recovers resolution that was never recorded — so the number quantifies the gap between this experiment and a dedicated one, and is not a limit of any kind.

Every spectral window of an observation enters the spectral-carrier search, capped at 8 per target to bound compute cost; the cap bound one target/band, 61 Vir Band 7, and three further targets are represented by their deepest single window and marked in the released catalogue. Each searched window yields its own $\text{EIRP}_{\text{trig}}$ row, and Figure 1 plots only the deepest per star. When a crossing is checked against the masked species the operative discriminant is velocity coincidence, since circumstellar emission sits at rest offset by the gas’s kinematics while a technosignature may appear at any frequency

and drift rate.

Both carrier-lane thresholds and continuum fluxes are corrected for ALMA’s primary-beam response at the star’s offset from the phase centre, negligibly for most targets (median factor 1.00003, 90th percentile 1.002), though 87 of the 1655 retained windows exceed 2 per cent and 73 exceed 10 per cent. Retention is decided by an explicit response floor rather than case by case: a window is kept only where the primary-beam response at the stellar position is at least 0.5, a correction of at most $\times 2$, because at low response a small error in the beam model becomes a large error in inferred flux. *The offsets are bimodal and the floor sits in the gap*: the last retained window has response 0.55 at $0.46\theta_{\text{PB}}$ (LP 736-15 Band 7, $1.81\times$), the first window beyond it has response 0.35 at $0.62\theta_{\text{PB}}$ ($2.88\times$), and nothing lies between them, so the rule binds prospectively without having been drawn around the present sample. It excludes none of the 1655 released windows for that reason and not because it was chosen not to. Beyond it are the 12 ϵ Eri Band 6 windows at $1.0\text{--}3.5\times$, which is also past the first sidelobe transition where the Gaussian form fails: the two criteria are independent and select the same 12 windows. We *withhold* them, and they enter no number here and are not in the released catalogue. Under the same nominal rule they would have covered $3.7 \times 10^{13}\text{--}1.1 \times 10^{15}$ W at 3.22 pc, which at the lower end is among the deepest thresholds in the survey, and that is exactly why we decline to quote them on a beam model we cannot defend. The annulus geometry keeps the pipeline’s own $\theta_{\text{PB}} = 1.22\lambda/D$, an Airy first-null coefficient; re-computing them on ALMA’s FWHM $1.13\lambda/D$ at each window’s own dish diameter moves 79 of 1655 windows by more than 1 per cent, extremes -7.3 to $+10.3$ per cent, and no headline threshold moves.

4.2.1. Transmitter power benchmarks

The natural benchmark here is frequency-matched and purely geometric. For a filled circular aperture of diameter D radiating P_{tx} at wavelength λ ,

$$G = \frac{4\pi A_{\text{eff}}}{\lambda^2}, \quad \text{EIRP} = P_{\text{tx}}G, \quad (5)$$

As an illustration, a 12-m aperture radiating 1 MW at 230 GHz with a realistic aperture efficiency of 0.7 has $G = 5.9 \times 10^8$ and an EIRP of 5.9×10^{14} W. For a perfect aperture, $\eta = 1$, the figures would be 8.4×10^8 and 8.4×10^{14} W; we quote the realistic case. The median Class A effective threshold, 1.9×10^{15} W, misses such a transmitter by a factor 2.3. The ν^2 scaling used below is the same equation at fixed A_{eff} .

We also compare thresholds with an Arecibo-like planetary radar (Drake 1974; Sheikh et al. 2025, EIRP 2.0×10^{13} W). This is an 2.38 GHz *power* scale and carries no transmitter model, so the figures label it as a power benchmark and nothing more. Gain scales as ν^2 , so the same aperture and transmitter power at 230 GHz would radiate $2.0 \times 10^{13} \times (230/2.38)^2 \simeq 2 \times 10^{17}$ W. The comparison is between physical transmitter powers, so it must use P_{eff} and not the trigger: that difference carries the deepest window from below the Arecibo figure to 3.7×10^{13} W above it (§6.1).

4.3. Ancillary and validation analyses

Nothing here generates candidates; it is validation, and the detail is in the appendices. Two demonstrations carry the weight. Synthetic tones of known amplitude were injected into real visibilities and recovered through the unmodified pipeline — 1200 trials across six window configurations, three bands and both classes (Appendix G) — which establishes that the machinery recovers a drifting narrowband carrier, the searched morphology. And β Pictoris CO is recovered and localised in two transitions and 8 blocks (§D.0.3), which establishes that it finds a real feature at a known stellar position. The baseline step is a per-integration median along *frequency* (Table 3, step 4) and so cannot remove a one- or two-channel feature. Appendix L reports the three quantified checks that bound undetected pipeline faults.

4.4. Dwell-fraction completeness

How long a carrier occupies one native channel defines the searched class, so we measure recovery against that directly. A carrier was injected into the retained per-integration spectra of 12 real windows, six coarse and six fine, at amplitudes of 2–20 times the per-integration noise and dwell fractions $f_{\text{dwell}} = 0.10\text{--}1.00$, six realisations each: 3888 trials, released with one row per injection (Table 4). Its detection criterion, recovered when the logged per-integration S/N reaches five, was re-derived from the released rows and agrees with the logged flag 3888 times in 3888 trials, with 0 mismatches across a clean separating gap. Recovery rises monotonically with dwell. Pooling amplitudes at or above 4σ , coarse windows recover 100 per cent of continuous carriers, 100 per cent at $f_{\text{dwell}} = 0.25$ and 83 per cent at 0.10; fine windows give 100, 91 and 71 per cent.

The end-to-end check through the unmodified pipeline, the scope of what the campaign does and does not inject into, and the representativeness of the injected windows are in Appendix G.1. The campaign measures completeness at the survey threshold and across configurations. The trigger acts on the stacked statistic and the de-drift step is an inverse-variance-weighted sum over integrations, so a carrier of per-integration amplitude A present for a dwell fraction f_{dwell} of a track of N_{int} integrations accumulates to

$$a = A f_{\text{dwell}} \sqrt{N_{\text{int}}} \quad (6)$$

in units of the per-integration noise. Recovery depends only on the combination a , so binning the released trials in a unifies the two axes and places all 12 injected windows on one axis. Those windows span a factor of 63 in integration count over 11 targets, two resolution classes and Bands 3, 6 and 7, so configuration-independence is measurable. In the 6 windows whose amplitude grid straddles the trigger, the 50 per cent recovery point lies at $a = 5.3\text{--}8.3$. That is the trigger itself, independent of channelisation, integration count and band. In the other 6 the weakest injection already stacks to $a = 3.4$, so they are consistent but uninformative about the curve’s position. The pooled curves (Table 4) agree between classes within the sampling noise.

One caveat attaches to the drift ceiling. The searched value is $12.0\text{--}13.3 \text{ Hz s}^{-1} \text{ GHz}^{-1}$, line-of-sight $|a| = 3.60\text{--}4.00 \text{ ms}^{-2}$, the generic value everywhere except the 56 windows where a known planet’s own bound raises it.

TABLE 4

COMPLETENESS MATRIX. (a) WHAT IS MEASURED FOR EACH SEARCHED MORPHOLOGY AND WHERE: ‘e2e’ IS INJECTION INTO VISIBILITIES THROUGH THE UNMODIFIED PIPELINE, ‘POST’ INJECTION INTO RETAINED PER-INTEGRATION SPECTRA AFTER THE BASELINE STEP, AND ‘CFG’ THE NUMBER OF WINDOW CONFIGURATIONS THE CELL SPANS. THE TWO CAMPAIGNS MEASURE DIFFERENT QUANTITIES: 1200 INJECTION TRIALS (APPENDIX G) AND 3888 DWELL TRIALS OVER 12 REAL WINDOWS. *The 0 of 500 coarse-window entry is a property of the test, not of the pipeline:* THE RECOVERY CRITERION REQUIRES THE RECOVERED DRIFT TO MATCH THE INJECTED DRIFT, WHICH A CHANNEL WIDER THAN THE WHOLE DRIFT TRACK CANNOT SATISFY. SCORED ON DETECTION ALONE THE SAME WINDOWS RECOVER ESSENTIALLY EVERYTHING (§4.4).

Morphology	e2e	cfg	thresh.	dwell	drift
Class A drifting	yes	1	yes ^a	no	yes
Class A near-static	post	6	no ^b	yes ^c	n/a ^c
Class B stationary	yes	6	yes ^d	yes ^e	none
Class B drifting	yes	6	yes ^d	yes ^e	none ^c
Intermittent, either class	post	12	no	yes ^e	by class

^a42 per cent at the trigger, 83 at twice it, one configuration only (AU Mic, Band 6, 488 kHz), transfer to the other 403 Class A windows measured at 0.64–1.13× on 28 configurations (Fig. 12).

^bAfter the baseline step only. ^cSub-channel over the track by definition. ^dHalf recovery at stacked $a = 5.3$ –8.3 in the 6 windows straddling the trigger. ^e83–100 per cent (coarse), 71–100 per cent (fine), over $f_{\text{dwell}} = 0.10$ –1.00.

(b) completeness at the trigger, against stacked amplitude a (Eq. 6), pooled over the 12 injected windows; trial counts in parentheses.

stacked a	Class B (coarse)	Class A (fine)
< 3	0% (30)	0% (72)
3–4	0% (12)	7% (42)
4–5	3% (30)	12% (42)
5–6	46% (24)	33% (48)
6–8	58% (60)	70% (96)
8–12	100% (108)	99% (174)
> 12	100% (1680)	100% (1470)

It clears Earth-rotation and Earth-orbit drifts by two orders of magnitude but is marginal against a close-in planet on a TRAPPIST-1 b-like orbit, so the limits of §6 are conditional on it (Figure 6).

Interference rejection, which is an operative gate. Satellite constellations grew across the 2013–2025 archival baseline. The highest frequencies allocated to any constellation are the E-band feeder links of the second-generation systems, 71–76 and 81–86 GHz (SpaceX 2024), below the 89.6 GHz start of the searched union, and ITU Radio Regulations No. 5.340 prohibits all emissions in bands covering 13.3 GHz (11.2 per cent) of it. But an allocation table excludes neither harmonics nor intermodulation products nor unconventional emitters, so *no exclusion claim here rests on one*. The defence is statistical: the control ensemble and the time domain. A satellite enters the star and the control extractions alike within an integration; the mismatch between its bandwidth and ours, together with the drift-stacked sum over the track, dilutes a transient pass; and a second epoch at the same tuning tests whether anything reproduces. What none of that reaches is a near-field emitter, whose interferometric response need not resemble a celestial source’s at all, which is one reason a per-integration RFI screen and a visibility-domain check belong in any successor analysis.

Scope of the constraints. Seven limitations bound what this null result covers, and they are collected here so that no reader has to assemble them.

1. The spatial screen is empirically calibrated and not exactly exchangeable, and the radial correction of §C.0.3 is derived on the calibration sample and therefore *not validated out of sample*. No disposition here depends on it.
2. That calibration is trustworthy for isolated point sources and not for beams filled by resolved circumstellar or foreground emission: in the HD 48370 case 73 per cent of null draws exceed every control, so for disc hosts whose CO happens to be fainter the screen is only approximately controlled.
3. T_{trig} is a trigger level, not a significance.
4. The drift ceiling excludes all but the most close-in orbits.
5. The $\pm 50 \text{ km s}^{-1}$ line exclusion removes 4.1 per cent of the union from the inference.
6. The limits assume near-continuous emission; duty cycles below ~ 0.1 are essentially unconstrained.
7. The sample is 0.51 per cent of the 40-pc population and is archive-selected toward disc hosts.

What was searched, by class. Class A, the fine-channel search: 26 frequency islands totalling 47.7 GHz between 114.0 and 873.1 GHz, channels 0.02–1.95 MHz, line-of-sight accelerations $|a| \leq 3.6$ –4.0 m s^{-2} . Class B, persistent channel-confined excess with no drift discrimination: 35 islands totalling $\Delta\nu_B = 90.5$ GHz between 89.6 and 682.8 GHz — the *full* Class B union, which overlaps Class A by 20.1 GHz and so adds $\Delta\nu_{B\text{-only}} = 70.4$ GHz — channels 7.81–31.25 MHz. Both: only the archival epochs observed, only carriers one native channel or narrower, and 113.3 GHz of the 118.1 GHz union after molecular-line masking, distributed as Fig. 4 shows and not uniformly, at nominal trigger thresholds whose measured recovery is 42 per cent, with the measured 0.64–1.13× transfer applying to the *power* at which a given recovery is reached and not to the recovery itself. Inside that box the non-detection is a constraint. *Nothing in this paper bounds the prevalence of:* (i) drift-rate structure on the 1252 coarse windows, which carry no drift discrimination and no survey-threshold completeness curve (§4.4); (ii) transmitters emitting *outside the searched spectral windows*, meaning those 46 islands, not ALMA’s frequency range (Fig. 4); (iii) transmitters active *outside the archival epochs*: the limits assume near-continuous emission and are only weakly constraining for intermittent transmitters; (iv) drift rates *above* the searched ceiling $|a| = 3.60$ –4.00 m s^{-2} , excluding the most close-in orbits (Fig. 6); (v) carriers placed *inside* the $\pm 50 \text{ km s}^{-1}$ molecular-line mask, which is frequency space excluded from the experiment and not searched-and-empty; (vi) emission *broad*er than one native channel, for which the thresholds must be rescaled; (vii) anything below the nominal threshold, where measured recovery is 42^{+16}_{-14} per cent (Appendix G); (viii) any transmitter-frequency prior spanning ALMA’s tuning range, under which the conditional bound vanishes (§6.2); (ix) *harmonics and intermodulation products*, which a search keyed to one channel cannot assemble across a spectrum; (x) *polarisation*: the search is Stokes- I only, so it uses none of the discrimination a polarised carrier would offer. Total power is unaffected — a fully polarised carrier appears in I at

full strength — but a per-hand search would gain up to $\times 1.41$ on such a carrier, and 198 of the 200 blocks whose states the archive resolves deliver both parallel hands, so it could be done (§6.4, Appendix L).

4.5. The experiment in summary

Everything above is the experiment in detail. Here it is in eleven steps, with the number surviving each one. The right-hand panel of Fig. 3 is this list, drawn, and Table 7 carries every stage-1 event through the last four.

1. 656 execution blocks. Every public ALMA execution block whose field contains a catalogued star within 40 pc.
2. 484 processed. Those delivered with a usable calibration, all put through one frozen pipeline with no per-target tuning. The 177 not processed are repeat coverage of a block already in hand.
3. 404 in the science sample. 77 blocks are withheld by a rule fixed before any of them was looked at — a hash of the block identifier — and are never used to set a threshold, only to check one. 3 yielded no usable window.
4. 1655 spectral windows. Each window of each retained block, de-duplicated by identifier.
5. 403 Class A windows. Those whose channels are fine enough to resolve a drifting carrier. This is the primary experiment; the remaining 1252 support only a search for unresolved spectral excess.
6. 75 threshold crossings. Windows in which the matched-filter statistic at the stellar position reaches $T_\star \geq 5$ somewhere on the drift grid.
7. 13 stage-1 windows. Of those, the ones in which the star also outranks all 512 spatial control positions in the same window. This is a prioritisation, not a test: the control ensemble is not exactly exchangeable with the star, and §C.0.2 measures by how much.
8. 10 survive radius correction. Standardising each control probe within its own radius bin removes the one gradient we can demonstrate in the control ensemble. 4 windows lose the flag and 1 gains it; all are marginal, and no disposition changes.
9. 6 localised at the star by the drift-following visibility fit, which recovers β Pictoris where it should.
10. 4 without an astrophysical attribution. 9 of the 13 are carbon monoxide, 8 of them the β Pictoris debris disc. Chance alone predicts 1.3 unattributed events at this stage against 4 observed under the primary frozen screen ($\mathcal{P}_{\text{scr}} \approx 0.04$) and 2 under the radius-corrected normalisation ($\mathcal{P}_{\text{scr}} \approx 0.4$). Neither is a calibrated significance, for the reason §5.2.3 gives, and neither disposes of an event (Table 5).
11. 0 independently confirmed candidates. Each of the 4 has at least one further execution block of the same star covering its frequency, 23 in all, and the feature recurs in none of them: the largest stellar statistic anywhere in those repeats is $T_\star = 5.42$ and 0 exceed their own controls. The separations are 0–2 d, so this tests persistence over days and not over years. A drift-following, continuum-subtracted fit to the visibilities — the estimator that recovers β Pictoris in 6 windows at $6.1\text{--}9.5\sigma$ — finds nothing at the stellar position in any of the four, the largest $|\text{Re}/\sigma|$ being 0.93. Nothing

is independently confirmed.

A reader who wants to check any single line can: the survivor count at every step is a column filter on the released catalogue, and `reproduce_from_catalogue_v385.py` performs exactly those filters.

5. RESULTS

Which of the two searches is the experiment. The 1655 windows are the size of the processing exercise, not of the technosignature search. The primary experiment is the fine-channel search: 403 Class A windows toward 60 systems, in which a drifting narrowband carrier can be resolved and its recovery has been measured by injection. The 1252 Class B windows, 76 per cent of the total, are a complementary search for channel-confined excess power; their channels are 7.8–31.25 MHz wide and in most of them drift cannot be discriminated at all. Results are reported per class, and the combined count appears only where it describes the archive processed rather than the sensitivity achieved.

The choice of detection statistic makes the non-detection more conservative, not less: the symmetric form of Eq. 3 flags a strict subset of what the region-maximum form flags, 13 windows against 66, with none flagged by the symmetric form alone (§L.2, Table 19).

Per-target values, including each band’s searched frequency range, are in the released catalogue (Data Availability).

5.1. Barnard’s Star and Wolf 359

Barnard’s Star and Wolf 359 are the second- and fourth-nearest stellar systems. We find no prior mm or submm technosignature limit for either, so these appear to be the first. Both are non-detections in public Band 6 archival data. The nominal EIRP_{trig} trigger thresholds are 6.2×10^{13} – 9.3×10^{13} W for Barnard’s Star over 12 windows and 7.3×10^{13} – 1.1×10^{14} W for Wolf 359 over 12, with continuum upper limits of 0.575 and 0.453 mJy.

5.2. Technosignature search

All 113 star/band datasets yielded at least one carrier-lane result, 1655 spectral-window measurements after the catalogue audit removed 45 duplicate rows, Bands 6 and 7 dominating (Table 14). Nominal trigger thresholds span 1.6×10^{13} – 2.8×10^{17} W at a median of 7.1×10^{14} W (Figs. 4 and 1), the deepest being reached toward the nearest systems, 1.6×10^{13} W for the UV Ceti pair at 2.7 pc, though depth also tracks integration time and channelisation.

The statistics come strictly before the astrophysics, in three named stages that do not promote one another. A *crossing* is one channel $\times$ drift cell at the stellar position reaching $T \geq 5$: 75 of the 1655 windows contain one, and 95 per cent of them are Class A. That is a trials effect rather than a physical one: a median Class A window holds 2.6×10^5 channel $\times$ drift cells against 468 for a median Class B window, a factor 559, so a cell at the trigger is correspondingly rarer in the coarse class. A *stage-1 flagged window* is a window whose stellar statistic also exceeds all 512 controls: 13 do, in 6 systems and 7 star–band pairs. That screening device selects a window for examination (§4.1). Each of them is then carried past that floor by a local scramble test ((v) below), which asks

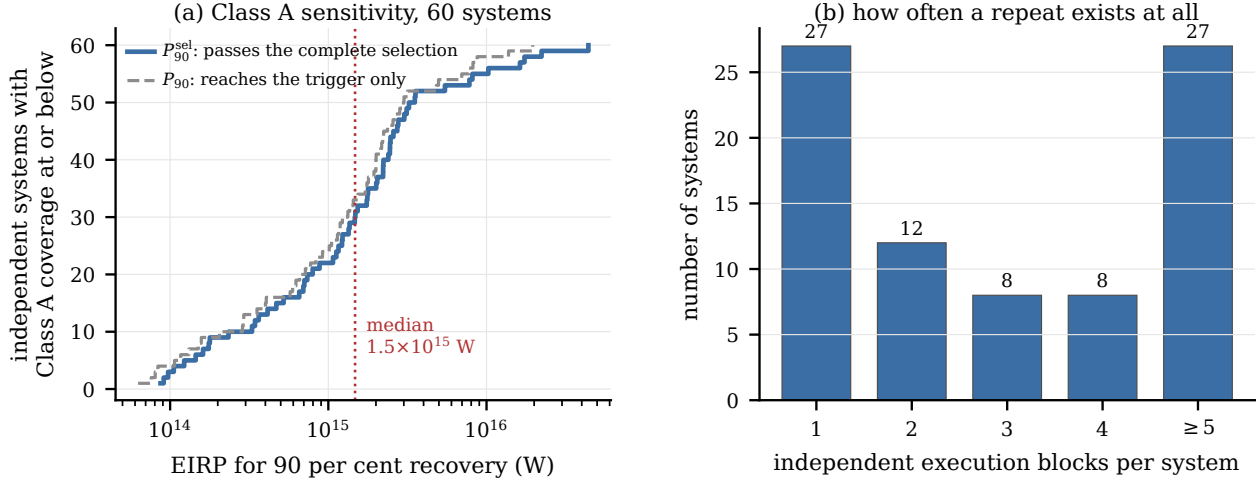


FIG. 7.— What the primary experiment could have found, and how often it could have confirmed it. (a) Cumulative number of independent systems whose best fine-channel window reaches a given power, in 90 per cent of injection trials. The solid curve is P_{90}^{sel} , the power that passes the *complete* automated selection — detection threshold *and* the window’s own control-ring gate — and is the survey’s sensitivity: 8.7×10^{13} to 4.4×10^{16} W, median 1.5×10^{15} W. The dashed curve is P_{90} , which reaches the matched-filter trigger only and is an internal completeness, spanning 6.3×10^{13} – 2.0×10^{16} W with a median of 1.3×10^{15} W. A single quoted sensitivity would describe almost none of the survey; 19 systems lie within a decade of the deepest. (b) Independent execution blocks per system. 27 of the 82 systems have only one — and restricted to the primary experiment, 42 of the 60 Class A systems have more than one block, 70 per cent — so for the rest the recurrence test that defines a confirmed technosignature cannot be applied at all; 67 per cent have two or more.

whether a feature exceeds the noise at its own position without discriminating spatially; in 3 of the 4 windows the control-ring maximum clears it too, so candidacy still turns on the first-stage star-versus-control comparison. A *confirmed technosignature* is a stage-1 flagged window that survives every stage of §4.1 and then recurs or is independently confirmed. There are none.

Three terms carry this section, in the sense fixed by Table 12: a *crossing* is a channel $\times$ drift cell at the star reaching $T_* \geq 5$; a *stage-1 flagged window* is a window whose T_* also exceeds all 512 controls, which prioritises it and establishes nothing; a *candidate* survives every stage of §4.1 and recurs or is independently confirmed.

The outliers fall into two groups. 9 of the 13 sit within $\pm 50 \text{ km s}^{-1}$ of a molecular transition in the star’s own frame and carry independent CO evidence: 8 are β Pictoris, in 5 Band 3 and 3 Band 6 windows drawn from 8 execution blocks, at -29 to -14 km s^{-1} and T_* from 8.0 to 30.8; one is the published foreground emission at HD 48370. That is 6 astrophysical sources, not 13 events, and the recurrence is the point: a frozen pipeline recovers the same disc in two transitions and 8 independent blocks without being told it is there.

The remaining 4 have no astrophysical attribution. How many should chance produce? This paper has quoted several expectations, and Table 5 collects them with their samples and denominators, because they are not all answers to the same question.

Five of them answer the wrong question. An expectation built on the rate at which a no-signal position ranks first among the control ensemble — whether that rate is the exchangeable $1/(512+1)$, the measured pseudo-star rate, or either of those with the tail excess folded in — predicts the number of windows in which the star would *rank first*. A stage-1 event needs more than that: the position must also reach the trigger. Comparing a rank-first expectation with a count of stage-1 events is comparing two different quantities, and it is the source of the apparent disagreement between 3.1, 3.2, 3.9, 4.4

and 4.7.

The correctly conditioned null is measurable from the stored control vectors. In a given window a control probe reproduces a stage-1 event only if it both reaches $T_{\text{trig}} = 5$ and exceeds every other probe; exactly one probe per window is the maximum, so the probability that the stellar position would have done so, treated as exchangeable with the probes, is $1/(512+1)$ in those windows whose control maximum reaches the trigger and zero in the rest. That condition is restrictive and informative: only 460 of the 1614 windows with a stored ensemble have a control maximum that reaches the trigger at all, 29 per cent, and 391 of those are Class A. *This is why every stage-1 event in the survey is Class A*, and it also means the choice of reference class, which earlier versions of this analysis treated as an open question, largely dissolves once the trigger condition is imposed.

Resampling execution blocks with replacement, and applying the measured non-exchangeability of the stellar position as a factor $\times 1.4$, gives the expectation the observed count should be read against. Probabilities are quoted to one significant figure throughout: the count is small and the tail factor that normalises the rate is itself known only to 1.2–1.9, so further digits would be spurious. That uncertainty is carried in the quoted range rather than added afterwards as a caveat.

- Primary, radius-corrected statistic: 1.3 expected over the 1614 windows with stored control ensembles, of which 460 can reach the trigger from noise at all; 95 per cent interval 0–4 against 2 observed, giving $\mathcal{P}_{\text{scr}} \approx 0.4$ (range 0.3–0.5 across the tail-factor uncertainty).
- Robustness, originally frozen screen: the same expectation against 4 observed, giving $\mathcal{P}_{\text{scr}} \approx 0.04$.

The statement to carry forward is therefore that 2 unattributed events were observed against 1.3 expected under the calibrated, block-clustered null, $\mathcal{P}_{\text{scr}} \approx 0.4$, which is consistent with chance. The frozen screen’s $\mathcal{P}_{\text{scr}} \approx 0.04$ on 4 events is the reason the radial normali-

sation was examined in the first place, and §C.0.3 shows that the apparent excess is an artefact of that screen’s position-dependent rank bias rather than a property of the sky. Neither number is evidence about any individual event.

What would absorb it, and what that would cost. An excess this size needs only a small unmodelled tail in the control ensemble, and §C.0.3 demonstrates that one exists. That is not a free explanation. The same noise model sets the recovery fractions of §4.4, so a tail large enough to remove the excess would also mean the quoted completeness is optimistic, by roughly the same factor, since both scale with the rate at which the ensemble produces an extreme value. The two cannot be traded independently, and we have applied such a correction to neither.

All four have a persistence test, all four fail it, and the test is short. Each has at least one further execution block of the same star whose tuning covers the event’s frequency: 1 to 16 of them, 23 in total (Table 8). The feature recurs in none. But a count of repeats is not a baseline, and the baselines here are days rather than years: of the 23 repeats, 4 are separated from their event by a resolvable interval and that interval runs 0–2 d, while the remaining 19 fall on the same day or within the same scheduling block. *So what these tests exclude is a transmitter persistent over hours to days.* A source with a duty cycle measured in months or years would not be caught by them, and nothing here bounds one. Across all 23 repeat windows the largest stellar statistic is $T_* = 5.42$, 4 cross the threshold at all, 0 exceed their own window’s control maximum and 0 are stage-1 flagged windows. Repeat coverage is matched on the window rather than on the crossing frequency, which the release records only where the pipeline stored a peak. CP–72 2713 is the case examined in detail (§A.3.2) because its repeat block is the deepest and closest in configuration.

5.2.1. The visibility-domain localisation test

Every stage-1 window has been tested in the visibility domain, twice, and none of the unattributed ones behaves like emission at the star. The image-plane rank is a screen; the physical question is whether the emission sits at the stellar coordinates, and the visibilities answer it directly. Phase-rotating every visibility to the star turns emission centred there into a constant real offset, so its real part measures the source and its imaginary part must be consistent with zero; emission elsewhere in the field is suppressed in the real part and in general leaves a non-zero imaginary part. The test therefore separates *at the star* from *displaced from the star*; it does not by itself separate unresolved from resolved, because a centrally symmetric resolved source is also centred on the star.

We recalibrated all 13 stage-1 execution blocks from the raw archive data — 4 unattributed and 9 attributed to carbon monoxide — and applied the test at the recorded peak frequency of each event, with eight control positions on the annulus the image-plane test uses (Table 21).

The attributed events are the test’s controls, and they are why the null result on the other four can be believed. 7 of the 9 return emission at the stellar position

at 10–59 σ , every one of them above its own controls: the β Pictoris CO belt is recovered in the visibilities in both transitions and in seven separate execution blocks. Of the 2 that do not, one is the strongest possible negative — HD 48370, whose real part is large (53 σ) but whose imaginary part is 7.9 σ from zero, exactly what emission displaced from the phase centre produces, and independently known to be a foreground cloud resolved several beams away. The other is a β Pictoris Band 3 block for which no measured peak frequency survives in the frozen release and the test had to be run at the frequency measured in its sibling blocks; it returns nothing, and we report it as a limitation of the substituted frequency rather than as a failure of the line.

Against that, the 4 unattributed events, panel by panel in Fig. 8: on this test 0 of them shows emission at the star. The largest real part is 2.5 σ ; 61 Vir does not exceed its own controls; and HD 14055 returns a 2.5 σ real part with a 3.2 σ imaginary part. A source centred on the star would leave the imaginary part at zero, so at face value that excess is displaced; but a phase-calibration residual or a small astrometric error produces the same signature, the uncertainties here are internal and assume independent visibilities, and 13 events give 13 looks. The defensible statement is that at this significance the uniform test does not localise the excess, not that it has localised it away from the star.

That comparison needs a caveat, because the test is not equally sensitive to the two groups. It uses a single channel and no continuum subtraction, identically for every event, which is what makes the thirteen comparable. A broad line fills that channel completely, so the test is fully sensitive to the attributed CO events. A narrowband carrier that drifts occupies the channel for only part of the track, and the tracks here sweep 4–23 channels for the unattributed events, so a uniform single-channel test can lose such a carrier by as much as $\times 2.0$ – $\times 4.7$ in signal-to-noise.

So we ran the sensitive estimator on all 13 events as well. For each event, and for each integration, we take the channel the carrier occupies at that moment following its recovered drift rate, subtract a local continuum estimated from reference channels either side of the track in the same integration, and phase-rotate to the star. Eight positions on the control annulus and four off-event frequencies in the same window give the comparison. Table 6 gives the result and it is decisive in both directions.

The estimator works: β Pictoris is recovered at the stellar position in 6 of its 8 windows at 6.1–9.5 σ , and HD 48370 comes out with a large real part, 5.9 σ , but an imaginary part 2.7 σ from zero — the displaced-source signature its foreground identification predicts, and the reason it is not counted as a recovery at the star. 1 window cannot be measured this way: its drift track sweeps more channels than the continuum reference leaves, and we report that as a limitation of the estimator rather than as a null.

And the four unattributed events show nothing. The largest $|\text{Re}/\sigma|$ among them is 0.93, 0 are consistent with a source at the star, and no control position or control frequency anywhere in the thirteen exceeds 2.14 σ .

One caution about the continuum applies to CP–72 2713 in particular. That window’s stellar continuum is 10.9 mJy at 5.5 σ , comparable to any narrowband

TABLE 5

EVERY FALSE-ALARM EXPECTATION QUOTED IN THIS PAPER, AND WHAT EACH ONE PREDICTS. THE FIRST SIX PREDICT THE NUMBER OF WINDOWS IN WHICH A NO-SIGNAL POSITION WOULD *rank first* AMONG THE CONTROL ENSEMBLE; THEY DIFFER IN SAMPLE, DENOMINATOR AND WHETHER THE MEASURED TAIL EXCESS AND THE WITHIN-BLOCK CLUSTERING ARE INCLUDED, AND THEY ARE PIPELINE DIAGNOSTICS. NONE OF THEM PREDICTS THE OBSERVED QUANTITY, BECAUSE A STAGE-1 EVENT ALSO REQUIRES THE POSITION TO REACH THE TRIGGER. THE LAST ROW IMPOSES THAT CONDITION AND IS THEREFORE THE ONLY ROW WHOSE EXPECTATION MAY BE COMPARED WITH AN OBSERVED COUNT; IT IS THE COMPARISON USED IN THE ABSTRACT, IN §5.2 AND IN THE CONCLUSIONS. THE TWO KINDS OF BRACKET ARE NOT THE SAME QUANTITY AND ARE PUNCTUATED DIFFERENTLY: SQUARE BRACKETS GIVE A BOOTSTRAP INTERVAL ON THE *expectation*, ROUND BRACKETS THE 95 PER CENT INTERVAL OF THE PREDICTED *count* UNDER THAT EXPECTATION.

Sample	Windows	Rate per window	Predicts	Expected	Obs.
All searched windows, exchangeable	1655	1/(512 + 1)	rank-first windows	3.2	–
Distinct (block, window) datasets	1616	1/(512 + 1)	rank-first windows	3.1	–
All windows, measured tail rate	1655	1.4/(512 + 1)	rank-first windows	3.9	–
All windows, block-resampled	1614	1.4/(512 + 1)	rank-first windows	4.4	–
Pseudo-star first-rank rate	1655	measured on inner probes	rank-first windows	4.7	–
Class A windows, block-resampled	394	1.4/(512 + 1)	rank-first Class A windows	1.1 [0.5–1.4]	–
Class A windows, block-resampled, trigger required	394	1.4/(512 + 1) if $\max c \geq 5$	unattributed stage-1 events	1.07 (0–4)	4

excess it could carry, so a fit that does not remove the continuum along the drift track will report the continuum instead. The fit used here removes it per integration, and returns -0.86σ at the stellar position with an imaginary part of 0.02σ . A simpler treatment of the same visibilities, which leaves the continuum in, returns 5.5σ ; the difference is the continuum, not the carrier.

Table 9 gives each star–band pair with the evidence dispositioning it.

5.2.2. Radio-frequency interference

Nothing above rules out terrestrial interference, so this subsection gives what does. Millimetre interferometry is an unusually poor environment for it, for three reasons. The searched range is 90–873 GHz, above every allocated terrestrial service and above the frequencies at which the RFI-mitigation literature of centimetre-wave SETI operates; the array is at 5000 m on a site chosen partly for its radio quietness; and, decisively, an interferometer is not a total-power detector. A signal that is not in the far field at the phase centre does not fringe-track: it acquires a rapidly varying geometric phase across the array and is suppressed in the correlation rather than imaged. Three tests are applied to every stage-1 event: the spatial control screen, which asks whether the excess is at the star or in the field; the visibility fit of §5.2.1, which asks whether it has the phase signature of a source at the phase centre; and recurrence at the same tuning in an independent epoch. Each is, for this reason, also an interference test.

Three measurements support that reasoning. First, interference is a property of the observatory and not of the star, so a terrestrial carrier should appear at the same *sky* frequency in every window that covers it. Of the 4 unattributed events, 1 has a released crossing frequency; 86 windows toward 6–39 other stars in 276 other blocks cover the frequencies of the four, and 0 of them carries a crossing in the same channel. For the other 3 the pipeline stored no peak frequency, so this test cannot be run on them and we do not claim it was. Second, over all 451 released crossing frequencies in the survey there is no clustering in sky frequency: binned at 1 MHz the largest multiplicity is 2, against 1.9 for a null that keeps each crossing in its own window and randomises its position within it ($p = 0.90$). Third, an artefact of the signal

chain would sit at a fixed intermediate frequency rather than a fixed sky frequency, and the crossings do not: their fractional positions within their own windows have median 0.50 and are uniform to $D = 0.023$ at $p = 0.97$ over 451 crossings. CP–72 2713’s crossing, the one with a released frequency, lies 35 channels from its nearest window edge.

None of this is a substitute for the ON/OFF cadence a dedicated search uses, and we do not present it as one. Accordingly we claim only that *no identified interference mechanism accounts for these events*; allocation tables cannot exclude harmonics, intermodulation products, unconventional emitters or near-field sources, and an archival dataset without a simultaneous off-source strategy cannot do better than that. What it establishes is that the crossing population carries none of the three signatures interference leaves: fixed sky frequency, fixed intermediate frequency, or presence in unrelated targets. The phase test each event failed is also the test a near-field interferer would fail.

HD 48370, in one line. $T_{\star} = 27.10$ against a ring maximum of 26.83, a margin of 0.27, the smallest in the survey; CO(2→1) at $v_{\text{bary}} = +41.2 \text{ km s}^{-1}$, matching the foreground cloud Cataldi et al. (2023) report; and the visibility fit puts the emission off the star, with an imaginary part 7.9σ from zero. *Disposition: foreground carbon monoxide*, on three independent lines of evidence, none of them the rank.

External checks, in brief. Three further checks are reported in Appendix D: a prospectively searched extension of 115 public blocks, which recovers the β Pictoris disc CO in three previously unsearched epochs and adds no unattributed event; the β Pictoris recovery itself, which is the survey’s astrophysical positive control for both the spatial and the spectral machinery; and the absence of any shared data with the only prior ALMA-archival search, which leaves the null result resting on internal validation alone.

5.2.3. Which normalisation, and why it does not change the result

The conclusion first. The choice of spatial normalisation does not change this paper’s confirmed-candidate count, which is zero under either form. The two differ only in the count of unattributed, non-recurring events: 4 under the originally frozen screen and 2 under the radius-corrected normalisation. Neither form is exchangeable,

TABLE 6

THE DRIFT-FOLLOWING, CONTINUUM-SUBTRACTED FIT AT THE STELLAR POSITION, FOR ALL 13 STAGE-1 WINDOWS. FOR EACH INTEGRATION THE CHANNEL THE CARRIER OCCUPIES AT THAT MOMENT IS TAKEN ALONG ITS RECOVERED DRIFT RATE, A LOCAL CONTINUUM ESTIMATED FROM REFERENCE CHANNELS EITHER SIDE OF THE TRACK IN THE SAME INTEGRATION IS SUBTRACTED, AND THE VISIBILITIES ARE PHASE-ROTATED TO THE STAR; ‘CTRL’ IS THE LARGEST REAL PART RETURNED BY EIGHT POSITIONS ON THE CONTROL ANNULUS OR FOUR OFF-EVENT FREQUENCIES IN THE SAME WINDOW. THIS IS THE SENSITIVE ESTIMATOR: IT RECOVERS β PICTORIS IN 6 WINDOWS AT $6.1\text{--}9.5\sigma$ AND STILL REPORTS HD 48370 AS DISPLACED, WHILE THE LARGEST $|\text{Re}/\sigma|$ AMONG THE UNATTRIBUTED EVENTS IS 0.93. THE ONE β PICTORIS WINDOW MARKED UNREACHABLE HAS A DRIFT TRACK WIDER THAN THE CONTINUUM REFERENCE LEAVES.

Star	B	N_{vis}	Re/σ	Im/σ	ctrl	verdict
<i>Unattributed</i>						
61 Vir	7	12 375	−0.93	+0.33	+1.18	nothing at the star
HD 14055	7	13 365	+0.56	−0.91	+0.38	nothing at the star
CP−72 2713	7	405 860	−0.86	+0.02	+0.98	nothing at the star
HD 23484	6	15 015	−0.57	−0.35	+1.30	nothing at the star
<i>Attributed to CO (positive controls)</i>						
β Pic	3	285 327	+8.13	+0.80	+1.14	source at the stellar position
β Pic	3	288 435	+9.52	−0.06	+2.14	source at the stellar position
HD 48370	6	9 832	+5.93	+2.74	+0.75	displaced: phase inconsistent
β Pic	3	421 890	+2.42	+0.43	+1.15	nothing at the star
β Pic	3	274 700	+6.12	+0.53	+1.84	source at the stellar position
β Pic	3	180 600	+8.01	−1.20	+1.63	source at the stellar position
β Pic	6	347 613	+8.47	+1.04	+1.49	source at the stellar position
β Pic	6	398 475	+6.34	−0.05	+1.39	source at the stellar position
β Pic	6	405 900	+0.00	+0.56	+1.53	track too wide to reference

TABLE 7

EVERY STAGE-1 EVENT AND ITS FATE. THE 13 WINDOWS THAT PASS THE DETECTION THRESHOLD AND THE SPATIAL SCREEN, CARRIED THROUGH THE FIVE STAGES OF §4.1 IN ORDER. *screen*: THE STAR OUTRANKS ALL 512 CONTROL POSITIONS — TRUE FOR ALL OF THEM BY CONSTRUCTION, SINCE THAT IS WHAT MAKES A WINDOW STAGE-1. *localised*: THE DRIFT-FOLLOWING, CONTINUUM-SUBTRACTED VISIBILITY FIT PLACES THE EMISSION AT THE STELLAR POSITION (§5.2.1); 6 DO. *attributed*: THE CROSSING LIES WITHIN $\pm 50 \text{ km s}^{-1}$ OF A CATALOGUED TRANSITION IN THE STELLAR FRAME AND INDEPENDENT EVIDENCE SUPPORTS IT; 9 DO. *recurs*: THE FEATURE REPEATS AT THE SAME TUNING IN ANOTHER EXECUTION BLOCK, WITH THE COUNT OF REPEAT BLOCKS AND THE LARGEST STATISTIC ANY OF THEM RETURNS; 9 DO AND 1 HAS NO REPEAT COVERAGE. 0 EVENTS PASS EVERY STAGE, WHICH IS THE RESULT OF THIS PAPER. ^a ONE β PICTORIS BLOCK HAS A DRIFT TRACK WIDER THAN THE CONTINUUM REFERENCE LEAVES AND CANNOT BE TESTED THIS WAY.

Star	B	T_*	screen	localised	attributed	recurs
β Pic	3	30.76	✓	✓	✓	✓ (4, $T_* = 27.7$)
β Pic	3	27.68	✓	✓	✓	✓ (4, $T_* = 30.8$)
HD 48370	6	27.10	✓	×	✓	no repeat
β Pic	3	17.91	✓	×	✓	✓ (4, $T_* = 30.8$)
β Pic	3	17.29	✓	✓	✓	✓ (4, $T_* = 30.8$)
β Pic	3	14.65	✓	✓	✓	✓ (4, $T_* = 30.8$)
β Pic	6	11.68	✓	✓	✓	✓ (5, $T_* = 10.2$)
β Pic	6	8.95	✓	✓	✓	✓ (5, $T_* = 11.7$)
β Pic	6	7.99	✓	n/a ^a	✓	✓ (3, $T_* = 10.2$)
61 Vir	7	6.16	✓	×	×	×
HD 14055	7	6.03	✓	×	×	✓ (16, $T_* = 5.4$)
CP−72 2713	7	5.81	✓	×	×	×
HD 23484	6	5.30	✓	×	×	×

TABLE 8

THE 4 STAGE-1 FLAGGED WINDOWS WITH NO ASTROPHYSICAL ATTRIBUTION, ONE ROW EACH. T_* IS THE SEARCH STATISTIC AT THE STELLAR POSITION AND ‘CTRL’ THE LARGEST OF THAT WINDOW’S 512 CONTROL STATISTICS; $|\dot{\nu}|_{\text{max}}$ IS THE WIDEST DRIFT THE STACK EVALUATED FOR THAT WINDOW; P_{eff} IS THE RESPONSE- AND SMEARING-CORRECTED THRESHOLD POWER. ‘REPEAT BLOCKS’ COUNTS OTHER EXECUTION BLOCKS OF THE SAME STAR WHOSE TUNING COVERS THIS WINDOW, THE LARGEST STATISTIC ANY OF THEM RETURNS AT THE STELLAR POSITION, AND THE RANGE OF TEMPORAL SEPARATIONS FROM THE FLAGGED BLOCK, TOGETHER WITH THE DEPTH OF THE DEEPEST REPEAT AND ITS SHORTFALL AGAINST THE EVENT, AND THE RECURRENCE VERDICT. ALL FOUR EVENTS HAVE AT LEAST ONE REPEAT, 23 IN TOTAL, AND NONE RECURS: THE DEEPEST REPEAT ANYWHERE IN THE SET REACHES $T_* = 5.42$, WHICH CROSSES THE TRIGGER BUT DOES NOT OUTRANK ITS OWN WINDOW’S CONTROLS, SO IT DOES NOT REPRODUCE A STAGE-1 EVENT. THE ELAPSED TIMES ARE THE LIMITATION, NOT THE DEPTHS — 4 OF THE SEPARATIONS RESOLVE AT ALL AND NONE EXCEEDS 2 d, SO THIS TEST BOUNDS SHORT-TIMESCALE INTERMITTENCY AND NOTHING LONGER. CP−72 2713’S IS EXAMINED IN DETAIL IN §A.3.2 BECAUSE IT IS THE DEEPEST AND THE CLOSEST MATCH IN CONFIGURATION. ALL FOUR HAVE ALSO BEEN TESTED IN THE VISIBILITY DOMAIN (TABLE 21) AND AGAINST THE INTERFERENCE SIGNATURES OF §5.2.2. ^a THE RELEASED CROSSING FREQUENCY IS STORED ONLY WHERE THE PIPELINE RECORDED A PEAK; FOR THESE THE WINDOW IS GIVEN INSTEAD. ^b THE ARCHIVE INDEXES 200 OF THE 404 SEARCHED BLOCKS BY EXECUTION BLOCK AND THE REST ONLY BY MEMBER OBSERVING UNIT; EPOCHS AND SEPARATIONS SO MARKED ARE THE UNIT’S, SO A SEPARATION WITHIN ONE UNIT CANNOT BE RESOLVED AND IS REPORTED AS UNDER A DAY. ^c THE DEEPEST REPEAT CROSSES THE TRIGGER, $T_* = 5.42$, BUT DOES NOT OUTRANK ITS OWN WINDOW’S CONTROLS AND SO DOES NOT REPRODUCE A STAGE-1 EVENT.

Star	EB	B	ν (GHz)	$ \dot{\nu} _{\text{max}}$ (Hz s ^{−1})	T_*	ctrl	P_{eff} (W)	epoch (UT)	N_{rep}	Δt	T_*^{rep} (Δ)	recurs?
61 Vir	Xc079b5_X82f	7	344.87–346.71 ^a	4150	6.16	5.88	9.4×10^{14}	2017-05-17 ^b	3	2 d ^b	4.83 (−1.32)	no
HD14055	Xfff3d8_Xd46	7	330.39–332.23 ^a	3976	6.03	5.96	1.9×10^{16}	2022-10-16 ^b	16	0–1 d ^b	5.42 (−0.61)	no ^c
CP−72 2713	Xff0235_X4a6d	7	344.2697	4141	5.81	5.68	1.9×10^{15}	2022-10-02	1	< 1 d ^b	4.70 (−1.11)	no
HD 23484	X10b22d7_X2445	6	229.61–231.45 ^a	2766	5.30	5.25	1.7×10^{15}	2023-08-06 ^b	3	2 d ^b	4.99 (−0.30)	no

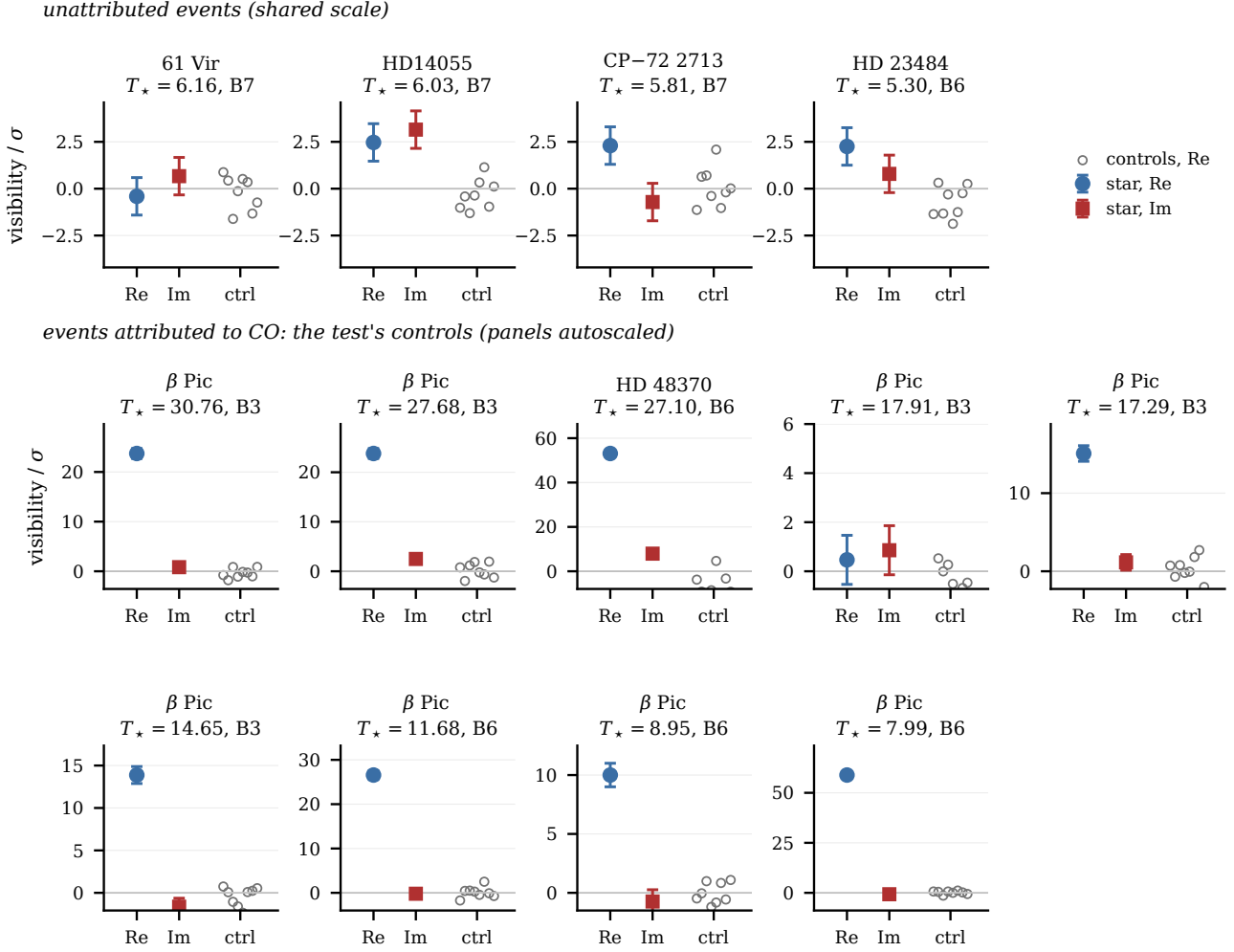


FIG. 8.— The visibility-domain test on all 13 stage-1 windows, one panel each. In units of the measurement uncertainty: the real part of the visibility phase-rotated to the stellar position (blue circle), which emission centred there makes positive and significant; its imaginary part (red square), which such emission leaves consistent with zero; and the real parts returned by eight control positions on the same annulus the image-plane screen uses (open circles). *Top row*: the 4 unattributed events, on one shared scale. None reaches 3σ ; 61 Vir does not beat its own controls; HD 14055 has a 3.2σ imaginary part, which a source centred on the star would not produce, but at this significance the test does not localise the excess either way. *Lower rows*: the 9 events attributed to carbon monoxide, which are the test's controls, each panel autoscaled because they reach 59σ . 7 are recovered at the stellar position; HD 48370 has a large real part but an imaginary part 7.9σ from zero, the displaced-source signature its foreground identification predicts; and one β Pictoris Band 3 block returns nothing, the one event whose test frequency had to be substituted from its siblings.

TABLE 9

EVERY STAGE-1 FLAGGED WINDOW OF THE COMPLETED SWEEP, ONE ROW PER STAR-BAND PAIR, GENERATED FROM THE RELEASED CATALOGUE. N_{win} IS THE NUMBER OF FLAGGED WINDOWS IN THAT PAIR AND THE REMAINING COLUMNS DESCRIBE ITS DEEPEST WINDOW: T_* IS THE SEARCH STATISTIC AT THE STELLAR POSITION (Eq. 3), ‘ctrl’ THE LARGEST OF THAT WINDOW’S 512 CONTROL STATISTICS, ν THE CROSSING CHANNEL, AND Δv THE OFFSET OF THAT CHANNEL FROM THE NEAREST LABORATORY REST FREQUENCY OF THE MASKED SPECIES *in the topocentric frame*, AS THE RELEASED CATALOGUE COLUMN STORES IT, SIGNED SO THAT A NEGATIVE VALUE PLACES THE OBSERVED CHANNEL BELOW THE REST FREQUENCY. THE DISPOSITION RULE IS APPLIED IN THE *stellar* FRAME, AFTER THE FRAME CHAIN OF §4.2, AND THE STELLAR-FRAME OFFSETS ARE GIVEN IN TABLE 18: FOR β PICTORIS THEY ARE WITHIN 0.46 km s^{-1} OF SYSTEMIC WHILE THE TOPOCENTRIC COLUMN READS ABOUT -28 , AND THE DIFFERENCE IS THE BARYCENTRIC AND SYSTEMIC TERMS, NOT A DISCREPANCY. A PAIR IS DISPOSITIONED AS CIRCUMSTELLAR OR FOREGROUND EMISSION WHEN THE STELLAR-FRAME OFFSET SATISFIES $|\Delta v| \leq 50 \text{ km s}^{-1}$ AND INDEPENDENT EVIDENCE EXISTS, AND AS UNATTRIBUTED OTHERWISE; THE 4 UNATTRIBUTED PAIRS ARE CONSISTENT WITH THE 3.2 EXPECTED BY CHANCE ($p = 0.40$). THE HD 48370 ROW REMAINS THE LIMITING CASE, A NEAR-TIE OF RANK DISPOSITIONED BY VELOCITY COINCIDENCE WITH INDEPENDENTLY PUBLISHED EMISSION.

Star	B	N_{win}	ν (GHz)	T_*	ctrl	Δv	disposition
β Pic	3	5	0.0000	30.76	7.34	-28	circumstellar CO
HD 48370	6	1	230.5117	27.10	26.83	-34	foreground CO
β Pic	6	3	230.5161	11.68	9.12	-28	circumstellar CO
61 Vir	7	1	0.0000	6.16	5.88	+449	unattributed
HD14055	7	1	0.0000	6.03	5.96	-12161	unattributed
CP-72 2713	7	1	344.2697	5.81	5.68	-1323	unattributed; absent in epoch 2
HD 23484	6	1	0.0000	5.30	5.25	+328	unattributed

so neither assigns those events a calibrated false-alarm probability, and the disposition of every one of them rests on the visibility-domain test and on recurrence rather than on either rank.

The supporting evidence is threefold and is set out in Appendix C. The stellar rank is displaced from uniformity in the pre-registered hold-out; the control level carries a measured radial gradient, which is the mechanism, because primary-beam-corrected images have noise rising with radius while the search divides every position by one global scale; and pseudo-stars reproduce the displacement where no star is present at all. A radius-corrected normalisation removes the radial component by construction, with no free parameter. Applied prospectively to 778 windows it played no part in building, at a level fixed in advance, the test returns FAIL: the corrected stellar rank remains inconsistent with uniformity and its median moves further from the exchangeable value. The correction therefore does not repair the statistic and is not adopted as primary.

5.2.4. Survey-level false-alarm calibration

The per-window rank probability, its survey-level counterpart, the pseudo-star and stratum tests that show the control ensemble behaves as an ensemble, and the trials budget in cells are all calibration rather than result, and are collected in Appendix K. What the main line of argument needs from them is the two statements above: the tail rate is measured, and no window in this survey approaches the survey-wide correction.

5.2.5. Dispositions of the remaining outlier windows

The third stage-1 flagged window, towards HD 48370 in Band 6, is the largest excess here in absolute terms and the smallest in relative ones, $T_\star = 27.10$ against a control-ring maximum of 26.83, a margin of 0.27. Star and ring rising together is what emission filling the primary beam looks like. The velocities agree immediately: Cataldi et al. (2023) report prominent CO(2→1) towards HD 48370 at $v_{\text{bary}} \approx 41 \text{ km s}^{-1}$ and flag it as possible foreground cloud contamination, and our own frame chain returns $v_{\text{bary}} = +41.2 \text{ km s}^{-1}$ for this crossing, so we recover their line and confirm it kinematically. Three checks close the circumstellar alternative. The same authors report the disc itself undetected in CO and in carbon. The sight line runs through the outer Galaxy, where a flat rotation curve reaches the measured velocity only at a kinematic distance of 2.07 kpc, which nothing local could produce. And the block's $^{13}\text{CO}(2\rightarrow1)$ window has a control maximum of 20.4 against 4.6 at the star, so the field carries bright ^{13}CO that the star does not, resolved 1.7 synthesised beams away. Appendix L gives the Band 8 [C I] tuning check, which says the same thing by omission. The rank statistic identifies nothing here, a stated limitation of the spatial test.

5.2.6. Line exclusion: definition and cost

The exclusion regions are defined before any crossing is classified, and those frequencies are *excluded from the experiment* rather than vetoed within it: the experiment declines to classify them instead of declaring whatever is found in them astrophysical. Each catalogued transition of 11 species is masked to $\pm 50 \text{ km s}^{-1}$ in the stellar

frame; Appendix I gives the kinematic justification for that half-width, the species criterion, and the counter-argument that a communicator might deliberately choose a natural reference frequency; Appendix L.4.3 gives the cost species by species.

The search statistic is computed inside the masked channels; the mask is applied afterwards, at the reporting stage. That wording matters, because “excluded from the experiment” could otherwise be read as meaning those channels were never examined. They were: the threshold and rank tests run over every delivered channel, and the mask then removes the masked intervals from the *reported* search domain, because a crossing coincident with a catalogued transition cannot be dispositioned as artificial on this data alone.

That ordering makes a completed robustness check possible, and we report it here rather than leaving it to an appendix, because it answers the most obvious objection to the mask. Removing the mask entirely adds no candidate. Of the 75 threshold crossings, 17 fall inside a tube: 9 are stage-1 flagged windows already dispositioned as CO in Table 9, and the remaining 8 do not exceed their own control ensembles. So the mask costs 4.1 per cent of the unique-frequency union and nothing else; had it been absent, the candidate list would be identical. The caveat that survives is one of principle rather than of outcome: a transmitter deliberately parked on a natural reference frequency, which is what Cocconi & Morrison (1959) argued a communicator might choose, would lie in unsearched reported space and this survey would not claim it either way.

5.3. What this experiment excludes, and what it does not

For Class A the complete automated selection recovers a carrier nine times in ten above $P_{90}^{\text{sel}} = 8.7 \times 10^{13} - 1.4 \times 10^{17} \text{ W}$ per window, median $2.8 \times 10^{15} \text{ W}$; per system, taking each system's best window, $8.7 \times 10^{13} - 4.4 \times 10^{16} \text{ W}$ with a median of $1.5 \times 10^{15} \text{ W}$. The per-system figure is the one to quote. Read every value as an upper bound on reach rather than a central estimate: the injected tones that calibrate them suffer no atmospheric decorrelation while a real carrier does, so they are optimistic by +5 to +20 per cent and cannot err the other way. Appendix E defines the chain from P_{trig} to P_{90}^{sel} .

Excluded. For a continuously transmitting, unresolved carrier lying outside the molecular-line mask, within the searched frequency range and inside the linear-drift grid, and present during one of the archival epochs, the fine-channel search recovers the signal at least nine times in ten above P_{90} . That threshold is $6.3 \times 10^{13} \text{ W}$ of EIRP in the deepest Class A window and $2.4 \times 10^{15} \text{ W}$ at the median one; it is a property of the individual window, and the per-window value is a column of the released catalogue.

P_{90} is not the survey's sensitivity, and the difference matters. It is calibrated against the trigger, but a window does not select a signal for technosignature vetting merely because the trigger fires: the stellar statistic must also exceed every one of that window's 512 control positions, and those maxima sit above the trigger. Across Class A they run $4.78 - 26.83\sigma$ with a median of 5.79σ , and only 3 windows (0.7 per cent) have a gate below T_{trig} .

The sensitivity of the experiment is therefore the power at which a carrier passes the complete automated selection, which is each window's P_{90} scaled by its own gate: $P_{90}^{\text{sel}} = 8.7 \times 10^{13} - 1.4 \times 10^{17}$ W, median 2.8×10^{15} W, and per system, taking each system's best window, $8.7 \times 10^{13} - 4.4 \times 10^{16}$ W with a median of 1.5×10^{15} W over the 60 Class A systems. That is $\times 1.16$ the median P_{90} , and it is the number quoted in the abstract, in Table 2 and in the conclusions. It inherits the one-sided atmospheric-decorrelation bias of +5 to +20 per cent that every injection-calibrated power in this paper carries, so it is an upper bound on the survey's reach and not a central estimate (Table 10). P_{90} remains in the catalogue and in the boxed rule above as the matched-filter completeness, which is what a reader comparing filters rather than surveys wants. Both are columns of the released file. Figure 9 shows the same completeness against duty cycle, and Fig. 7 the per-system spread and the epoch coverage that limits confirmation. The campaign covers 28 configurations rather than all 403. The limit is computational, not methodological: each injected window is re-searched end to end through the frozen pipeline over the full drift grid, so the campaign already costs a substantial fraction of the survey's own search time, and extending it to every window would cost more than the original reduction. That is why the window-to-window transfer is reported as a measured bracket rather than removed, and why widening the campaign is the recommendation of §6.4 that would most directly improve the headline number. The stratified campaign measures how far the recovery curve itself moves between configurations: across the 28 injected windows P_{90} ranges over $\times 0.48$ to $\times 1.35$ of its median-window value and P_{50} over $\times 0.64$ to $\times 1.13$, so a single quoted threshold should be read with a factor of about two of spread around it, and the catalogue carries each star's own bracket. No such carrier was found toward any of the 82 systems searched.

How well is it known? The dominant term is the window-to-window transfer of the injection calibration, $\times 0.48 - \times 1.35$, and on top of it sits a one-sided atmospheric-decorrelation bias of +5 to +20 per cent that can only make the figure optimistic; the ± 9 per cent calibration budget on any one window's flux scale is smaller than either. Table 10 gives the full breakdown. It is quoted to two significant figures. Table 10 is the budget. P_{90} is linear in every factor that builds it, and in d^2 , so each entry is a percentage of the answer. The independent errors combine to ± 9 per cent, dominated by the absolute flux scale: ALMA's delivered absolute calibration is a few per cent at the low bands and up to 20 per cent at Bands 9 and 10, and the Class A sample is weighted toward Bands 6 and 7. Parallax is negligible even though it enters squared, 0.2 per cent typically and 5.0 per cent for the one close binary with a poor solution. The injection campaign's own finite size contributes 2.0 per cent (400 bootstrap resamples, 95 per cent interval $\times 0.97 - \times 1.04$), and residual pointing at 0.6 arcsec contributes 1.9 per cent, the star being at the phase centre where the beam is flat.

One worked example makes the combination unambiguous. The median Class A window, Wolf 219 Band 6, has $P_{90} = 2.37 \times 10^{15}$ W. The two-sided terms combine in quadrature to ± 8 per cent, giving $2.18 - 2.55 \times 10^{15}$ W.

TABLE 10

SYSTEMATIC UNCERTAINTY BUDGET ON P_{90} . THE FIRST BLOCK ARE INDEPENDENT ERRORS ON THE QUOTED NUMBER AND COMBINE IN QUADRATURE; THE SECOND ARE CARRIED SEPARATELY: THE SUB-CHANNEL PHASE IS AN UNKNOWN ABOUT THE TRANSMITTER, THE DECORRELATION TERM IS A *one-sided bias* THAT MAKES P_{90} OPTIMISTIC AND CANNOT MAKE IT PESSIMISTIC, AND THE WINDOW-TO-WINDOW TERM IS A TRANSFER UNCERTAINTY FOR THE WINDOWS THE CAMPAIGN DID NOT INJECT INTO. FOLDING THEM INTO ONE TOTAL WOULD DOUBLE-COUNT THE PER-WINDOW BRACKET THE CATALOGUE ALREADY CARRIES.

Term	Origin	ΔP_{90} (%)
Absolute flux scale	ALMA calibration, band-weighted	± 7
Distance	Gaia parallax, as d^2	± 0.2
Injection statistics	4032 trials, bootstrap	± 2.0
<i>Combined</i>	in quadrature	± 8
Decorrelation	one-sided; injections suffer none	+5/ + 20
Pointing	one-sided; worst case, 0.6'' off axis	+1.9
Sub-channel phase	irreducible; median-corrected	-12/ + 17
Window-to-window transfer	uncertainty off-grid	-52/ + 35

The one-sided terms then apply to the upper end only, since they cannot make P_{90} smaller, taking it to 3.06×10^{15} W. And the window-to-window transfer bracket, a different kind of quantity again, spans $1.15 - 3.21 \times 10^{15}$ W about the same central value. The three are not added together.

Three further terms are listed separately, and they are not all of the same kind. A carrier at an unknown position within a channel is attenuated by a factor the analysis corrects at its median, leaving -12 to +17 per cent that no amount of calibration removes; that is an unknown about the transmitter, not about the telescope. Atmospheric decorrelation is a *one-sided bias* of +5 to +20 per cent: the injected tones that calibrate P_{90} are added to already-calibrated visibilities and suffer none of it, while a real carrier does, so every P_{90} in this paper is optimistic by that amount and cannot be optimistic in the other direction. And the window-to-window spread, -52 to +35 per cent, is a measured property of the survey for the 28 windows the campaign injected into — but for the 403 Class A windows it did not, it is a genuine transfer uncertainty, and it is the largest term in the budget by a factor of several. The practical consequence is that the second significant figure of P_{90} is meaningful for a single injected window and not for the survey, and that the quoted values should be read as an upper bound on the survey's reach rather than a central estimate.

Not excluded. A transmitter absent during the archival observation, or on for less than about a tenth of the time. Emission inside the ± 50 km s $^{-1}$ molecular-line mask, at any strength. Drift outside the searched grid, or drift curved enough to leave a channel within the track. Signals broad enough to be removed with the continuum baseline. Anything below the window's own completeness threshold. And any emission from the 17 566 stars within 40 pc that ALMA has not observed, which is almost all of them.

Two blind spots are worth naming. The search is Stokes I only, so it makes no use of polarisation: a fully polarised carrier still appears in I at its full strength and is not missed, but the discrimination a polarised signal would offer against unpolarised astrophysical emission is unavailable, and a per-hand search of the 198 blocks that deliver both parallel hands would gain up to $\times 1.41$ in signal-to-noise on such a carrier (§6.4). And the frequency coverage is the archive's, not a choice:

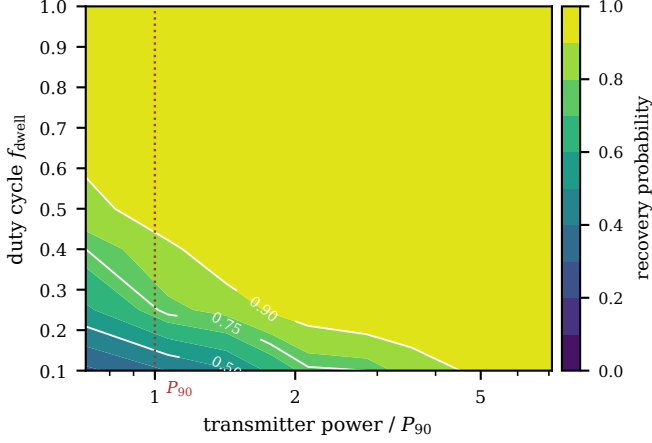


FIG. 9.— The duty-cycle selection function, measured. Recovery probability against transmitter power, in units of P_{90} , and duty cycle f_{dwell} , from 3888 injection trials. The abscissa is in units of P_{90} , which is the pipeline’s nominal trigger power multiplied by $\times 2.29$ for the spectral response and intra-integration smearing and then by $\times 1.2$ from the injection campaign, so a value of 1 here is $\times 2.81$ the trigger power and not the trigger power itself. The quoted P_{90} thresholds apply along the top of this plot. Recovery falls away below $f_{\text{dwell}} \simeq 0.3$ and a transmitter on a tenth of the time needs several times P_{90} to reach the same recovery, so the EIRP limits in this paper constrain persistent and high-duty-cycle emission and say progressively less about brief or scanning beacons. The dwell injections are zero-drift, so this selection function is measured for a stationary carrier and transferred to the drifting primary experiment; the joint drifting-and-intermittent cell is the one cell of Table 4 that remains unmeasured.

the survey has no sensitivity at all to the “magic frequency” arguments that have guided targeted searches — the 1420 MHz line, its multiples, the water hole, or π and e times them — because none of them falls in an ALMA band. A transmitter deliberately placed on such a frequency is outside this experiment by construction, and its absence here is not evidence against it.

6. DISCUSSION

Everything in this section is conditional on what the archive chose to observe, and two features of that choice bound every inference drawn below. 90 per cent of the searched stars were observed for debris-disc or planet-formation science, which skews the sample young, dusty and dynamically active, and puts circumstellar CO — the source of 9 of the 13 stage-1 events — in the field of a large fraction of the windows. And 21 of the 5908 M dwarfs within 40 pc carry a searched window, so the most numerous stellar class in the volume is the least represented here. Neither is a defect of the search; both are properties of an archive assembled for other purposes, and together they mean that the quantities below describe *these* stars and not the 40 pc population.

6.1. What signals could ALMA archival data actually detect?

Each morphology is illustrated with a real window, the completeness entries of Table 4 and Fig. 10. Denominators count the 82 systems whose deepest threshold in the class lies at or below the case power.

(i) *A continuous, frequency-stationary carrier* in a coarse channel is the best case. The UV Ceti pair at 2.7 pc has a deepest window whose trigger is 1.6×10^{13} W and whose effective threshold is 3.7×10^{13} W, and a

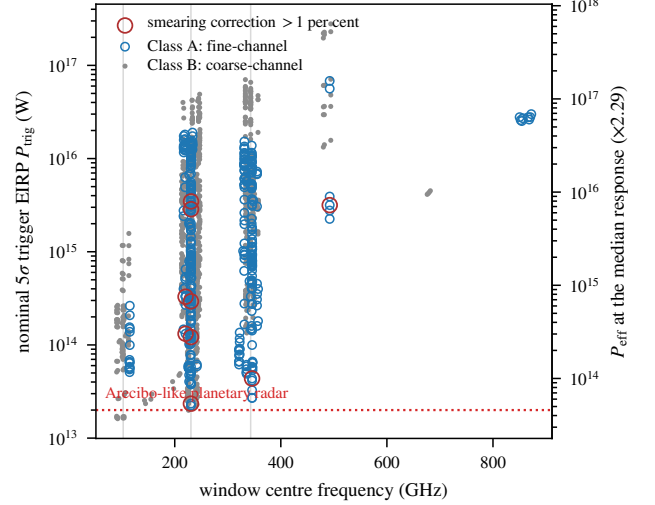


FIG. 10.— Two-dimensional sensitivity against window centre frequency for all 1655 searched windows, symbol by channelisation class, the axis along which completeness differs (Table 4). *Left axis*: the nominal trigger threshold P_{trig} , an internal detection statistic. *Right axis*: the effective threshold at the *median* response, $P_{\text{eff}} = 2.29 P_{\text{trig}}$, which is the power to compare with a transmitter (§4); the released per-window ratio $P_{\text{eff, total}}/P_{\text{trig}}$ runs 1.33–3.98, so read a single window’s threshold from the catalogue and not from this axis. Red rings mark the 9 windows whose intra-integration smearing correction exceeds 1 per cent. Grey lines mark the Bands 3, 6 and 7 concentrations; dotted, the Arecibo 2.38-GHz EIRP benchmark, a power scale only. Note the two decades of scatter at fixed frequency: there is no single survey sensitivity.

channel-confined carrier there is recovered at near-ideal $\sqrt{N}$ gain (§4.4). Read on the trigger, 1 systems of 82 reach the numerical EIRP of the Arecibo planetary radar and 4 reach twice it; read on the effective threshold, which is the comparison a transmitter power calls for, 0 do and 1 reach twice it. The survey’s best case is therefore a few times that numerical EIRP.

(ii) *A continuously drifting carrier* in a fine 488-kHz channel is the only morphology with a measured end-to-end recovery curve (Appendix G): above $\sim 8 \times 10^{13}$ W nominal it is recovered in 83 per cent of trials with its drift inside the searched ceiling, and 10 systems reach that depth in Class A. The measured $0.64\text{--}1.13\times$ transfer applies to the power at which that recovery is reached, not to the recovery fraction itself.

(iii) *An intermittent carrier, on for a quarter of the track*, above $\sim 2 \times 10^{14}$ W is detected toward 28 mid-distance systems. Recovery falls rapidly at smaller dwell fractions (Table 4): below $f_{\text{dwell}} \sim 0.1$ only the nearest windows carry information.

6.2. What does the null result mean?

No persistent, independently confirmed narrowband spectral technosignature has been found within the domain searched (§5.2); an intermittent emitter is not excluded. The EIRP_{trig} values are nominal trigger powers (§4), and completeness-calibrated exclusion holds only in Class A, where the measured recovery curve applies.

One exclusion deserves the converse emphasis. The line mask removes $\pm 50 \text{ km s}^{-1}$ stellar-frame tubes around the catalogued transitions, whose rest frequencies are the ‘magic’ channels Cocconi & Morrison (1959) argued a communicator might choose. This survey is blind by

construction to a transmitter parked there in the stellar frame, a false-positive control purchased with 4.7 per cent of gross bandwidth and one physically motivated class of signal. A carrier deliberately placed on CO(2→1) is not undetectable by ALMA, it is unsearched here, and those channels are retained in the release for anyone bringing independent spatial or time-domain discrimination. Of the 42 planets known around the 19 hosts, only TRAPPIST-1b reaches the drift ceiling in its edge-on worst case, on accelerations GM_*/r^2 from NASA Exoplanet Archive parameters maximised over phase and inclination. Figure 6 shows the ceiling as a selection function of period and host mass, the host-mass dependence being weak ($a \propto M_*^{1/3}$ at fixed period). Coverage eligibility and search completeness differ: §3 audits the 656 public blocks in scope against the 404 in the headline sample. The remainder is accounted for and not outstanding, and the split closes exactly: $656 = 401 + 78 + 177$, being the science-sample blocks that lie in the progenitor set, the reserved calibration hold-out (§C.0.2), and blocks never searched because they are duplicate coverage of the same star and tuning (§3), of which 20 carry no pipeline calibration in the archive at all. The 3 science blocks whose member-OUS link the archive does not expose sit outside this identity and are counted in §3 instead.

Against Wright et al. (2018)’s haystack, only Band 3 falls inside that axis’s 10 MHz–115 GHz span (20.6 of 115 GHz); the remaining 97.5 GHz in Bands 4–8 is new territory. The median threshold detects a 10^{13} W transmitter only within ~ 0.9 pc. Against the sole prior ALMA-archival search (Mason et al. 2024, 28 bycatch targets at ≥ 1.01 kpc), the 82 systems here reach median per-system selection sensitivities P_{90}^{sel} that are $\sim 459\times$ lower in EIRP (1.5×10^{15} W; $\sim 595\times$ if the weaker trigger-level P_{90} is used instead, which is closer to the definition their own published threshold carries, so the quoted ratio is the conservative one). That comparison is almost entirely the distance term and not a statement about the instruments: the ratio of d^2 alone is ~ 6563 , so in *received flux* the present survey is the shallower of the two. The EIRP ratio is the right comparison for a transmitter power and the wrong one for sensitivity, and we give both. We do not combine these fractions into one coverage number, because the axes are strongly coupled and a product of marginals would misstate the overlap by orders of magnitude.

Comparison with previous work. The meaningful comparison is axis by axis rather than on a single figure of merit: this survey differs from centimetre-wave searches in frequency, spectral resolution, number of systems, EIRP completeness, drift range and temporal coverage simultaneously, and Fig. 1 makes the central point, which is that its novelty is the frequency regime and not depth. For readers who want the conventional Continuous Waveform Transmitter Figure of Merit it is computed in Appendix B, together with the reason the survey-union convention flatters every archival survey and why we prefer the per-system value. On that metric this survey is about four orders of magnitude less competitive than major centimetre-wave programmes, which is the correct conclusion and not a criticism of either.

How far that null reaches is set by recurrence, not by the rank statistic. No single event in this design can clear

the survey-wide threshold (Appendix K), so a genuine but weak transmitter seen in one epoch is not separable from the 4.7 chance events the survey expects, which is exactly the position CP–72 2713 leaves us in. What separates them is a second epoch or an independent astrophysical identification, and here the distinction between a repeat block and a temporally independent epoch matters. Of the 82 systems, 27 were searched in a single execution block and 55 in more than one — but 13 of those 55 have all their blocks within 1 day of each other, which is one observing session and not an independent test of persistence. **Only 42 systems, 51 per cent, have epochs separated by more than 1 day:** 12 over days, 7 over months and 23 over years. Recurrence over years can therefore be tested for fewer than a third of the sample. A further 10 hold archived blocks that were not both processed, 65 in all, so the recurrence test could in principle be extended to 79 per cent of the sample with no new observing time.

In plain terms, ALMA has observed 90 nearby stars as a by-product of other science, across parts of the millimetre band no technosignature search had examined, and we found no channel-confined carrier the surrounding sky could not also produce. On the few gigahertz, few hours and single morphology class each star happened to be observed with, that says little about how common transmitters are.

6.3. What this sample can and cannot speak for

Selection is on archival coverage, and ALMA’s archive was not built to sample stars. 81 of the 90 searched stars were observed under proposals categorised *Disks and planet formation* (Fig. 2), and 75 carry the *Debris disks* science keyword (69 of 82 systems). Among the 53 searched stars with a Gaia temperature, 21 are M dwarfs, 40 per cent, against 69 per cent of the reference census (Table 13), so the sample is F- and G-enriched and M-dwarf deficient. Window counts are concentrated too: three systems — BD+05 1668, Proxima Cen and HD 33793 — carry 285 of the 1655 searched windows, 17.2 per cent.

Two consequences follow, and they run in opposite directions. A debris-disc-dominated sample is the worst case for astrophysical false positives at millimetre wavelengths, which is where the stage-1 flagged windows of §5.2 concentrate, so the vetting burden here is heavier than a blind sample would carry. And because M dwarfs host most of the habitable-zone planets in the solar neighbourhood, and only 21 of the 5908 within 40 pc carry a searched window: 0.4 per cent. This result therefore carries very little demographic weight for the local habitable-zone planet population. Both counts are restricted to the temperature-classified sample of Table 13, so they illustrate the selection rather than measure it.

The point is sharper stated in targets than in percentages. Five of the searched M dwarfs host a planet that some published analysis places in the habitable zone: Proxima Centauri, TRAPPIST-1, LHS 1140, GJ 581 and HN Lib. Habitable-zone membership is model-dependent and two of these are contested, so the number should be read as an order of magnitude and not a census. Five is what this survey covers of the class of target that most current technosignature programmes prioritise, against a local M-dwarf population of 5908 within

40 pc. Conversely, the over-representation of F and G stars means Sun-like stars are sampled relatively heavily, though still through a highly non-random, disc-selected archive. Either way it constrains transmitters around the stars ALMA happened to point at, not the local stellar population.

Useful subsamples still fall out of archival selection. 19 of the 90 processed stars (42 known planets) are confirmed exoplanet hosts, among them Proxima Centauri, τ Ceti, ϵ Eridani, β Pictoris and the seven-planet TRAPPIST-1 system, while HN Lib and LHS 1140, each with a habitable-zone planet, now carry an ALMA technosignature non-detection. Four searched stars are white dwarfs (Gentile Fusillo et al. 2021), all continuum non-detections and among the few mm/submm SETI constraints for that class (Huang et al. 2026).

6.4. A next-generation ALMA technosignature search

ALMA’s advantages here are its mm-wave sensitivity, an essentially unsearched tuning range, the interferometric spatial discrimination that supplies the control statistic of §4.1, and a public archive re-processable without new allocations. Six changes would enlarge what this experiment can say. They are given in priority order, and the criterion is explicit: how much of the signal space each buys per unit of effort, counting an archival reprocessing as cheap and new observing time as expensive. The first two would change a conclusion; the last four would widen a domain.

(1) *Use the visibilities* [changes what a candidate means] This is the principal methodological lesson of the survey, and §C.0.3 is why. A celestial point source at the stellar direction $\mathbf{x}_\star$ contributes

$$V_{ij}(\nu, t) = A(\nu, t) \exp(-2\pi i \mathbf{u}_{ij} \cdot \mathbf{x}_\star), \quad (7)$$

so rotating the phase centre to the star should leave a coherent signal entirely in the real part, with an imaginary component consistent with zero on every baseline. That test asks whether the data contain a point source *where the star is*; the control ranking asks only whether one sky extraction is unusually large compared with neighbouring ones, which is a weaker question and, as §C.0.3 shows, one whose null depends on where in the field the comparison positions sit. Applied to the one unattributed window (§5.2) the visibility fit was decisive; applied to every window it would demote the control ensemble to a calibration layer, remove the radius-matching problem entirely, and give each candidate a physically specific test rather than a rank. Doing so needs the visibilities retained, which this survey did not do, and it is the first change we would make, which is why it is ranked first here.

(2) *Search the remaining public blocks these stars already hold* [multiplies the epochs on a minority of systems, at no telescope cost] This recommendation has since been acted on in part, and §D.0.2 reports the result: 115 of the 177 blocks have been searched with the frozen pipeline. They cost no telescope time and no new proposal. What they buy needs stating precisely, because it is narrower than it sounds. Every one of them is repeat coverage of a star and a tuning already searched, so they add no star, no system and no frequency: the only axis they extend is time. They are also concentrated, not

spread — 135 of them attach to 29 of the 90 stars and 27 of the 82 systems, with Proxima Centauri, TRAPPIST-1 and HD 202628 accounting for the largest holdings. And they barely touch the confirmation gap that limits this survey: of the 27 systems with a single searched epoch, only 5 would acquire a second block. The extension bears that out, having raised the number of multi-epoch systems from 55 to 56 and moved 1 system out of the single-epoch class. A survey that keeps growing has no fixed denominator for its false-alarm accounting, which is why the analysis reported here has one. One caution carries over from the worked case, whose two blocks are repeat executions of a single scheduling block on one night: a second block often supplies a baseline of only hours, and which it is can be settled with one archive query before any download.

(3) *Retain more control spectra* [repairs the second-stage null, at storage cost only] Full dynamic spectra survive for only 8 of the 512 control positions, which is what breaks the second-stage null in a structured field and what limits the variance test of §L.1.2. Retaining all of them costs storage and nothing else.

(4) *Drift discrimination on coarse windows* [recovers an axis for 1252 of 1655 of the windows] Within the searched ceiling every in-grid drift on a 15.625-MHz window is sub-channel over a track, so 1252 of 1655 windows measure amplitude but cannot distinguish a drifting carrier from a stationary one (§4.4). Finer channelisation, or a drift model spanning several blocks, would recover that axis.

(5) *Extend the injection campaign and the polarisation axis* [tightens the transfer bracket, the largest uncertainty on the headline number] The stratified campaign of Appendix G now spans band, channel width, integration time and array, and what remains unspanned is other target environments and edge-phased drift grids. The polarisation axis buys discrimination against weakly polarised astrophysical backgrounds at no sensitivity cost.

(6) *A boundary on drift linearity* [bounds a modelling assumption that no observed target violates] The search assumes linear drift over a track, and curvature matters once $\frac{1}{2}\ddot{\nu}T^2$ exceeds half a native channel, which only the shortest-period orbits reach: $P \lesssim 1.6$ d for a typical Class A window (230 GHz, 488 kHz, $T \approx 2000$ s). Among the 42 known planets of the 19 hosts in the sample, 1 lies inside that boundary: TRAPPIST-1 b at 1.51 d, which is also the case that reaches the drift ceiling. The next shortest period is TRAPPIST-1 c at 2.42 d and the shortest outside that system is GJ 581 e at 3.15 d, both comfortably linear; periods are catalogued for 39 of the 42, the exceptions being directly imaged companions on orbits of decades, which are further from the boundary still.

Where the ceiling actually falls. The grid is built at constant acceleration rather than constant drift rate, so the boundary is one number across the whole band: 3.60 ms^{-2} , which is 1.4 kHz s^{-1} at 115 GHz and 10.5 kHz s^{-1} at 872 GHz (Fig. 6). It clears the accelerations a transmitter fixed to a rotating, orbiting planet cannot avoid by wide margins — Earth’s rotation by a factor of 106 — so for terrestrial-analogue geometries the grid is not the limiting choice. The grid takes 2 values,

the wider one used for the 5 windows of TRAPPIST-1 alone, and even that widened grid stops 0.01 per cent short of TRAPPIST-1 b at maximum line-of-sight acceleration. 1 of the 39 known planets is therefore outside the searched domain and 38 are inside it in every window.

Budget for the disc-CO false positives, but do not pre-judge the targets. 9 of the 13 stage-1 flagged windows are molecular emission toward disc hosts, arising in 2 of the 69 debris-disc systems searched: one flagged system in 34, or 2.2 per cent of Class A windows. That is the natural false-positive rate an archive proposed for disc science carries, and any future search on ALMA, NOEMA or the JVL A inherits it. The remedy is procedural: run the continuum lane (Appendix J) *before* promoting a window, which would have caught HD 48370 and both β Pictoris windows at the first gate. We do *not* recommend lowering the threshold toward disc hosts: a detection statistic should not know whether its target is astrophysically interesting, or a real transmitter would be penalised for the company it keeps. Disc membership belongs in classification, where it already is.

7. CONCLUSIONS

1. **Search performed.** 403 fine-channel spectral windows toward 60 independent stellar systems, covering $\Delta\nu_A = 47.7$ GHz of unique sky frequency, searched for an unresolved narrowband carrier whose frequency drifts. A secondary coarse-channel spectral-excess search covered a further $\Delta\nu_{B\text{-only}} = 70.4$ GHz in 1252 windows but cannot discriminate drift. Both come from a uniform reduction of 484 public execution blocks, 4.1 TB of raw data.
2. **Sensitivity.** Measured by injection and recovery through the complete automated selection, not inferred from a nominal threshold. Per system, on each system's best window, $P_{90}^{\text{sel}} = 8.7 \times 10^{13} - 4.4 \times 10^{16}$ W of equivalent isotropic radiated power with a median of 1.5×10^{15} W. This is an upper bound on reach rather than a central estimate: the window-to-window transfer of the calibration is uncertain by $\times 0.48 - \times 1.35$ and dominates the budget, and a one-sided $+5$ to $+20$ per cent decorrelation bias can only make the figure optimistic.
3. **Result.** 13 windows passed the automated screen; 9 are attributable to circumstellar or foreground carbon monoxide, among them the β Pictoris debris disc recovered blind in two transitions, which serves as the survey's positive control. **None of the remaining 4 is localised as a point source at the stellar position in the visibilities, and none is independently reproduced in a repeat observation.** The screen itself assigns them no calibrated false-alarm probability, because its null distribution is empirically non-uniform under both the frozen and the radius-corrected normalisation (§5.2.3).
4. **Interpretation.** No persistent, independently confirmed unresolved narrowband carrier was found within the frequency, drift-rate, duty-cycle and sensitivity domain searched. **This cannot be converted into an occurrence constraint on technological civilisations within 40 pc.** The sample is archive-selected and strongly incomplete: 90 per cent of the searched stars come from disc and planet-

formation programmes, only 21 of 5 908 catalogued M dwarfs within 40 pc are covered, and the median system is searched over 1.84 GHz rather than the survey's 47.7 GHz union. Any occurrence figure quoted in §6.2 is conditional on that sample, those frequencies and those epochs, and on nothing wider.

AUTHOR CONTRIBUTIONS

G.J.W. conceived the survey, defined the target sample and the selection criteria, and wrote the manuscript. R.D. built and operated the archival retrieval, calibration and search pipeline, implemented the spatial control statistic and the injection-recovery campaigns, and produced the released catalogue and its regeneration scripts. Both authors designed the candidate vetting procedure, carried out the statistical validation reported in §4.1 and Appendix K, and agreed the dispositions of Table 7.

FUNDING

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COMPETING INTERESTS

The authors declare no competing interests. VBRL Holdings is a privately held company through which R.D. conducts independent research, and is named here solely as his affiliation. It is a private company and had no role in the design of the survey, the selection of targets, the analysis, the decision to publish, or the content of this paper; it provided no funding for the work.

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This paper uses ALMA data from the public ALMA Science Archive. The 404 execution blocks searched span 2013 October 6 to 2025 June 11 across 36 proposal codes; their project codes and epochs are listed in the machine-readable release and should be cited from it in the standard JAO.ALMA#XXXX.X.NNNNN.S form. ALMA is a partnership of ESO (representing its member states), NSF (USA) and NINS (Japan), together with NRC (Canada), NSTC and ASIAA (Taiwan), and KASI (Republic of Korea), in cooperation with the Republic of Chile. The Joint ALMA Observatory is operated by ESO, AUI/NRAO and NAOJ. This work has made use of data from the European Space Agency (ESA) mission *Gaia*, processed by the *Gaia* Data Processing and Analysis Consortium (DPAC). This research has made use of the SIMBAD database (CDS, Strasbourg, France).

DATA AVAILABILITY

Every reduced data product, the full target-selection and analysis pipeline, the per-window catalogue, the injection records and the control statistics are archived at <https://github.com/Tilanthi/SETI> and are sufficient to reproduce every number, table and figure in this paper. Appendix A describes the release and the checks that demonstrate its sufficiency.

SOFTWARE

The pipeline depends on CASA 6.7.6 (casatools/casatasks 6.7.6-14; CASA Team et al. 2022), Python 3 with astropy (Astropy Collaboration et

al. 2022), `numpy` (Harris et al. 2020) and `scipy` (Virtanen et al. 2020), and the ALMA Science Archive and SIMBAD TAP interfaces through `astroquery` (Ginsburg et al. 2019); all are free and open-source, with exact versions recorded in the frozen release.

ABOUT THIS PAPER

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APPENDIX

APPENDIX ROADMAP

The appendices are reference material; none is needed to follow the result. They answer, in order: how frequency, linewidth and smearing are defined and corrected (F); how the injection campaign was built and what it measures (G); the per-window catalogue (H); the velocity-space line mask (I); the ancillary continuum lane (J); false-alarm accounting under the symmetric criterion (K); provenance, extraction geometry and the robustness checks that are summarised rather than narrated (A); the spatial normalisation, its evidence and its prospective validation (C); the conventional figure of merit (B); and the notation used throughout (E).

A. REPRODUCIBILITY AND PROVENANCE

The reduced data products underlying this paper, with the full target-selection and analysis pipeline, are at <https://github.com/Tilanthi/SETI>. All numerical values here come from the *submitted-analysis snapshot*, the repository commit tagged at submission, which freezes pipeline code and input data as run and reproduces every number, table and figure from the regeneration scripts shipped with it.

The machine-readable catalogue, `per_target_results_v3.92.csv`, has one row per searched window, 1655 rows in 50 self-described columns. They cover four things. *What was observed*: star, system, distance, band, execution block, searched frequency range, channel width, exposure and per-channel rms. *What the search could reach*: the nominal EIRP_{trig} trigger with both response-corrected values (§4), P_{90} for that window with the $\times 0.48$ – $\times 1.35$ bracket the injection campaign measures, the drift ceiling, the trial count, η_{drift} and η_{smear} , and the resolution and search class. *What the search found*: the stellar and control peak statistics, the add-one rank under both the released and the radius-corrected normalisations with the stage-1 flag each implies, the crossing frequency, the nearest catalogued transition, and — for the 10 rows that have one — the disposition verbatim from Table 9, since only a window that reaches stage 1 and is dispositioned carries a verdict. *How the controls were placed*: ring centre, inner and outer radius, probe count and seed.

For the referees’ use, and released with the paper, the file `REFeree_VERIFICATION_v392.txt` carries the verbatim output of the regeneration and audit scripts: every quantity they reproduce from the released catalogue

alone, every cross-check they assert, and the candidate-selection counts under both the primary and the frozen statistic. It exists so that the density of cross-referenced counts in this paper can be checked mechanically rather than by eye.

The prospective extension of §D.0.2 is released as a *separate* file, `export_extension.json`, with its own 463 window records in the same schema and the full 512-element control vector for each. It is deliberately not merged into the catalogue: those blocks were searched after the statistic and the mask were fixed, so pooling them would silently change the denominator on which every rate in this paper is calibrated. Anyone wishing to pool them can, but it is then their choice and not a hidden one. That list is chosen so that the catalogue is sufficient, not merely informative: the sample sizes, the window and system counts, the candidate list under either statistic, the frequency union, the control geometry and every quoted P_{90} — per window and per system, best, median and worst — are derivable from this one file. The release ships `reproduce_from_catalogue_v385.py`, which opens the catalogue and nothing else and re-derives 41 of the paper’s headline numbers from it, comparing each against the value typeset here; it exits non-zero if any disagrees. The statistical unit is `system_id`, with 8 bound pairs sharing one identifier. Deleting every generated table and figure and rebuilding from the scripts alone reproduces all 83 products byte for byte. Blocks searched by the epoch-extension campaign of §C.0.2 lie outside the frozen release and ship in the accompanying epoch-extension record. The release also builds diagnostic figures the paper does not print, among them the per-band noise-quality plot and the illustrative occurrence curve; they are products of the same generators and carry no number quoted here.

The version of record is a frozen archive, not a branch: the Zenodo deposit carries the exact commit that produced every number, table and figure in this paper, the 1655-row catalogue, the control statistics, the injection records and the calibration sample as a separately labelled product, and the repository tag resolving to that commit is named in the deposit. A reader who regenerates from the deposit reproduces this paper; the GitHub branch may move on. The deposit carries a DOI minted on acceptance. It holds the per-window search products for all 1655 windows, each with its 512 control statistics, so every rank and p -value in this paper is recomputable. It also holds the line mask in frozen and repaired forms, the execution-block metadata and pointing ledger, the 1200 injection, 3 888 dwell and 4032 stratified trials, the local-scramble code and outputs, and the repeat-block and visibility products behind the CP–72 2713 disposition. Finally it holds the frozen pipeline and data snapshot, with the change-log that timestamps the promote and retire criteria in the repository before the data were seen.

Data products carry the Creative Commons Attribution 4.0 International licence (CC BY 4.0) and the pipeline code the MIT licence. Raw ALMA visibilities are public from the ALMA Science Archive at <https://almascience.org>.

A.1. Extraction, control geometry and position-correlated systematics

The baseline filter takes the median in contiguous blocks of 65 channels and interpolates linearly between block centres, using $\max(2, n/65)$ blocks over n usable channels. Its effective width is therefore 65 channels in the 134 windows long enough to hold more than two blocks, and $n/2$ in the short coarse windows where the block count collapses to two. This sets a ceiling on the linewidth the filter passes: ≈ 32 MHz at the fine class’s modal channelisation, and ≈ 0.9 GHz at the coarse class’s (Appendix F). The crossing channel is *not* excluded from the scale. The median is taken across positions at a fixed channel, so what matters is how many of the 512 controls carry the feature. A compact feature at the star appears in none of them and the 50-per-cent breakdown point holds; emission filling the primary beam appears in all of them, which is the HD 48370 case (§5.2). $T_\star \equiv T(\mathbf{x}_\star)$ at the proper-motion-propagated stellar position, and the same operation, drift grid, channelisation and noise map give $\{T_i\}$ at all 512 control positions.

The controls are placed by `probe_positions()` in the released extraction code, which adds its offsets to the star’s own direction cosines, so the annulus is centred on the *star* itself, wherever the star sits relative to the field phase centre. It holds 512 positions drawn uniform in area, $r = \sqrt{U(r_{\text{in}}^2, r_{\text{out}}^2)}$, and uniform in azimuth, from the fixed seed 20260825 re-applied per window. All windows therefore share one pattern in units of θ_{PB} , and two windows of one tuning sample identical sky positions. Radii run $0.14\text{--}0.78\theta_{\text{PB}}$, with θ_{PB} evaluated at each window’s own median frequency, so the annulus scales window by window. A separate inner check-ring of 135 positions serves the point-source test, and the full per-window geometry is released as catalogue columns.

The frozen products carry none of that geometry, so it is reconstructed from the code, and two checks confirm the layout that ran: the inner check-ring it generates has the 135 positions the retired region-max statistic maximised over, and where extended CO fills the beam the control statistics fall with reconstructed radius, at Spearman $\rho = -0.49$ in the HD 48370 Band 6 window. Neither would survive a wrong position-to-index mapping.

Placing controls off-centre costs sensitivity. A star near the pointing centre sits at primary-beam response ≈ 1 while the annulus samples 0.95 at $0.14\theta_{\text{PB}}$ falling to 0.19 at $0.78\theta_{\text{PB}}$, area-weighted mean 0.47. Star and controls are therefore not exchangeable in flux density, and they remain exchangeable in the signal-to-noise ratio the statistic actually ranks, because $T(\mathbf{x})$ is formed on uncorrected amplitudes against one shared noise scale. Extended sky brightness does not cancel: smooth emission lives on short baselines whose phases barely change under a rotation of a few arcsec, so star and annulus rise *together*, as they do for HD 48370, while emission offset from the star raises particular controls above it, as in the same star’s ^{13}CO window (ring maximum 20.4 at 9.2 arcsec against 4.6 at the star). Two asymmetries remain and they run in opposite directions: a suppressed control can only make a star-exceeds-ring outcome more likely under the null, whereas a multiplicative calibration error scales with the continuum flux at the extracted position and so favours the star, where that flux is not zero.

Neighbouring controls are correlated below the synthesised-beam scale, so a window’s annulus carries $N_{\text{eff}} \simeq 30\text{--}43$ independent spatial trials in the compact configurations where the symmetric reprocessing runs measure it. That figure is an analytic beam-geometry estimate, not a measured correlation length, and it is the worst case: the annulus contains far more resolution elements than the 512 positions drawn in all but those configurations (Appendix K). The rank test needs exchangeability rather than independence in any case, and correlation among controls costs resolution without touching the $1/513$ floor. The frozen products do not carry the synthesised beam or the array configuration, so radii are quoted in arcsec and in θ_{PB} ; the ALMA archive carries both, and what they say about this geometry, including the 64 per cent of windows taken with the ACA 7 m array against a θ_{PB} defined on 12 m, is in Appendix K.

The consequences of that resolution for survey-level significance are in §4.1, Eq. 2. Table 3 allows the statistic to be reimplemented from the main text alone.

Residual continuum after baseline removal, self-calibration residuals and position-correlated calibration error all concentrate at or near the phase centre, where the pointed science target sits, and σ is built from the annulus and carries none of it. These stars are not all blank there, the continuum lane detecting 17 of 57 target/bands (Appendix J). A multiplicative spectral calibration error carries the same asymmetry in the unfavourable direction, since a residual bandpass ripple of fractional amplitude ϵ leaves a channel-confined artefact of order ϵS_{cont} at the extracted position and essentially nothing on the annulus. The direct test of the premise is §L.1.2, which compares the star’s own residual variance with the controls’ in the flagged windows of the released catalogue and finds them equal to better than a per cent, and two indirect ones agree with it. We also recomputed the scale locally, in a band around each feature and excluding the feature itself. The local-to-global ratios are 0.97 (CP–72 2713), 1.29 (HD 48370), 0.93 (β Pic B3) and 1.02 (β Pic B6). They agree with the reproducible R_σ test above and carry the same conclusion: the global scale is accurate to a few per cent except towards HD 48370, where the field is filled with CO and rescaling both the statistic and its ensemble leaves the window flagged ($T_\star = 21.0$ against 20.8), which is the disposition it already has. And because the same estimator runs at the star and at every control, a badly misstated σ would displace the control distribution, which the uniform pseudo-star ranks exclude. None of this reaches a per-crossing local scale inside the search, which remains an open item (§6.4).

A rank means something only when the identical operation runs at the star and at every control: under a pure-noise null a region maximum over n_{src} positions at the star exceeds all n_{ctrl} single controls with probability $n_{\text{src}}/(n_{\text{src}} + n_{\text{ctrl}}) = 20.9$ per cent for the 135-position check-ring, whatever the window and whatever the noise distribution. The released statistic therefore evaluates one position at the star and one at each control; §L.2 compares the two forms and audits every window the asymmetric one flags.

A.2. Statistical units and independence

Six units appear below and they are not interchangeable, so we fix them here and say which one carries which probability. A *catalogue entry* is one star as SIMBAD designates it, 90 of them searched; a *physical system* is the set of entries gravitationally bound together, 82 of them, seven designation-linked pairs accounting for the difference. A *target/band* dataset is one entry in one ALMA band, 113 in all. An *execution block* is one archival observation, 404 searched, and it is the unit that shares weather, calibration and correlator setup. A *spectral window* is one window of one execution block: 1655 catalogue rows, which reduce to 1616 distinct (execution block, window) datasets once repeated archive products are collapsed. An *independent frequency trial* is neither of those: within a window the channel $\times$ drift grid is oversampled on both axes, so its 177–6 869 740 cells carry about a factor 5.3 fewer independent trials than cells (Appendix K). An *independent epoch* is a separate execution block at the same tuning, and blocks of one scheduling block executed on one night are not independent epochs in any useful sense.

Which unit enters a population-level statement therefore depends on the statement. Per-window rank probabilities are conditional on the window and use no denominator at all. The survey-level flag expectation uses 1655 windows, which is conservative because the 1616 distinct datasets are the smaller number and because windows within a block are correlated; 1655 windows are not 1655 independent trials, and no probability here should be read as though they were. Anything about how common transmitters are must use 82 systems, never 90 entries and never 1655 windows. Anything about persistence uses epochs, of which most stars here have one.

A.3. Data-quality exclusions and open items

113 target/bands are searched to completion and reported here. Four classes of entry are absent from the released catalogue and are itemised in Appendix L.

A further 68 candidate target/bands were selected but never delivered a searchable product, and the cause lies in the selection step. Re-running the crossmatch shows that 60 of them were never observed at all, no observing unit offered to those entries containing the star: they sit a median 1.6 degrees from the nearest field centre, because the observations were pointed at solar-system targets whose quoted field of view is enormous, and a crossmatch that trusts that quantity admits stars far outside any real primary beam. The remaining 8 entries did not yield a search product. A positional crossmatch must not use the quoted field of view of an ephemeris or solar delivery as a search radius; the per-star pointing ledger is released so the selection can be audited. 53 of the 168 stars admitted to the candidate list fail on pointing in every band, so the sample here is 90 of at most about 115 genuinely covered stars rather than 90 of 168, and the coverage completeness of Table 2 should be read against the smaller denominator.

A.3.1. Data-quality screening of the noise estimate

One automated quality check is applied before any sensitivity number is reported. The radiometer relation makes $q \equiv \sigma\sqrt{t_{\text{on}}\Delta\nu_{\text{ch}}} \propto \text{SEFD}/\sqrt{N_{\text{bl}}}$ a quantity

physics predicts, so a window whose q falls far below the sample median cannot be a noise fluctuation. 13 coarse time-division windows fail it by about three decades, all of them channel-averaged auxiliary windows riding on a science baseband and carrying a constant WEIGHT column, so they never received that baseband’s calibration. Left in, they would have supplied the deepest apparent thresholds in the survey. The deflation is common mode in the data and in the weights alike, so it misstates a threshold but cannot manufacture a flag: the 13 windows are excluded from the sensitivity accounting only, and their signal-to-noise statistics and ranks were carried through candidate generation unchanged. The full diagnosis, including the sibling-window comparison that identifies the cause, is in Appendix L.

45 of the 1725 extracted rows are duplicate catalogue entries for one physical window, the same (star, execution block, frequency range, channel width) key appearing twice in the archive query, so one row per key is the right statistical unit. The retention rule is deterministic and quantity-independent, the key’s first-listed row in the query’s own identifier order, so it consults no measured value and cannot prefer a crossing or a deeper threshold. Repeats need not agree: two reductions of the same τ Ceti Band 6 window differ by 44 per cent in threshold, which exceeds the flux-scale systematic of §4.2 and is the most direct empirical bound here on threshold reproducibility, and the two HR 1010 Band 6 rows straddle the trigger at $T_{\star} = 5.10$ and 4.95, so the crossing count is rule-dependent at the level of one window. The four flagged windows are not, no discarded row being both a crossing and a stage-1 flagged window (Appendix L).

A.3.2. CP–72 2713: an unconfirmed first-block event, absent in the repeat block

CP–72 2713 is the worked example of how a marginal event is dispositioned, and it is not a special object: at the measured false-alarm rate the survey expects 4.7 such events and has one. In Band 7 it reached $T_{\star} = 5.81$ against a ring maximum of 5.68, never acquired an astrophysical attribution, and was recorded as an unattributed stage-1 flagged window under promote and retire criteria committed on 2026 September 11 (119a68a6ab50), 9 hours before the second block was searched.

That second block matches the first in tuning, channel width, channel count, integration count and on-source time, registers to 0.18 of a channel, and is the deeper of the two, so a persistent emitter at the first block’s flux would have returned $T_{\star} = 6.21$. Pinned to the first block’s own frequency and drift it returns 3.60, and read as two measurements of one constant amplitude the pair is inconsistent at 4.6σ . The two blocks are 2.05 h apart, so this is a within-night test and not an independent epoch; it excludes a persistent emitter of that strength on that baseline and excludes nothing about an intermittent one. Appendix L gives the frequency, drift, control and continuum diagnostics in full, and §5.2.1 the visibility result.

CP-72 2713, in one line. $T_\star = 5.81$ against a ring maximum of 5.68 in one Band 7 window; no line within $\pm 50 \text{ km s}^{-1}$ in the stellar frame; no localisation in the visibilities ($\text{Re}/\sigma = -0.86$); absent in the deeper repeat block 2.05 h later, where a persistent source of the same flux would have given $T_\star = 6.21$. *Disposition:* unattributed stage-1 flagged window, not a candidate, on the recurrence test and not on the rank.

A.4. Ancillary results

The line-excluded continuum lane is exploratory and bears on no technosignature limit (Appendix J).

A.5. Sample bookkeeping: entry counts, coverage criteria and pointed versus serendipitous

The 168 stars are complete against, and conditional only on, the 40-parsec reference catalogue intersected with the public-archive snapshot of 2026 September 8. Against that Gaia DR3 catalogue of 17 566 stars they are $\lesssim 1$ per cent and the 82 independent systems searched are 0.47 per cent, fewer still once §A.3 removes those never observed (Fig. 3). The two parallax cuts are not interchangeable, the sample's binding condition being $\sigma_\varpi/\varpi < 20$ per cent against the census's 10 per cent, and the realised sample is better than its condition, reaching 2.48 per cent at worst. Because the census is incomplete at both ends, the fraction shows only that coverage is small.

Stellar counts throughout are catalogue-entry counts. The 90 searched entries comprise 82 independent systems, differing by seven designation-linked component pairs (SIMBAD designations; per-entry rows, distances and unresolved-pair notes in the machine-readable catalogue). The 113 star/band datasets, 1655 windows and 404 execution blocks therefore derive from 82 systems. The 1655 rows comprise 1616 distinct (execution block, spectral window) datasets. We use the larger number as the trials denominator throughout, which is conservative and makes no practical difference: it predicts 3.2 chance outliers against 3.1 for the smaller.

1. At least one public execution block whose visibilities are the archive's delivered QA2 products, taken as delivered and not re-calibrated. Calibration vintage therefore varies across the 2013–2025 span; the flux-scale term this contributes is small beside the $\times 2.29$ spectral-response correction. The block must cover the star's epoch-propagated position. There is no minimum integration, sensitivity, antenna count, mosaic-tile count or array configuration beyond QA2.
2. That position inside the field's primary beam. Extraction aborts beyond $2\theta_{\text{PB}}$ and the largest retained offset is $0.46\theta_{\text{PB}}$, so nothing approaches the bound.
3. Each spectral window admitted individually, on QA2 vetting alone.
4. No programme-category restriction: science, calibration and observatory projects enter on identical terms.
5. A fixed archive snapshot date, 2026 September 8.

Completeness with respect to the work list holds in targets and tunings, not in epochs or integration time: the survey takes at most 25 blocks per target and deduplicates spectral windows by identifier, so 404 of the 656 public execution blocks those targets hold are in this release, 61.6 per cent. Every epoch statement in this paper concerns analysed epochs, and the archive would support a stronger one. Appendix K gives the full block

accounting and the 326 further blocks reserved as an external calibration sample.

ALMA project metadata quantify pointed versus serendipitous coverage: almost every spectral-window-covered star is the pointed science target in at least one execution block, leaving Wolf 219 the sole serendipitous star, so the residual selection function of a work-list-complete sample is the union of ALMA target lists.

B. THE CONVENTIONAL FIGURE OF MERIT

The standard occurrence metrics, and what they are worth here. A reader placing this survey on the axes the literature uses needs the same figures of merit everyone else quotes, so we compute them. Following Enriquez et al. (2017), the Transmitter Rate is $(N_\star \nu_{\text{rel}})^{-1}$ and the Continuous Waveform Transmitter Figure of Merit is

$$\text{CWTFM} = \zeta_{\text{AO}} \frac{\text{EIRP}}{N_\star \nu_{\text{rel}}}, \quad \nu_{\text{rel}} = \frac{\Delta \nu_{\text{tot}}}{\nu_{\text{mid}}}, \quad (\text{B1})$$

with $\zeta_{\text{AO}} = 5.0 \times 10^{-11}$ fixed by their normalisation (CWTFM = 1 at EIRP = 10^{13} W , $\nu_{\text{rel}} = 1/2$, $N_\star = 1000$) and lower values better. On the Class A experiment, $N_\star = 60$ systems and $\Delta \nu_{\text{tot}} = 47.7 \text{ GHz}$ about $\nu_{\text{mid}} = 494 \text{ GHz}$ give $\nu_{\text{rel}} = 0.097$, a Transmitter Rate of 0.17 and, at the median per-system $P_{90}^{\text{sel}} = 1.5 \times 10^{15} \text{ W}$, CWTFM = 1.3×10^4 (3.8×10^5 at the worst-case system). That figure is reported for comparison with the literature and is not our preferred measure. It uses the survey-wide frequency union, which no single star was observed over; the quantity that answers whether a transmitter's frequency was actually observed is the per-system version below. For scale on the conventional axis, Enriquez et al. (2017) reach 0.85 and Price et al. (2020) 0.11 at the GBT. This survey is therefore some four orders of magnitude worse than cm-wave SETI on the standard continuous-carrier metric, and that is the correct conclusion: 60 systems at a 10^{15} W floor cannot compete with hundreds of stars at 10^{13} W . What the metric does not score is the frequency axis, the one on which this survey is first, since Eq. B1 credits fractional bandwidth and not location, and 47.7 GHz at 494 GHz is a small fraction of a large number.

Two caveats are not presentational. First, the transmitter-hosting fraction. Zero confirmed detections in 60 Class A systems bounds the fraction hosting a continuously transmitting carrier above each system's own P_{90}^{sel} at $f < 5.0$ per cent, the 95 per cent Poisson limit; the $1/N$ convention common in the literature would say 1.7 per cent, and is optimistic. That bound holds only for carriers inside the searched frequency islands, drift grid, duty-cycle domain and archival epochs, on a sample ALMA selected for disc science, and it is not a statement about the 40 pc population.

Second, and more seriously, ν_{rel} in Eq. B1 is a survey-level quantity, and *a civilisation around one star does not have access to the union of frequencies observed toward all 82 systems*. System by system the median Class A coverage is 1.84 GHz, from 0.11 to 19.08 GHz over the 60 systems that have any — 3.9 per cent of the survey's own union. Recomputing on that median gives $\nu_{\text{rel}} = 0.0037$ and CWTFM = 3.3×10^5 , worse by $\times 26$. We therefore take the per-system value as the scientifically meaningful one and the survey-union figure as a literature conven-

tion: it measures aggregate bandwidth, not the probability that a given transmitter frequency was observed. Under any frequency prior broader than the searched islands the occurrence bound vanishes, which is a property of the archive and not of the method.

C. THE SPATIAL NORMALISATION: EVIDENCE AND VALIDATION

This appendix collects the three measurements behind §5.2.3 and the prospective test of the correction.

C.0.1. *Prospective validation of the radius correction*

The radius-corrected normalisation was derived from the calibration data, so on its own it establishes nothing about data it has not seen. We therefore froze it — the probe radii, the bin edges and the standardisation are those of Appendix L, with no free parameter to tune — and applied it unmodified to 778 windows that played no part in its construction: the pre-registered hold-out and the prospectively searched extension of §D.0.2. The level for consistency with uniformity was fixed at $p \geq 0.05$ before the test was run, and nothing was retuned afterwards.

Two rank distributions can be tested and they answer different questions. The *pseudo-star* distribution ranks an inner-annulus probe against the other probes; it passes both before and after correction (uncorrected $D = 0.005$ at $p = 0.16$, corrected $D = 0.006$ at $p = 0.09$, over 49792 probes). That is expected and is not evidence for the correction: every probe is drawn from the same radial distribution, so probe-against-probe ranks are near-uniform however they are normalised. The discriminating test is the *stellar* distribution, because the star lies at the phase centre, inside the probe annulus, and that asymmetry is what the correction exists to repair.

On that test the correction fails. Out of sample the uncorrected stellar rank has median 0.409 against the expected 0.5, with $D = 0.105$ at $p = < 10^{-4}$; after correction the median is 0.407 with $D = 0.097$ at $p = < 10^{-4}$. The departure from uniformity is essentially undiminished, and the median moves further from the exchangeable value. The result holds separately in both samples (hold-out 0.405→0.392, extension 0.409→0.417).

The interpretation we draw is narrow and, we think, unavoidable. The radial gradient is real and is one mechanism behind the displacement, but removing it does not leave an exchangeable statistic, so the residual is not radial. Neither normalisation yields a calibrated tail probability, and the spatial screen is therefore a candidate-generation device under both. extbfWe consequently retain the frozen screen as primary — it at least has the virtue of having been fixed in advance — and rest every disposition on the physical evidence rather than on either rank.

C.0.2. *The external calibration sample*

A sample of 326 execution blocks toward 56 stars was reserved by a rule fixed and committed before the archival search finished and before any reserved block was searched, so that the statistic could be checked on data it had never seen. Appendix K gives the reservation rule and its commitment dates, and Appendix L.4.1 the

full result block by block. Three findings carry into the main line of the paper.

First, the pipeline behaves on unseen data: over 1322 windows it returns 7 stage-1 flagged windows, of which 4 are β Pictoris CO and 3 are unattributed, against 3.7 expected at the measured tail rate. Second, and adversely, the stellar rank is *not* uniform out of sample: its median is 0.405 where 0.5 is expected, with a block-clustered 95 per cent interval of 0.341–0.458 and $D = 0.109$ at $p0.0011$ over 315 windows. A displacement that survives a hold-out chosen before the data is a property of the statistic and not of this sample, which is why the spatial rank is used as a screen and never as a significance (§4.1). It reproduces a third time, at 0.409 over 463 prospectively searched windows (§D.0.2). Third, the false-alarm tail factor — the ratio of the measured rate at which a no-signal position ranks first to the rate exchangeability would give — is 1.2 on these blocks, with a bootstrap over the 77 reserved blocks giving 0.6–1.9. That factor and its width set the accuracy of every chance expectation in this paper.

C.0.3. *Where the displacement comes from: radius, not stars*

The displacement has a mechanism, and it is instrumental. Images are corrected for the primary-beam response, so the true noise rises with distance from the pointing centre, while the search divides every position by one scale estimated over the whole window. The control probes therefore sit at systematically different effective noise from the star, which is at the phase centre. Appendix L.4.2 sets out the evidence: the control level varies with radius across the annulus, pseudo-stars carrying no star at all are displaced as far as real ones, and rescoring the survey with the gradient removed changes which windows are flagged but not what the paper concludes.

Two numbers matter here. Standardising each control probe within its own radius bin — the probes come from one fixed seed, so their radii are known, and the correction has no free parameter — takes the stage-1 list from 13 windows to 10, losing 4 and gaining 1, all of them marginal and all already dispositioned on other grounds. And the survey median add-one rank does *not* move toward 0.5 under the correction: it goes from 0.445 to 0.432, slightly further away. The radial term is one demonstrated mechanism for the displacement, not the whole of it, so we do not adopt the corrected statistic as primary and we do not present it as a validated null.

D. EXTERNAL CHECKS AND THE POSITIVE CONTROL

D.0.1. *No independent pipeline check is available, and why*

Every validation reported here tests this pipeline against itself: the injection campaign, the radius correction, the region-maximum audit and the false-alarm calibration all share one codebase, so none of them can catch a design fault common to the search and the injection code. The obvious external check would be a comparison against the only prior ALMA-archival technosignature search, Mason et al. (2024), on shared data.

There is no shared data. Their 28 “stellar bycatch” targets all lie at or beyond 1.01 kpc, while every star here

is within 40 pc — the two distance ranges are disjoint by a factor of about 25 — so no target, and therefore no execution block selected by target, is common to both surveys. We searched 112 Band 3 windows, which is their band, but in different fields.

We record this as a real limitation rather than an absence of concern. The null result rests on internal validation alone, and the strongest available external check — an independent re-implementation of the search statistic, or a blind injection by a third party before delivery — has not been performed. Of those, a blind third-party injection would be the more informative and is the check we would recommend to anyone extending this work.

D.0.2. *A prospective epoch extension*

After the analysis above was frozen we found that the work list had been built against an earlier archive map, which had not exposed 177 public in-scope execution blocks. 115 of them have now been searched with the frozen pipeline, *after* the detection statistic and the molecular-line mask were fixed (snapshot 2026-09-22 08:18 UTC). They are reported here rather than merged into the science sample, and the reason is the point of them: the statistics above rest on a denominator and a candidate list fixed before the data were examined, and merging would destroy that property. The frozen sample is unchanged.

What the extension is. 463 windows toward 16 stars in 16 systems, 67 fine-channel and 396 coarse, 57.6 GHz of frequency of which 13.2 GHz is Class A. Every block is repeat coverage of a star and a tuning the survey had already searched, so it adds no star, no system and no frequency; what it adds is epochs. 32 further windows in 8 blocks were withheld under the survey’s own rule, being ϵ Eridani Band 6, where the star sits near 0.7 primary-beam FWHM and the Gaussian beam form used for the correction does not hold.

The epoch gain is real but small, and we say so. The number of systems with more than one searched epoch rises from 55 to 56 of 82, and the number with a single epoch falls from 27 to 26: 1 system acquires a second epoch. The extension is concentrated on systems that already had several, so it deepens the time coverage of a minority rather than repairing the confirmation gap of §6.3.

Three results carry. First, and most usefully, the departure from exchangeability reproduces on data the statistic had never seen: the median add-one rank is 0.409 over 463 windows where 0.5 is expected, with $D = 0.116$ at $p = < 0.01$. That is a third independent measurement of the effect §C.0.2 and §C.0.3 report, and it is why the spatial rank is used as a screen and never as a significance. Second, the positive control recovers: 3 windows reach stage 1 and all 3 of them are the β Pictoris disc in CO(1–0), at $T_\star = 23.17, 19.39$ and intermediate against ring maxima below 6.21, within 8.8–9.0 MHz of the transition, in three *different* previously unsearched execution blocks. Third, the extension adds no unattributed event: 13 threshold crossings (2.81 per cent of windows), 4 rank-first against 1.3 expected, and 0 unattributed stage-1 events against 0.21 expected. That expectation is computed on the 78 extension windows whose control maximum reaches the trigger, the conditioning §5.2 establishes as correct. The count of

TABLE 11

THE PROSPECTIVE EPOCH EXTENSION AGAINST THE FROZEN SCIENCE SAMPLE. EVERY FROZEN-COLUMN NUMBER IS COMPUTED FROM THE RELEASED CATALOGUE BY THE SAME DEFINITION APPLIED TO THE EXTENSION, AND THE GENERATOR CROSS-ASSERTS THEM AGAINST THE MACROS THE REST OF THE PAPER USES, SO THE TWO COLUMNS ARE LIKE FOR LIKE. THE EXTENSION IS *not* POOLED INTO THE SAMPLE: IT WAS SEARCHED AFTER THE STATISTIC AND THE MASK WERE FIXED. THE THREE RESULTS THAT MATTER ARE ON THE LAST THREE LINES: THE ATTRIBUTED EVENTS ARE ALL THE β PICTORIS DISC, THE UNATTRIBUTED COUNT IS ZERO, AND THE ADD-ONE RANK IS DISPLACED FROM 0.5 BY ABOUT AS MUCH AS IN THE SAMPLE ITSELF, ON WINDOWS THE STATISTIC HAD NEVER SEEN.

	frozen sample	extension
execution blocks	404	115
windows	1655	463
Class A	403	67
stars	90	16
systems	82	16
threshold crossings	75	13
rank-first windows	16	4
stage-1 events	13	3
attributed to CO	9	3
unattributed	4	0
median add-one rank	0.446	0.409

unattributed events in this paper therefore remains 4 (Table 11).

D.0.3. *An astrophysical positive control for spatial and spectral recovery: β Pictoris CO*

A search that finds nothing must show that it could have found something. Two different demonstrations are needed and the paper keeps them apart: the injection campaign of Appendix G establishes that the pipeline recovers a *drifting narrowband carrier*, which is the searched morphology, and β Pictoris establishes that it recovers, localises and re-finds a real astrophysical feature at a stellar position across epochs. β Pictoris CO is resolved, several km s^{-1} wide and effectively undrifting, so it is a positive control for *localisation and recurrence* and not for the carrier morphology. On that reading β Pictoris is the survey’s *astrophysical* positive control — it demonstrates recovery, localisation and recurrence of a real narrow feature at a stellar position, and does *not* demonstrate recovery of an artificial drifting carrier, which is what the injection campaign of Appendix G is for: a real, spatially localised, narrow spectral feature at a stellar position, recovered blind by the frozen pipeline, in two transitions, in 8 execution blocks, over 8.4 yr from 2013-10-06 to 2022-03-03, and at a drift consistent with none: the Band 3 crossings agree across blocks to 1.00 channel once each block’s own tuning is removed, which bounds any apparent drift far below the grid the window searches. HD 48370 is the complementary case, an extended foreground line that fills the annulus and so is rejected by the same spatial statistic that promotes β Pictoris; CP–72 2713 and HD 23484 (Table 25) are the third regime, marginal single cells that do not recur.

8 of the 13 flagged windows are towards β Pictoris, a debris disc with resolved CO emission (Matrà et al. 2017; Dent et al. 2014): 5 in Band 3 and 3 in Band 6, from 8 execution blocks (Table 9). The Band 6 window is a line tuning, 53.9 MHz wide at 15.26 kHz, so it holds 3533 channels like a wide window and searches 1944 trial drift rates against a survey median of 4, the largest trials

load in the survey at $\eta_{\text{drift}} = 422$: this rank is harder to earn than a typical one, not easier. Both windows sit at the same place in velocity. Carried through the frame chain, the Band 3 crossing lies -0.39 and the Band 6 crossing -0.29 km s^{-1} from the stellar systemic velocity, in the two lowest CO rotational transitions of a star with mapped circumstellar CO, and a third, unflagged window six years earlier shows the same component at -0.46 km s^{-1} . All three lie within 0.46 km s^{-1} of systemic. We attribute both crossings to this kinematically offset CO. The full frame audit, the flux ratio and the excitation argument are in Appendix L, with the per-window numbers in Table 18.

Attributing the pair to CO does not diminish it. Against the survey background the Band 3 peak $T_{\star} = 30.76$ is matched or exceeded by 0 of the 1655 per-window control maxima and the Band 6 peak $T_{\star} = 11.68$ by 16 of them. Those are ranks in a reference set this paper shows is not exchangeable with the stellar position, so we quote them as ranks and not as p -values; the attribution rests on the velocity agreement and the recurrence, neither of which needs them. The alternative reading needs a transmitter placed by coincidence at matched offsets from two CO transitions in one star. The outliers also concentrate where the astrophysical foreground is richest: β Pic, AU Mic and CP-72 2713 are all β Pictoris moving-group members with debris discs, and HD 48370 is a disc host behind a molecular cloud (§6.3).

The recurrence test, run on emission we already believe in. A non-recurrence elsewhere means nothing unless the same machinery recovers a repeat when one exists, so we applied it here. Every β Pictoris CO feature this survey flagged is recovered in an independent execution block: the Band 3 unit is exhaustive at 4 of 4 blocks with the line present in every one, the Band 6 recurrence unit likewise, and the stage-1 Band 6 window recurs in its unit's second block at $T_{\star} = 8.95$. Three things make this a recurrence and not a coincidence: the peaks agree to about one channel once each block's own tuning is removed, the barycentric velocities agree to 0.7 km s^{-1} of systemic across 8.4 yr, and the lines are $10.8\text{--}15.2 \text{ km s}^{-1}$ wide, an order of magnitude broader than the single channel a carrier occupies. Appendix L gives the block-by-block numbers, including the warning that the spatial screen loses power when the CO belt fills the control annulus.

E. NOTATION

Table 12 collects every power scale and every term this paper uses in a particular sense. It is placed here rather than in the main text so that the reader meets each quantity in context first; the table is for reference when one is needed again.

F. FREQUENCY CONVENTIONS, LINEWIDTH READING, AND SMEARING

Reference frames. The line mask's chain runs topocentric $\rightarrow$ barycentric (Earth ephemeris at the execution-block mid-time) $\rightarrow$ stellar rest frame (catalogued systemic radial velocity), with JPL/CDMS rest frequencies (Pickett et al. 1998; Müller et al. 2005); in the LSR frame it stops at the standard solar motion, not the systemic velocity (§5.2.6). The residual error is the catalogue radial-velocity uncertainty alone, a few km s^{-1} within

TABLE 12

NOTATION. EVERY POWER SCALE AND EVERY TERM THIS PAPER USES IN A PARTICULAR SENSE IS DEFINED HERE AND NOWHERE ELSE; LATER SECTIONS USE THEM WITHOUT RE-EXPLANATION. THE ABSTRACT AND CONCLUSIONS QUOTE P_{90}^{sel} ; THE CATALOGUE CARRIES ALL OF THE POWER QUANTITIES PER WINDOW.

Quantity	Meaning and use
$T_{\star}$	the search statistic at the stellar position (Eq. 3): a matched-filter amplitude divided by a noise estimate. Dimensionless, and <i>not</i> quoted in units of σ , because its null distribution is neither assumed nor shown to be Gaussian
T_{trig}	the trigger, $T_{\text{trig}} = 5$: a fixed level on $T_{\star}$, not a tail probability. Throughout this paper σ is reserved for genuine noise scales (injected amplitudes, image rms, fit uncertainties) and never used for the trigger (§4.1)
<i>The power hierarchy, in order. Only the last is a sensitivity.</i>	
P_{trig}	the power at which the search statistic reaches T_{trig} in one native channel: an internal trigger, not a detection significance and not a completeness level (§4)
C_{resp}	spectral-response correction, $\times 2.29$ median, window-specific (Appendix F)
C_{smear}	intra-integration drift smearing, $1/\eta_{\text{smear}}$ (Appendix F)
P_{eff}	$P_{\text{trig}} C_{\text{resp}} C_{\text{smear}}$: the power a transmitter must radiate to reach the trigger (Fig. 5). Thus every EIRP quoted here is the <i>total</i> radiated power of an unresolved monochromatic carrier, not the power falling in the peak ALMA channel
P_{50}, P_{90}	the injected amplitudes the <i>trigger</i> recovers half and nine-tenths of the time, $1.2 P_{\text{eff}}$ at 90 per cent: completeness of one stage, not of the selection (§4.4, Appendix G)
P_{90}^{sel}	the amplitude the complete automated selection recovers nine-tenths of the time, trigger <i>and</i> control-ring gate together (§5.3). The headline sensitivity: this is the quantity every limit in the abstract, discussion and conclusions is expressed in
Class A/B	fine-channel / coarse-channel windows; different experiments, never pooled in a completeness statement (§4, Table 1)
dwel	recovery against the fraction of the track a signal is present (§4.4)
candidate	a stage-1 flagged window that also recurs or is independently confirmed; this survey reports none (§4.1)
pseudo-star	an inner-annulus control probe ranked against the others; it contains no star, so a first rank is a false alarm by construction (§C.0.3)
debit $m(0)$	the radial correction subtracted from the stellar statistic before ranking (§C.0.3)
stratum	a subset of windows sharing an instrumental property (band, array, channelisation) analysed separately (Appendix G)
ring max	the largest control statistic in a window's annulus (§4.1)
R_{σ}	the star's robust residual scale divided by the controls' median; unity means the star carries no noise peculiar to its position (§L.1.2)
hold-out	blocks withheld from the analysis by a rule fixed in advance, used only to check a threshold and never to set one (§C.0.2)
drift	the rate at which a carrier's observed frequency changes, $\dot{\nu} = \nu a/c$ for line-of-sight acceleration a . Quoted in three units: Hz s^{-1} or kHz s^{-1} for one

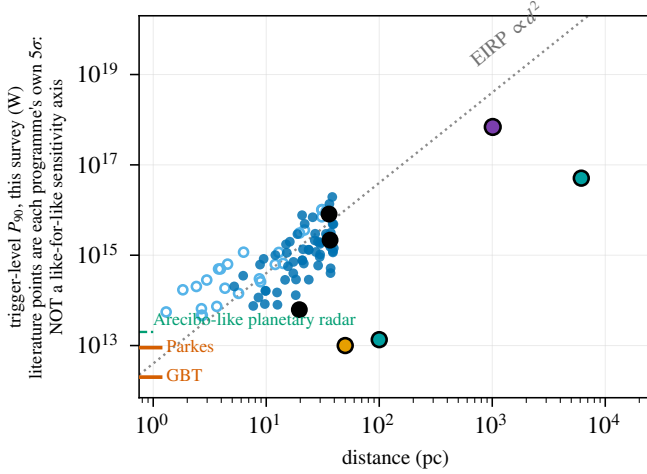


FIG. 11.— EIRP against distance for this survey and for published centimetre-wave searches, retained here for completeness. Blue: this survey, plotted as the trigger-level P_{90} for the deepest window per star; orange: Enriquez et al. (2017); aqua: Margot et al. (2023); violet: Mason et al. (2024); left-edge ticks: Price et al. (2020). Grey dotted: $\text{EIRP} \propto d^2$ at constant received flux. Green dashed: the Arecibo 2.38-GHz benchmark, a power scale only. The vertical coordinates are not comparable completeness limits: each programme’s value is its own threshold statistic over its own channel width, and those widths differ by about seven orders of magnitude (Fig. 1).

40 pc; second-order relativistic terms at the 50 km s^{-1} scale are $\lesssim 5 \text{ kHz}$ at 350 GHz, below the finest native channel searched (15.3 kHz). The β Pictoris and AU Mic attributions use the same chain, so their offsets are comparable to the exclusion tolerance.

Channel widths, and the instrumental response. These come from the measurement sets’ own spectral window tables (CHAN_WIDTH). A missing “EFFECTIVE_BANDWIDTH” column is not evidence of an absence of online spectral averaging, because MS v2 defines no such column; we therefore make no claim that the quoted channelisations are noise bandwidths.

ALMA applies online Hanning smoothing by default, so the spectral response is ~ 2 channels wide and neighbouring channels are correlated. For the 0.25/0.5/0.25 kernel the coefficients are $\rho_1 = 2/3 = 0.667$ and $\rho_2 = 0.167$, with $\sigma_{\text{smoothed}}/\sigma_{\text{raw}} = 0.612$: the correlation is not confined to adjacent channels, the lag-two term being a sixth. A sub-channel tone keeps 0.50 of its power in the peak channel at the channel centre and 0.38 on a boundary, median 0.44 over uniform placement. Every $\text{EIRP}_{\text{trig}}$ here is therefore optimistic by $\times 2.00$ to $\times 2.67$, median $\times 2.29$; the released catalogue carries the nominal threshold and both corrected values per window, and the sensitivity summary quotes the worst case. The injections deposit an unsmoothed delta-in-channel tone, so the recovery curves of §4.4 and Appendix G carry the same bias and the end-to-end check (Eq. 6) cannot see it. The retained per-integration residuals of the second-stage local null (§5) have a measured lag-one spectral autocorrelation of 0.65, against the Hanning prediction 0.667, so that correlation is instrumental and the residual spectra are smoothed and un-averaged, which is the independent confirmation of the smoothing state. The correlation model is truncated at lag two; the baseline step is a per-integration median in blocks of up to 65 channels, which could imprint correlation on

longer scales, and the released products retain no residual spectra with which to measure it. If such a term is present the independent-trials count of Appendix K is an over-estimate and the achievable per-window resolution is coarser still, which strengthens rather than weakens the conclusion that rank alone cannot authenticate a candidate. The archive’s reported effective spectral resolution per window, against each measurement set’s own channel separation, is exactly 2.00 in 1509 of the 1655 windows, in both correlator modes and including 1220 of the 1252 coarse time-division-mode (TDM) ones; the other 146 report a smaller ratio, the signature of online channel averaging, where the peak-channel loss is smaller, and the 0 Band 9 and 10 windows, absent from the metadata snapshot, take ALMA’s default. The blanket factor is right for the great majority of both classes. We still do not rescale, so each threshold here is nominal and up to $\times 2.67$ optimistic.

Linewidth convention. The statistic is the single-channel flux density after drift correction, so the quoted $\text{EIRP}_{\text{trig}}$ applies to any transmitter narrower than the native channel and is flat in the assumed linewidth. Two sub-channel qualifications attach. Flatness is exact only for a centred tone, a boundary-straddling one splitting its power between two channels at worst by a factor of two. And the channel-to-channel transition follows the correlator’s passband shape, which the frozen products do not retain. Thresholds therefore carry a sub-channel-position systematic of tens of per cent, and EIRP values are quoted to two significant figures.

Transmitters wider than one channel. The baseline filter (§4.1) is flat for features narrower than roughly half its effective width W and falls to zero above W , so the survey’s sensitivity is to channel-confined morphologies and W is the linewidth ceiling: 65–74 channels over the fine class and 30–59 over the coarse one, that is $\approx 32 \text{ MHz}$ at 488 kHz and $\approx 0.9 \text{ GHz}$ at 15.62 MHz. Within that ceiling, a top-hat transmitter spanning N_{ch} native channels has per-channel flux the total divided by N_{ch} at unchanged noise, so to first order

$$\text{EIRP}_{\text{trig}}(N_{\text{ch}}) \simeq N_{\text{ch}} \times \text{EIRP}_{\text{trig}}(1), \quad N_{\text{ch}} \lesssim W/2, \quad (\text{F1})$$

the rescaling the main text applies to signals broader than one channel.

Worked example: from flux to EIRP. Take UV Ceti (G 272-61B) Band 3, the deepest of its 4 windows in that band (execution block A002_X877e41_X14c1, lower edge 91.565 GHz). Its per-channel rms is 0.241 mJy, so $S_{\text{min}} = 5 \times 0.241 = 1.21 \text{ mJy}$ in one 15.625-MHz native channel, or $1.21 \times 10^{-29} \text{ W m}^{-2} \text{ Hz}^{-1}$ at 2.67 pc, and $\text{EIRP}_{\text{trig}} = 4\pi d^2 S_{\text{min}} \Delta\nu = 1.6 \times 10^{13} \text{ W}$.

Intra-integration smearing. The frequency excursion at the maximum searched drift rate across one integration, $\nu_{\text{max}} t_{\text{int}}$, is at most 1.10 channels across the 1655 windows (median 0.001, 90th percentile 0.06). A linear drift smeared by Δ channels within an integration keeps a fraction $\eta_{\text{smear}} = \text{sinc}(\pi\Delta/2)$ of the peak-channel amplitude, so the effective threshold is $\text{EIRP}_{\text{trig}}/\eta_{\text{smear}}$. The 9 windows with $\eta_{\text{smear}} < 0.99$, as star, band, (η_{smear} , threshold multiplier), are HIP 10679 Band 6 (0.574, $\times 1.74$), β Pic Band 6 (0.574, $\times 1.74$), HD 172555 Band 6 (0.574, $\times 1.74$), HIP 10679 Band 6 (0.606, $\times 1.65$), HD 172555 Band 6 (0.606, $\times 1.65$), η Crv Band 8 (0.865,

$\times 1.16$), η Crv Band 7 (0.932, $\times 1.07$), HD 48370 Band 6 (0.979, $\times 1.02$) and CP-72 2713 Band 6 (0.979, $\times 1.02$). The remaining 446 agree with their nominal thresholds to better than 1 per cent. Compounding the worst smearing penalty with the worst Hanning factor gives $\times 4.6$ on the nominal threshold for those five windows, and no released column carries that product. The $\text{sinc}(\pi\Delta/2)$ form is the conservative reading: a pure occupancy argument would retain $\min(1, 1/\Delta)$, giving 1.0 rather than 0.64 at $\Delta = 1$.

G. INJECTION-RECOVERY SENSITIVITY VALIDATION

To test whether the trigger behaves as idealised Gaussian noise predicts, we inject synthetic tones of known amplitude into real visibilities and measure what the unmodified pipeline returns. The injection uses the extraction code's own mechanism, as far upstream of the statistic as the retained products allow. Six configurations span three bands and both classes: three coarse Band 7 spectral windows of AT Mic, one coarse window each in Bands 3 and 6 of AU Mic, and AU Mic's 488-kHz flag-band window with its 49-trial drift grid. Each receives a complex tone at the star position at amplitudes $4\text{--}10\times$ the statistic's own noise. The grid covers drift fractions $f_{\text{drift}} = 0$ to 1.0 of the ceiling, on- and half-grid rates, and channel-centred and boundary placements, 200 trials per configuration and 1200 in all. The recovery criterion was frozen in advance: $T \geq 5$, peak channel within 3 of the injection, peak drift within one grid step.

The coarse-window variant recovered 0 of 500 injections from $T_\star = 4$ to 10 under that criterion, and the criterion rather than the pipeline is what produced that number. Its drift-matching clause cannot be satisfied on coarse windows, where every in-grid drift is sub-channel over the track, so the reported peak drift is arbitrary. Scored on detection alone the same windows recover essentially everything, an end-to-end injection through the released pipeline returning $T_\star$ of 20.5, 65.3 and 217.7 at 1, 3 and 10 times the per-integration noise against an ideal stacking gain of $\sqrt{639} = 25$. Class B limits therefore bound carriers of any dwell fraction, with the completeness of §4.4, and what those windows lack is drift discrimination. The fine window fails the same criterion the same way.

The second result is the completeness curve. On the fine window, drifting carriers ($f_{\text{drift}} \geq 0.5$, pooled over grid phase and sub-channel placement) are recovered 17, 42, 58, 75 and 83 per cent at $T_\star = 4, 5, 6, 8$ and 10, saturating by twice threshold power (Fig. 12). The machinery alone, with the tone injected after the baseline step, is amplitude-faithful to within $-9/+14$ per cent for channel-centred tones. Boundary-straddling placement costs 0.6–0.8 in the statistic and recovers 0 per cent at the trigger against 83 per cent for centred ones, crossing 50 per cent near $1.4\times$ threshold amplitude. Survey completeness uses the placement-pooled curve, conservative for the boundary-straddling case: $\text{EIRP}_{50} = 1.10 \text{EIRP}_{\text{trig}}$ and $\text{EIRP}_{90} > 2.0 \text{EIRP}_{\text{trig}}$, a bound only.

This is a demonstration that the pipeline recovers a carrier in a representative configuration. It is not a survey completeness function. The curve is measured on one window; the stratified campaign below measures how far it travels to the rest of the 403 fine-channel windows. Of

the 403 Class A windows, 197 (49 per cent) differ from that configuration on at most one axis, the mismatch being dominated by band (200 windows) and drift-trial count (167) rather than channel width (30), integration time (87) or beam offset (42). Those are the axes the campaign below spans. Not spanned at all are other target environments and edge-phased drift grids.

The stratified campaign. Those axes have since been measured. A drifting carrier was injected into the retained per-integration spectra of 28 further windows, one per star, spanning Bands 3–10, 15.3–977 kHz channels, 45–485 integrations, 7–1186 drift trials and both the 12 m array and the ACA, pushed through the pipeline's own baseline removal and de-drift stack on the search's drift grid: nine amplitudes, 4032 trials. Recovery is a threshold crossing at the injected frequency, not the global maximum of the grid: requiring an injection to beat a window's brightest cell measures its worst artefact, the conflation behind the withdrawn result above.

Pooled recovery is 34, 61, 89 and 99 per cent at $T_\star = 4, 5, 6$ and 8, $P_{50} = 4.6\sigma$, $P_{90} = 6.1\sigma$ (Fig. 12). The spread across configurations is what the transfer bracket needed: P_{50} runs $0.64\text{--}1.13\times$ its median and P_{90} $0.48\text{--}1.35\times$ its own, so the assumed $0.5\text{--}2\times$ bracket is conservative. Band and array matter least, P_{50} varying only $4.4\text{--}4.8\sigma$ across bands.

Drift rate, resolved. Every one of the 4032 trials carries a non-zero drift, drawn over $9\text{--}10344 \text{ Hz s}^{-1}$, so the pooled curve above is already a drifting-carrier curve and not a stationary-tone one. A pooled curve could still hide a deficit at the edge of the grid, where a carrier crosses the most channels and the de-drift stack has the least signal in any one of them, so Table 23 splits the campaign by drift rate as a fraction of each window's own ceiling. P_{90} is 6.1σ in the inner third, 6.5σ in the middle and 5.9σ in the outer third: a spread of 10 per cent, with no trend, and the outer third is not the worst. At a fixed 6σ injection the three recover 89, 84 and 93 per cent. The search also recovers the rate, not merely the frequency: 92 per cent of detections above 6σ return the injected drift trial itself, with a median error of 0.00 grid steps and 0.5 at the ninetieth percentile. Carrier sub-channel phase is stratified separately, in the response campaign below, over 5 positions and 400 trials; it is the one axis that does matter, and it is the reason the threshold carries the $\times 2.69\text{--}\times 2.02$ response correction rather than a single factor.

How far the grid reaches. 28 configurations are a sparse grid for 403 windows, so we state plainly which windows the bracket interpolates and which it extrapolates. The grid spans Bands 3, 6, 7, 8 and 10, channel widths 15.3–977 kHz, 45–485 integrations and both arrays. Every Class A window lies in a band the grid samples. Of the 403, the configuration needed to make the comparison — band, channel width and integration count together — is recorded for 132; the peak-frequency freeze that carries the integration count does not cover the 271 blocks searched since, and those are counted here, never assumed to be inside the grid. Of the 132 that can be judged, 113 lie inside the grid on all three axes at once, 86 per cent of them and 28 per cent of all 403. The 19 that can be judged and are not inside are extrapolations, 18 of them in integration count alone (shorter or longer tracks than any calibrated configuration) and 1 in

channel width. The transfer bracket should be read as measured for the 113 windows and assumed for the rest. The five deepest Class A thresholds in the survey are all interpolated, so the headline sensitivity does not rest on an extrapolation.

The spectral response, measured. The $\times 2.29$ correction of §4 was analytic because the injected tones were unsmoothed. Injecting a tone at fractional position ϕ , split between neighbouring channels as the delivered channelisation requires and convolved with the 0.25/0.5/0.25 kernel, and comparing its peak with a channel-centred unsmoothed tone of equal total flux, returns $\times 2.02$ at $\phi = 0$ and $\times 2.69$ at the channel edge, median $\times 2.31$ over 400 trials, every point within 1.1 per cent of the analytic value. Both campaigns ship trial-level records.

G.1. Dwell campaign: end-to-end check, scope and representativeness

An end-to-end check confirms this through the unmodified pipeline. Injecting a zero-drift, always-on tone into the calibrated visibilities of a coarse Band 7 window at 1, 3 and 10 times the per-integration rms returns $T_\star = 20.5$, 65.3 and 217.7, each within one channel of the injection. The ideal stacking gain over its 639 integrations is $\sqrt{639} = 25$, and the measured 20.5 is that value less weighting and flagging losses.

Two distinctions apply. The dwell campaign injects into retained per-integration spectra, after the baseline step, while the end-to-end check injects into visibilities before it; and one cell, the through-pipeline recovery of near-static carriers on Class A windows, is measured only in part.

How representative are the injected windows of the survey they stand for? The end-to-end demonstration is made on one configuration, and 197 of the 403 Class A windows — 49 per cent, so about half and not most — differ from it on at most one axis. The stratified campaign of Appendix G extends the measurement over 28 further configurations and reports the spread directly, and the appendix also states which windows the resulting bracket interpolates and which it extrapolates. The reader should take the quoted completeness as measured for roughly half the Class A sample and transferred, with a measured bracket, to the rest.

One coarse-window injection result belongs here in the main text, because at face value it looks alarming. Under the recovery criterion frozen in advance, which requires the recovered peak drift to match the injected drift to within one grid step, the coarse variant recovered *none* of 500 injections between $T_\star = 4$ and 10. That criterion cannot be satisfied on a coarse window by construction: every drift in the grid is sub-channel over the whole track, so the reported peak drift is arbitrary and the test measured its own inapplicability. Scored on detection alone the same windows recover essentially everything, reaching $T_\star = 20.5$, 65.3 and 217.7 at one, three and ten times the per-integration noise (Appendix G). This is what the class is for throughout the paper: Class B detects excess power. It cannot tell you whether the excess drifted.

The injected carriers do not drift, so what is measured across 12 configurations is the persistent and partial-dwell non-drifting class. On Class B windows that is the searched class; on Class A windows the searched class is the drifting carrier, whose curve rests on the single cali-

TABLE 13

SELECTION FUNCTION OF THE SEARCHED SAMPLE AGAINST A VOLUME-LIMITED REFERENCE CENSUS: GAIA DR3 SOURCES WITH $\varpi > 0$, $\sigma_\varpi/\varpi \leq 0.1$ AND $1/\varpi \leq 40$ pc (17566 OBJECTS, OF WHICH 8594 CARRY A **TEFF_GSPHOTO** ESTIMATE, 49 PER CENT), AGAINST THE 90 STARS WITH AT LEAST ONE SEARCHED WINDOW. TEMPERATURES ARE AVAILABLE FOR 53 OF THE 90 (ALL 90 MATCHED TO THE 168-ENTRY ALMA-COVERED CENSUS, 55 BY EXACT NAME; 53 OF THOSE MATCHES CARRY A TEMPERATURE), SO THE TEMPERATURE PANEL IS NORMALISED OVER CLASSIFIED OBJECTS ONLY. “RATIO” IS THE SAMPLE COLUMN FRACTION OVER THE CENSUS COLUMN FRACTION; > 1 IS OVER-REPRESENTATION.

Property	Census	Searched	Ratio
Distance (all 90 searched stars; 17566 census objects)			
$d \leq 5$ pc	48 (0.3%)	12 (13.3%)	48.8
5–10 pc	240 (1.4%)	13 (14.4%)	10.6
10–20 pc	2069 (11.8%)	18 (20.0%)	1.7
20–30 pc	5391 (30.7%)	19 (21.1%)	0.7
30–40 pc	9818 (55.9%)	28 (31.1%)	0.6
T_{eff} class ^a (53 searched; 8594 census)			
A (≥ 7500 K)	24 (0.3%)	1 (1.9%)	6.8
F (6000–7500 K)	402 (4.7%)	9 (17.0%)	3.6
G (5300–6000 K)	860 (10.0%)	16 (30.2%)	3.0
K (3900–5300 K)	1400 (16.3%)	6 (11.3%)	0.7
M (< 3900 K)	5908 (68.7%)	21 (39.6%)	0.6
Other axes			
Known planet hosts ^b	20 (11.9%)	19 (21.1%)	1.8
Searched stars / systems	–	90 / 82	–

^aBins are temperature bins, not MK types: A ≥ 7500 K, F 6000–7500, G 5300–6000, K 3900–5300, M < 3900 K. Gaia **teff_gspphoto** is unavailable for about half the reference census, preferentially for the coolest and faintest objects, so the census M fraction is a lower limit and the earlier-type ratios are correspondingly upper limits. White dwarfs are not separable in this parameterisation and fall wherever their (unreliable) photometric temperature places them.

^bThe Gaia reference census carries no planet information; the census column here is instead the 168-entry ALMA-covered census, of which 20 entries are known planet hosts. The 90 searched stars include 19 hosts of 42 known planets.

bration configuration of Appendix G. The quantity a is a derived combination, justified by the form of the statistic and by the collapse of 12 configurations onto one curve rather than measured independently at every (A, f_{dwell}) pair.

H. PER-TARGET RESULTS

The per-window results ship as the comma-separated catalogue of the Data Availability statement. Its EIRP_{trig} column uses the definition of §4, on the same threshold and primary-beam correction as the continuum column, so a row’s two limits are comparable; its per-channel flux-density threshold uses the carrier lane’s per-channel noise rather than the tabulated rms_mJy, the two differing by $5 \times G(r)$. Bands 6 and 7 supply 91 per cent of the windows.

I. THE VELOCITY-SPACE LINE MASK

A one-channel line check is too tight for circumstellar gas: a native channel spans a fraction of a km s^{-1} , while real emission is shifted by many channels (§5.2). The mask tolerance is set by the kinematic budget of circumstellar gas: Keplerian motion where these molecules survive ($\lesssim 13 \text{ km s}^{-1}$ beyond 10 au), linewidths $\lesssim 1 \text{ km s}^{-1}$, and an uncorrected barycentric term bounded by $\pm 30 \text{ km s}^{-1}$. The result is empirically conservative: the largest attributed circumstellar component (β Pic, within 0.46 km s^{-1} of systemic)

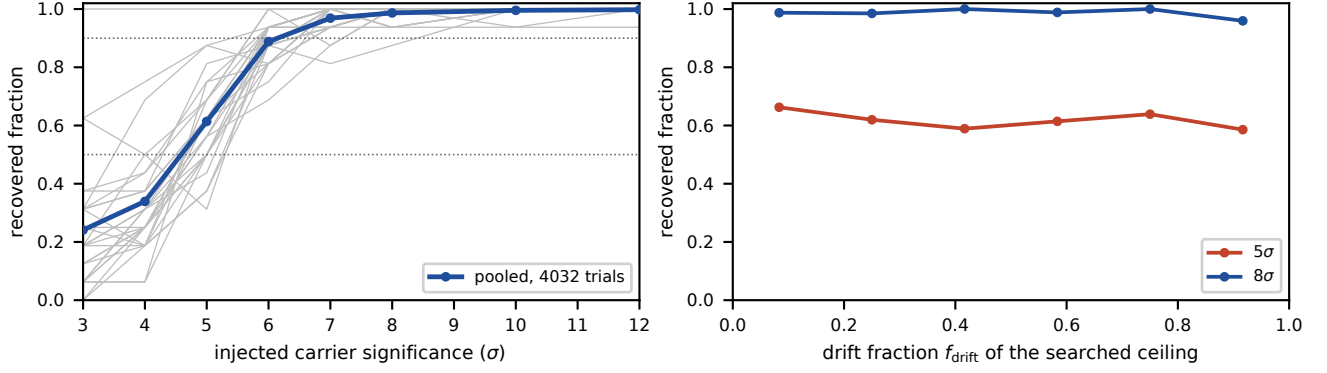


FIG. 12.— Stratified injection recovery. Grey, each of the 28 configurations separately; blue, the pooled curve over 4032 trials. Dotted lines mark the 50 and 90 per cent levels. The spread between grey curves is the completeness transfer, measured rather than assumed. Right, recovery against drift fraction of the searched ceiling at $T_\star = 5$ and 8, the axis the single-configuration measurement used to carry alone.

TABLE 14

PER-BAND SURVEY CHARACTERISATION OF THE 1655 SEARCHED WINDOWS. “SYS.” COUNTS UNIQUE STELLAR SYSTEMS SEARCHED IN THE BAND, SO A SYSTEM OBSERVED IN TWO BANDS IS COUNTED IN BOTH ROWS (82 SYSTEMS, 90 STARS IN TOTAL); “EBs” THE CONTRIBUTING EXECUTION BLOCKS, ATTRIBUTED BY THE PROVENANCE OF THE DE-DUPLICATED WINDOWS; $\Delta\nu_{\text{ch}}$ THE NATIVE CHANNEL WIDTH IN MHz (MEDIAN, RANGE IN BRACKETS); “UNION” THE UNIQUE SKY FREQUENCY SEARCHED IN THE BAND AFTER MERGING OVERLAPPING WINDOWS; $\text{EIRP}_{5\sigma}$ THE MEDIAN NOMINAL THRESHOLD ACROSS THE BAND’S WINDOWS; $\dot{\nu}_{\text{ceil}}$ THE RANGE OF DRIFT-RATE CEILINGS IN Hz s^{-1} PER GHz OF CARRIER FREQUENCY. THE “ALL” ROW IS COMPUTED OVER THE WHOLE SAMPLE, NOT SUMMED DOWN THE COLUMNS: EBs AND SYSTEMS RECUR ACROSS BANDS AND THE FREQUENCY UNIONS ARE DISJOINT BY CONSTRUCTION.

Band	Sys.	EBs	Win.	$\Delta\nu_{\text{ch}}$ (MHz)	Union (GHz)	Med. $\text{EIRP}_{5\sigma}$ (W)	$\dot{\nu}_{\text{ceil}}$
3	8	27	112	15.63 (0.244–15.63)	20.6	1.3×10^{14}	12.0–13.3
4	1	1	4	15.63	6.9	2.6×10^{13}	12.0
5	1	1	4	15.63	6.9	4.4×10^{13}	12.0
6	55	265	1072	15.63 (0.015–31.25)	29.4	5.2×10^{14}	12.0–13.3
7	35	102	427	15.63 (0.061–15.63)	26.7	3.6×10^{15}	12.0–13.3
8	3	6	24	15.63 (0.061–15.63)	6.7	4.2×10^{16}	12.0
9	1	1	4	15.63	6.9	4.3×10^{15}	12.0
10	1	1	8	0.977	13.8	2.7×10^{16}	12.0
All	82	404	1655	15.63 (0.015–31.25)	118.1	7.1×10^{14}	12.0–13.3

TABLE 15

BAND-BY-BAND COST OF THE MOLECULAR-LINE EXCLUSION. UNION: UNIQUE SKY FREQUENCY SEARCHED IN THE BAND. MASKED: THE PART OF IT INSIDE A $\pm 50 \text{ km s}^{-1}$ STELLAR-FRAME TUBE AROUND A CATALOGUED TRANSITION OF ONE OF THE MASKED SPECIES. LEFT: WHAT A TECHNOSIGNATURE SEARCH CAN STILL OCCUPY. THE MASK IS EXCLUDED SEARCH SPACE AND NOT A VETO (§5.2.6); THE LOSS IS CONCENTRATED IN BAND 6, WHERE $\text{CO } 2 \rightarrow 1$ SITS.

Band	Union (GHz)	Masked (GHz)	Left (GHz)	Lost (%)
3	20.64	0.272	20.37	1.3
4	6.94	0.049	6.89	0.7
5	6.94	0.065	6.87	0.9
6	29.45	1.374	28.07	4.7
7	26.74	1.589	25.15	5.9
8	6.68	0.164	6.52	2.5
9	6.94	0.454	6.48	6.5
10	13.80	0.844	12.96	6.1
All	118.13	4.811	113.31	4.1

sits at 0.9 per cent of it, and *zero* crossing frequencies lie $13\text{--}50 \text{ km s}^{-1}$ from their nearest *masked* transition (the wider catalogues are another matter: §5.2), so a $\pm 13 \text{ km s}^{-1}$ Keplerian-only mask would have suppressed and released the same windows: the wider tolerance costs no yield and buys robustness against radial-velocity catalogue error. The alternative selection quoted in §5.2.6 is every Splatalogue transition over the 46 islands with $E_u \leq 150 \text{ K}$ and $\log A_{ij} \geq -5$, 5,114 of 235 carriers; HCN, CS, SiO, CN and H_2CO are in the adopted list already, and no coverage figure and no disposition changes.

The repaired mask. The four defects and their re-

pair are in §5.2.6. The rounding defect is why the released $\text{CO}(1 \rightarrow 0)$ offset column, referred to the frozen mask’s rounded 115.271000 GHz entry, reads 0.202 MHz (0.53 km s^{-1}) smaller than the laboratory-referred value of Table 18. A cheaper rule exists: the archive records the rest frequency each execution block was tuned to, so masking each window’s own tuned line, inside an LSR-frame tube of a few $\times 10 \text{ km s}^{-1}$, catches the β Pictoris, HD 48370 and [C I] windows automatically.

The crossing ledger, once: of the 75 windows with an on-star crossing, 13 also beat their control ring and three of those are accounted for by the mask. Two further crossing windows fall inside the mask without beating their ring (the unflagged β Pic Class A window at -0.46 km s^{-1} ; HD 285968 at $+8.7 \text{ km s}^{-1}$, already rejected by its ring, maximum 10.9 against a star peak of 6.5), and the remainder lie far outside any circumstellar tolerance (AU Mic $+379$, LHS 1140 $+115 \text{ km s}^{-1}$). The other 16 are ordinary noise crossings: in each, the window’s own control ring reaches or exceeds the stellar value (ring-maximum median 5.84 against a crossing median 5.18), the expected Gaussian-tailed population of moderate $T_\star = 5\text{--}5.5$ crossings with comparably strong peaks among 512 controls. But the transitions that make astrophysical false positives cheap are also natural reference frequencies for a deliberate communicator (Cocconi & Morrison 1959), so the released products keep a known-line-coincident list: every on-source threshold crossing inside the masked intervals, with its offset from

each nearby transition.

J. THE ANCILLARY CONTINUUM LANE

The continuum lane is an exploratory by-product: 57 measurements, 17 detections and 40 limits. A millimetre excess is at best an indirect and model-dependent observable of waste-heat technology. An “upper limit” here is $5\times$ the primary-beam-corrected map rms at the star’s position, and the deepest non-detection is towards TRAPPIST-1 in Band 3. Under the combined error model sixteen of the seventeen detections stay above the trigger, the exception being η Corvi Band 7 at $T_\star = 4.7$.

Every detection is astrophysically unsurprising: the unresolved UV Ceti binary; six systems of disc or chromospheric emission (β Pic, AU Mic, AT Mic and τ Cet among them); six stars at photospheric flux; and two late-M dwarfs of documented magnetic activity, LP 349–25 and the auroral emitter LSR J1835+3259 (Hallinan et al. 2015). Closure phase discriminates structure for none of them at these flux densities.

The anomaly screen. Each star’s flux is tested by a robust median/MAD statistic against its own photospheric expectation and against same- T_{eff} stars in coarse bins of at least five. The error model, fixed before any flag, adds the map noise, a calibration term of 5 per cent in Bands 3–6 and 10 per cent in Bands 7–8, and a model term propagated from the Gaia temperature and radius errors. It has *no intrinsic-variability term*. That is adequate for a photosphere and wrong for an active M dwarf, whose millimetre flux can vary by factors of several, so the screen is uninformative for exactly the stars a radio-stellar reader cares about. It flags for inspection and tests no per-star hypothesis. One outlier results, χ^1 Ori at Band 3, consistent with a spectroscopic G0V binary’s unresolved companion or a chromosphere.

K. FALSE-ALARM ACCOUNTING UNDER THE SYMMETRIC CRITERION

The per-window rate is $1/513$ and the survey expectation $1655/513 = 3.23$ flags. The floor falls only if the control ensemble grows, which the primary beam bounds. Windows are not independent trials either, grouping into 404 execution blocks that share weather, calibration and correlator setup, and that clustering is measured here rather than assumed. Drawing one control index per execution block and applying it to all of that block’s windows gives the clustered distribution in closed form: mean 3.23 rank-first windows, and under the trigger 0.92 stage-1 flagged windows with $P(\geq 1) = 0.60$ for the single unattributed one. Clustering changes little because the *position* of the control maximum is not shared across a block’s windows.

The trials budget, in one set of units, is Table 16. A window holds $177\text{--}6869740$ channel $\times$ drift cells (median 468), the survey 1.32×10^8 in total, so a one-sided Gaussian tail at $T = 5$ predicts 37.8 chance cells across the survey, an expectation in cells and not in windows. Against that, 229 on-star cells reach the trigger in 75 windows, 174 of them in the stage-1 flagged windows; the remaining 55 stand against 37.8 expected. Correlation cannot move the expected *number* above the trigger, which is Np whatever the correlation, only its clumping, so the factor decomposes into cell multiplicity within a crossing

TABLE 16
THE ON-STAR CELL EXCESS, DECOMPOSED. TOP: WHAT IS OBSERVED AGAINST AN IID-GAUSSIAN BUDGET. MIDDLE: THE TWO MULTIPLICATIVE PARTS OF THAT FACTOR, THE FIRST PREDICTED BY THE HANNING CHANNEL CORRELATION AND THE SECOND A WINDOW COUNT CONSISTENT WITH CHANCE. BOTTOM: THE SAME CORRELATION CORRECTION APPLIED TO THE CONTROL MAXIMA, WHICH ARE MEASURED IN THE SAME UNITS AND ARE WHAT CALIBRATES THE BUDGET EMPIRICALLY.

Quantity	Value	Factor
On-star cells $\geq T_{\text{trig}}$	229	
in the stage-1 windows	174	
remaining, unattributed	55	
iid-Gaussian expectation (cells)	37.8	1.5
Cells per crossing window (obs./exp.)	3.4/1.1	3.0
predicted by Hanning correlation	2.67	
Crossing windows (obs./exp.)	62/33.2	1.9
one-sided Poisson p	<0.01	
Frequency-axis over-count $1 + 2\rho_1 + 2\rho_2$	2.67	
Drift-grid oversampling	2	
Combined cell over-count		5.3
Control max., fine: pred. $\rightarrow$ corrected	5.74 $\rightarrow$ 5.45	
observed	5.79	+6 %
Control max., coarse: pred. $\rightarrow$ corrected	4.47 $\rightarrow$ 4.10	
observed	4.44	+8 %

window and the count of crossing windows.

The control ensembles calibrate this without a Gaussian assumption: the same arithmetic predicts 477 of the 1655 windows carrying a control above T_{trig} and 471 do, agreeing to 1 per cent, with per-window control maxima tracking the iid-Gaussian prediction at a median ratio 0.996 (5th–95th percentile 0.90–1.12). The residual bookkeeping is that n_{cells} counts grid cells and not independent trials: Hanning smoothing correlates neighbouring channels at $\rho_1 = 2/3$, $\rho_2 = 1/6$, so $1 + 2\rho_1 + 2\rho_2 = 2.67$ grid channels carry one channel’s information, and the drift grid is oversampled two to one, giving a combined over-count $2.67 \times 2 = 5.3$. Dividing n_{cells} by it lowers the predicted maxima from 5.74 to 5.45 in the fine class and from 4.47 to 4.10 in the coarse one, against 5.79 and 4.44 observed: a residual +6 and +8 per cent in the two classes independently. Non-Gaussianity of that size is why the operative statement is the empirical rank, not the Gaussian budget.

The same correction disposes of most of the on-star cell excess. Of the factor 1.5 above, the part contributed within a crossing window is 3.4 cells observed against 1.1 expected under independence, a factor 3.0, which is what an over-count of 2.67 on the frequency axis predicts: one real threshold excursion is counted about three times. What that leaves is the window-level factor, 62 windows carrying at least one unattributed on-star crossing against 33.2 expected, a factor 1.9 whose one-sided Poisson probability is <0.01. Once the four dispositioned windows are set aside, the residual excess is consistent with chance at that level. Table 16 collects the decomposition, so that the accounting can be audited at a glance rather than followed through four paragraphs.

It is not zero, runs one way, and carries the same sign at the controls, so we report it as an open systematic. The crossing list is dominated by CO(2 $\rightarrow$ 1) at 230.5 GHz toward disc hosts, and that a residual excess of on-star chance crossings at the tens-of-per-cent level cannot be excluded. The operative false-alarm statement remains the rank against the control ensemble.

This fixes what a single window can ever mean. The rank floor is $1/(N_{\text{ctrl}} + 1) = 1.95 \times 10^{-3}$ against a survey-

TABLE 17

ARRAY-SPLIT ROBUSTNESS. THE CONTROL GEOMETRY USES A 12-M θ_{PB} FOR EVERY WINDOW, SO THE ACA 7-M STRATUM CARRIES A KNOWN GEOMETRIC ERROR. EXPECTED RANK-FIRST COUNTS ARE $N_{win}/(512 + 1)$ UNDER EXCHANGEABILITY, AND ARE THEREFORE LOWER BOUNDS: THE MEASURED TAIL RATE IS A FACTOR 1.4 HIGHER (§L.1.3), WHICH BRINGS THE 12 M STRATUM’S 9 OBSERVED WITHIN ONE OF EXPECTATION. NO DISPOSITION AND NO HEADLINE THRESHOLD DEPENDS ON THE ACA WINDOWS.

	12 m	ACA 7 m
Windows	601	1054
Execution blocks	139	265
Stars; systems	70; 62	38; 38
Threshold crossings	22	53
Rank-first windows	9	7
expected	1.17	2.05
Stage-1 outliers	9	4
Stellar-rank $D(p)$	0.027 (0.77)	0.10 (<0.001)

wide Bonferroni scale of 3.0×10^{-5} , a factor 65, and correlation widens it: with $N_{eff} \simeq 30$ –43 independent spatial trials the ensemble resolves only 0.023–0.032. Drawing more controls would not help, the primary beam bounding the annulus. This is the design’s central limitation and is why authentication rests on recurrence and astrophysical attribution rather than on rank.

The control geometry uses a 12-m θ_{PB} for every window, so the 1054 ACA 7-m windows (64 per cent of the survey, array read from the archive’s antenna list rather than assumed) carry a known geometric error. It runs in the conservative direction: on the true 7-m beam those annuli span 0.08–0.46 of θ_{PB} at a median area-weighted gain of 0.75 against 0.47 for the 12-m windows, so exchangeability in flux density improves there. Of the stage-1 flagged windows only HD 48370 Band 6 is ACA.

Dropping the ACA windows tests what that geometry costs, and Table 17 gives the two strata side by side. The 12 m stratum holds 9 of the 13 stage-1 flagged windows, the one lost being HD 48370’s foreground CO; the deepest trigger and the deepest P_{eff} are both 12 m windows and do not move, so no survey-level statement changes. The ACA stratum alone departs mildly ($D = 0.10$, $p < 0.001$), opposite in direction to what a mis-scaled annulus gain would produce. Re-extracting the 1054 ACA windows on the correct 7 m beam would remove the error rather than bound it; it needs the raw visibilities and a full re-search, which is beyond this cycle, and the table is what we offer in its place. Excluding the 15 β Pictoris windows leaves one unattributed crossing among 416, against 3.13 expected ($P(\geq 1) = 96$ per cent).

K.1. The window-level and survey-level rank calibration

The drift grid and the thousands of channels apply identically at the star and at every control, so the per-window trials factor is absorbed empirically: under exchangeability one designated position of 513 ranks first with probability exactly $1/513$ whatever the noise distribution, and a heavy tail cannot by itself produce an excess of rank-first windows. The survey expectation is 3.23 and 16 are observed; the execution-block clustered permutation returns $P(\geq 16) = 2.2 \times 10^{-1}$, and the excess is astrophysical, celestial line emission sitting at the stellar position in 3 of the 16 (Table 9) and leaving 2 against 3.23 expected. A prespecified Monte Carlo that promotes one control per window to pseudo-star and runs it through the operative gates expects 0.20 flagged

pseudo-target windows, agreeing with the rank budget and the Bonferroni scale.

Ranking against 512 controls cannot resolve a probability below $1/513$, so a local scramble test was built to carry the four stage-1 windows past that floor. It scrambles each integration independently in channel and re-runs the identical search with the same noise map, weights and drift grid, which destroys anything coherent along a drift track while preserving each integration’s own spectral autocorrelation and trials factor; its validation criteria and failure mode are in Appendix L. Three of the four windows of the released catalogue pass validation; the diagnostic has not been re-run on the windows the completed sweep added, which is stated as a limitation in §6.4. The two β Pictoris windows behave as positive controls, no draw of 10^5 reaching T_* in either ($p < 1 \times 10^{-5}$ and $p < 5 \times 10^{-5}$), the empirical positive control expected where a genuine localised signal is present. For CP–72 2713 the null resolves the value itself, 157 draws of 10^5 equalling or exceeding T_* , and an extreme-value fit to that window’s own control maxima agrees to a factor 1.3; both are used with that window below. HD 48370 Band 6 fails validation outright, 73 per cent of its real controls exceeding the maximum of 10^5 null draws, because the field carries extended emission and imaging residuals that are coherent across integrations and that the scramble destroys by design, so we quote no scrambled-noise probability there. That failure is a direct consequence of retaining full dynamic spectra for only 8 of the 512 control positions, and no number quoted here rests on a parametric extrapolation of a null tail.

Exchangeability is the statistic’s central assumption, and it can fail through spatially varying noise, sidelobes, correlated pixels or calibration residuals at the phase centre. For a debris-disc archive the dominant violation is none of those but resolved astrophysical emission filling the annulus, seen here in HD 48370’s CO ring and in β Pictoris’ recurring Band 6 window, where the ring outshines the star by $1.5\times$; that suppresses a star-exceeds-ring outcome instead of manufacturing one, so it costs sensitivity and nothing else; the repeat block of the *flagged* Band 6 window measures that cost (§D.0.3). The primary beam is settled by construction, since the statistic is formed on *uncorrected* amplitudes whose thermal noise does not depend on position and $\sigma(t, \nu)$ is one scale shared by the star and every control (§4.1), so the 513 positions are *approximately* exchangeable under a *restricted* noise-only null. Both qualifiers carry weight, because that argument covers the thermal noise and the beam and does not cover residual calibration or continuum systematics, which can occur preferentially where the pointed science target sits.

Four measurements bound what is left, and all four are null. Ranking each control against the other 512-1 by the rule used for the star gives 220 672 pseudo-star ranks, uniform to the resolution the ensemble can express ($p = 0.37$) in every band and at both channelisations; that geometry cannot by itself detect a centre-versus-annulus asymmetry, so the test that can is the two-sample comparison of the 1655 real stellar ranks against the pooled pseudo-ranks, and it returns $p = 0.38$. On that comparison the stellar position behaves like one more position in its own annulus, on these data; out of

sample it does not (§C.0.2). The Spearman correlation of control statistic against reconstructed radius has median 0.006 over the 1165 crossing-free windows, bounding any systematic radial suppression of the controls below 13 per cent in the mean. And the star’s own residual variance is measured directly against the controls’ in §L.1.2.

One stratum departs, and we report it separately. The fine-channel windows fail the $U(0,1)$ rank test at $p=0.015$ ($n = 403$), because 13 of them contain an on-star peak above their own control maximum against 0.79 expected, and those are precisely the flagged windows of Table 9. The coarse stratum, which is three times larger and predicts 2.44 such windows, contains 0: every stage-1 flagged window in the survey is drift-resolving. A 15.6–31.25 MHz channel dilutes a narrow circumstellar line, so the class that can resolve one is the class that finds it; the same reasoning says a genuine narrowband emitter would appear in Class A first. That fits the astrophysical line emission identified below, with no misbehaving ensemble, but the departure is genuine and it is not confined to the windows carrying a crossing. Removing all 71 fine windows with a stellar crossing at the trigger reduces the Kolmogorov–Smirnov statistic only from 0.150 to 0.103 and leaves $p = 0.002$ over the remaining 332 windows, so fine-window rank probabilities are optimistic for the class as a whole and not only where a crossing sits. The coarse class departs too, $D = 0.050$ at $p = 0.004$ over 1252 windows. This is the quantitative reason the spatial comparison is used as a prioritisation and never as a test of the null (§4.1), and Appendix K measures what it costs. Two stratifications the frozen products cannot support, per-control noise records and gain, are bounded only through these empirical rank distributions. Figure 13 shows the by-stratum behaviour: median per-window control maxima are flat across bands and the fine class sits higher than the coarse, 5.79 against 4.44, as its far larger channel $\times$ drift trial count predicts.

A fully symmetric region-max variant, which maximises identically at the star and at every control, can be run on only seven locally reprocessed windows because the frozen release does not carry the products it needs. In all seven the star is exchangeable with its ring (Appendix L). The single-position calibration above remains the operative release-wide criterion, and the two are never interchanged.

L. ANALYSIS PROVENANCE AND ROBUSTNESS CHECKS

L.1. *The β Pictoris frame audit*

Table 18 carries the frame chain for the line-attributed windows. It recovers the whole check list untuned: frequencies to the frame chain’s stated tolerance, velocity concordance across epochs and transitions, localisation against the control ensemble, and drift class zero to within the grid.

Two further properties fit CO and not a carrier. The stage-1 Band 6 window carries 114 formal threshold crossings where a channel-confined carrier admits one. The retained products do not record whether those channels are contiguous, so the count sets a floor of 2.26 km s^{-1} on the feature’s extent without measuring it. The peak flux densities, 76 and 78 mJy per channel, are the right order for this belt’s spatially integrated

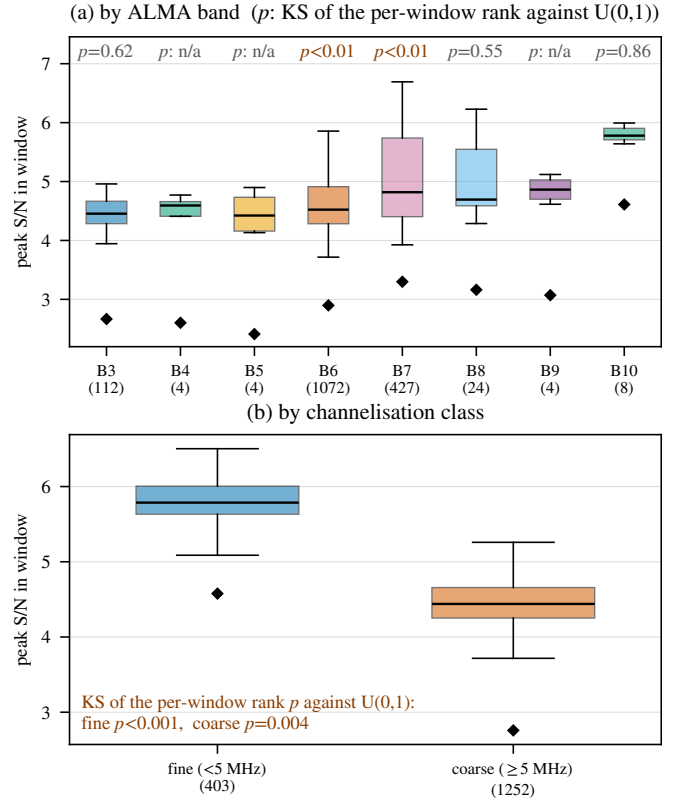


FIG. 13.— Exchangeability of the 512-position control ensemble by stratum, over 1655 windows. Boxes give the per-window control maximum, black diamonds the median on-star peak, and p the Kolmogorov–Smirnov test of the per-window rank against $U(0,1)$. (a) by ALMA band, with the window count under each label; (b) by channelisation class, fine (< 5 MHz) against coarse. The ensemble tracks the on-star statistic in every band, but both channelisation classes depart from uniformity: fine $D = 0.150$, $p < 10^{-4}$ over 403 windows, and coarse $D = 0.050$, $p = 0.004$ over 1252. The departure is not the flagged windows’ doing — removing all 71 fine crossings leaves $p = 0.002$ — and §C.0.3 identifies the radial noise gradient as the cause.

CO. They are per-beam peaks on emission resolved in both beams, and Matrà et al. (2017) find the gas subthermally excited, so beam dilution and excitation both depress their ratio below optically thin LTE.

Stellar emission is a competing explanation for any stage-1 flagged window towards an active young star, and the search suppresses it twice. Gyrosynchrotron and free-free emission from chromospheres and coronae are broadband, while the block-median baseline of §4.1 removes everything broader than its 65-channel passband, ~ 32 MHz at the fine class, a fractional bandwidth of $\sim 10^{-4}$; MacGregor et al. (2020)’s AU Mic flares are coherent across the full 8 GHz. Flares are also time-confined, so the inverse-variance stack over all integrations dilutes them. What survives is the multiplicative route of §4.1, a flare continuum times a spectral calibration residual, bounded there and measured for CP–72 2713 in §5.2. The search is blind to the time axis by construction, and the retained dynamic spectra are the only handle on it.

HD 48370 in Band 8. The carbon tuning says the same thing by omission. All 24 Band 8 windows here belong to 3 stars — η Crv, HD 48370 and HD 61005 — and are $[\text{C I}]^3 P_1 - ^3 P_0$ (492.160651 GHz) tunings. HD 48370’s carries an on-star crossing at $T_\star = 5.05$ against a ring maximum of 6.23, so it is not one of the stage-1 flagged

TABLE 18

LINE-ATTRIBUTION AUDIT THROUGH THE FRAME CHAIN OF §4.2: THE THREE LINE-ATTRIBUTED WINDOWS, AND THE CLASS A β PICTORIS BAND 6 WINDOW THAT IS NO STAGE-1 FLAGGED WINDOW YET SHOWS THE SAME CO COMPONENT. TOPOCENTRIC OFFSETS ARE OBSERVED MINUS CATALOGUED LABORATORY REST FREQUENCY (PICKETT ET AL. 1998), SO A NEGATIVE OFFSET MEANS THE OBSERVED FREQUENCY LIES BELOW REST. THE BARYCENTRIC CORRECTION IS THE EARTH-EPOCHERIS TERM AT THE EXECUTION-BLOCK MID-TIME. THE AU MIC SYSTEMIC VELOCITY IS CATALOGUED (FOUQUÉ ET AL. 2018). FOR β PICTORIS WE ADOPT $v_{\text{sys}} = +20.0 \pm 0.7 \text{ km s}^{-1}$ (MATRÀ ET AL. 2017) RATHER THAN THE *Gaia* DR3 VALUE OF $+16.84 \text{ km s}^{-1}$ (GAIA COLLABORATION ET AL. 2022), BECAUSE THE STAR IS AN A6V ROTATOR INSIDE THE HOT-STAR TEMPLATE REGIME IN WHICH DR3 RADIAL VELOCITIES CARRY A KNOWN SYSTEMATIC (BLOMME ET AL. 2023).

Flag	UTC start	f_{obs} (GHz)	Transition	Topo. offset (MHz; km s^{-1})	Bary. corr. (km s^{-1})	v_{sys} (km s^{-1})	Stellar-frame offset (km s^{-1})
β Pic B3	2022-03-03 21:49	115.260326	CO(1 $\rightarrow$ 0)	-10.88; -27.46	-7.90	+20.0	-0.39
β Pic B6	2019-03-12 00:14	230.516128	CO(2 $\rightarrow$ 1)	-21.87; -28.44	-8.15	+20.0	-0.29
AU Mic B6	2014-08-18 04:01	230.825230	CO(2 $\rightarrow$ 1)	+287.23; +373.51	-10.02	-4.71	+378.82
β Pic B6 [†]	2013-10-06 06:23	230.528134	CO(2 $\rightarrow$ 1)	-9.87; -12.83	+7.63	+20.0	-0.46

[†]Class A Band 6 window of a different execution block, without a stage-1 flag (Appendix H), shown because it independently shows the same CO component.

windows. Its crossing channel sits -129 km s^{-1} from [C I], so it is not [C I] emission from the disc, the cloud or the star, and its own ring dispositions it as an ordinary noise crossing. What the window does show is no line at the star in this disc’s carbon tuning, which is what Cataldi et al. (2023) report. The crossing peak in the flagged window itself is 0.82 Jy in one 122-kHz channel with the ring at the same level.

L.1.1. The survey rescored with a repaired screen

Candidate generation was rerun under two corrections, both built from the stored control vectors. Both can only act where a control vector was stored, so this subsection covers 447 windows of the 1655 searched and the 4 stage-1 windows that fall inside them; the corrected statistic applied to the survey as a whole is in §C.0.3, where it takes the stage-1 list from 13 to 10. *Radius matching* ranks the star against only the 54 probes with $u \leq 0.30$; its cost is resolution, so the screen admits 8.1 windows by chance and returns 5. *Detrending* keeps all 512 probes and subtracts the pooled radial profile $m(u)$ measured on the calibration windows, leaving the resolution unchanged. We adopt detrending, because the trend belongs to the field and not the star: pseudo-stars carrying no star at all are displaced as far as real ones (§C.0.3), so a measured profile uses every probe where radius matching discards nine in ten to approximate the same thing.

The stellar debit $m(0)$ is the one free choice, between 0.15σ (the innermost measured bin, conservative) and 0.32σ (the value centring the calibration ranks on 0.5). Rather than argue for one, Table 20 reports the debit each flagged window survives: 10.80 and 25.77σ for the two β Pictoris windows, 0.61σ for CP-72 2713, and only 0.21σ for HD 48370, whose emission fills the annulus, which is why it is dispositioned as foreground CO.

No window is promoted by either repair. Within this subset the adopted screen returns the released set unchanged, 4 windows, with 0 added and 0 lost. Under radius matching the one addition is a Class A window at $T_\star = 5.11$, against 8.1 expected by chance at that screen’s resolution. Under the strongest debit the set loses HD 48370, an attributed line, and keeps the rest. Within this subset the number of *unattributed* stage-1 flagged windows is one for every repair and every debit between zero and 0.61σ ; over the whole survey §C.0.3 reaches the same conclusion on 2 against 4. No disposition in this paper depends on the defect.

What the repair does not do is make the rank distribu-

tion uniform. Detrending moves the calibration-sample median from 0.429 to 0.437 ($D = 0.091$ to 0.081) and the survey median from 0.466 to 0.474 ($p = 0.26$ to 0.51): radius is part of the story and not all of it. Nor would fitting and testing the profile on the same windows validate it. Leaving each of the 27 calibration stars out of the fit in turn and scoring it against a profile it did not contribute to returns median 0.437 and $D = 0.081$, against 0.437 and 0.081 in sample: the profile is not over-fitted, and it does not restore exchangeability either. We therefore do not present the detrended statistic as a validated null. It is a robustness check on the dispositions, and the survey’s false-alarm budget is taken from the measured tail rate below instead. A parametric beam model would not settle it: the windows mix 12-m and 7-m primary beams (Appendix K), and a displacement that survives where no star is present is not the beam acting on a star. The residual is why §6.4 puts visibility-domain localisation, which compares no sky positions at all, ahead of any further repair of a position-ranking statistic.

L.1.2. Does the stellar position carry its own noise?

The noise scale of Eq. 3 is built across the annulus, so by construction it carries no variance peculiar to the stellar position, where residual continuum, deconvolution or self-calibration residuals and position-correlated calibration error would concentrate. That is the largest untested assumption in the method, and the retained products bound it. R_σ , the star’s robust residual scale over the 4 stage-1 windows of the released catalogue for which the full control vectors were retained against the median of the 8 control positions the release keeps (Appendix L), runs 0.995 to 1.001 , so the star agrees with its ring to better than 0.7 per cent at 95 per cent confidence. That matters most for CP-72 2713. Because $T_\star$ scales inversely with the noise the star is divided by, an unmodelled broadband excess of 2.2 per cent at the stellar position would bring that window’s $T_\star$ below its own ring maximum and dissolve the outlier; the bound is a factor of three smaller, so that explanation is disfavoured without being formally excluded. The same test on a seeded random sample of 25 *unflagged* windows returns 0.997 (range 0.968 – 1.004), every one within 3 per cent of unity, so the no-excess-at-the-star assumption holds generally and not only where a disposition needed it. It is a test of variances and not of means, and it is blind to the one- or two-channel artefact the search is built to find, which the local null and recurrence address instead.

TABLE 19

THE STAGE-1 STAR-BAND PAIRS, SCORED UNDER BOTH STATISTICS, SO THE DIFFERENCE BETWEEN THEM CAN BE AUDITED. THE SUPERSEDED REGION-MAXIMUM STATISTIC FLAGS 66 WINDOWS IN 41 STAR-BAND PAIRS ACROSS THE SURVEY, FAR MORE THAN THE SYMMETRIC STATISTIC; THIS TABLE GIVES THE PAIRS EITHER STATISTIC PROMOTES TO STAGE 1, NOT ALL 66 OF THEM. S_{reg} : REGION MAXIMUM OF EQ. 3 OVER $n_{\text{src}} = 135$ POSITIONS; T_* : THE SYMMETRIC SINGLE-POSITION STATISTIC; ‘CTRL MAX’, ‘RING MAX’: THE LARGEST CONTROL STATISTIC UNDER EACH. THE ν COLUMN IS EACH WINDOW’S *centre*; THE CROSSING FREQUENCY DIFFERS BY UP TO 0.85 GHz, AND TABLE 9 CARRIES ν_{cross} . FOR CP-72 2713 THE REGION MAXIMUM ALREADY LAY ON THE STAR, SO THE TWO COINCIDE.

Star	Band	ν (GHz)	region-max statistic		symmetric statistic		Symmetric outcome
			S_{reg}	ctrl max	T_*	ring max	
HD 48370	6	230.7305	27.95	26.83	27.10	26.83	flagged
β Pic	3	115.2605	20.31	7.27	14.65	7.27	flagged
β Pic	6	230.5164	12.78	9.12	11.68	9.12	flagged
AU Mic	6	230.5384	5.97	5.85	5.22	5.85	not flagged
ALMA J1537-3319	6	217.0183	5.83	5.63	5.09	5.63	not flagged
CP-72 2713	7	345.1152	5.81	5.68	5.81	5.68	flagged
HD 14055	7	345.1377	5.80	5.74	5.22	5.74	not flagged

TABLE 20

THE SURVEY RESCORED UNDER BOTH REPAIRS. “SURVIVES” IS THE LARGEST STELLAR DEBIT $m(0)$, IN UNITS OF EACH WINDOW’S OWN CONTROL SCALE, AT WHICH THE STAR STILL EXCEEDS EVERY DETRENDED CONTROL; IT IS A PROPERTY OF THE WINDOW AND ASSUMES NEITHER DEBIT. FOR REFERENCE THE CONSERVATIVE DEBIT IS 0.15σ AND THE DEBIT THAT CENTRES THE CALIBRATION-SAMPLE RANK DISTRIBUTION IS 0.32σ , AND A WINDOW SURVIVES WHICHEVER DEBIT IS SMALLER THAN ITS ENTRY. NO WINDOW IS ADDED BY EITHER REPAIR.

Window	T_*	Survives (σ)	Rad.-m.	Detr.
β Pic B3	14.65	25.77	yes	yes
β Pic B6	11.68	10.80	yes	yes
HD 48370 B6	27.10	0.21	yes	yes
CP-72 2713 B7	5.81	0.61	yes	yes
Flagged, whole survey			5	4
Added			1	0
Expected by chance			8.1	0.9

L1.3. The false-alarm rate, measured instead of assumed

Because the body of the rank distribution is displaced, $1/(512+1)$ is the rank *resolution* of the ensemble and not a per-window null probability. The survey’s false-alarm budget is therefore quoted from measurement. Two independent measurements agree. Of the 599 calibration windows carrying no attributed line, 1 exceeds all 512 of its controls, one in 599 or 0.9 times exchangeable; and the pseudo-stars, which cannot contain a signal at all, rank first in 0.28 per cent of trials out of sample against 0.19 per cent exchangeable, a factor 1.4. The uncertainty on that factor is not binomial: the 16 inner probes of one window share its noise realisation and its execution block. Recomputing the test from the stored control vectors over all 1326 calibration windows gives 63 first ranks in 21 216 trials, a factor 1.5, and resampling whole windows, or whole execution blocks (327 of them), widens the 95 per cent interval barely at all: 1.2–1.9 and 1.2–1.9 against 1.2–1.9 binomial. Every window ever searched retains its 512-probe vector, so this rests on 21 216 trials over 1326 windows and 327 execution blocks. The factor is therefore *between 1.2 and 1.9*, with the released-product value 1.4 inside that interval, and no disposition in this paper changes anywhere in it. The tail is *mildly* anti-conservative where the body is *strongly* displaced, and it is the tail the candidate screen uses. Propagated over the survey’s 1655 windows, the pseudo-star rate predicts 4.7 unattributed stage-1 flagged windows, with $P(\geq 1) = 99$ per cent, against 3.23 predicted under exchangeability and one observed. Every per-window probability quoted in this paper should be read as multiplied by 1.4, which changes no disposition here: no window was claimed sig-

nificant after the survey-wide correction of Appendix K, and this rate predicts about one CP-72 2713. The displacement is widespread rather than one configuration’s doing, 23 of 27 stars having a median rank below 0.5 (Fig. 14).

The calibration sample covers 56 stars and windows within a block are not independent epochs, so the sizes above carry that caveat; at 578 null windows the test resolves a shift of this size ($D \geq 0.06$ against 0.08 observed). On data that played no part in building it the frozen pipeline recovers the signal present, raises no candidate, and ranks the stellar position above its annulus more often than exchangeability predicts. Counts are those at this build date (2026 September 13, 07:44 UTC).

L.2. Audit against a region-maximum statistic

The statistic of Eq. 3 evaluates star and controls identically. A region maximum taken over $n_{\text{src}} = 135$ positions at the star, ranked against single-position controls, is not exchangeable with them and ranks first with probability $n_{\text{src}}/(n_{\text{src}} + 512) = 20.9$ per cent rather than $1/(512+1)$; that is a property of the estimator class, derivable without reference to any window’s outcome. The primary analysis uses the symmetric form throughout and all 1655 windows are scored under both. Table 19 audits the star-band pairs either form promotes. Over the whole survey the symmetric criterion flags a strict *subset*: 13 windows against 66 in 41 star-band pairs, and not one window is flagged by the symmetric form alone. Those that drop out do so because the stellar position never exceeded its own control ensemble. Both columns are released for every window, so the dependence of the result on the choice can be checked directly. The clearest single case is AU Mic, where the region form flags a crossing that the symmetric form does not, because the stellar position never exceeded its own control ensemble: the excess is in the field around the star, not at it.

50 stars hold further public blocks this survey did not take (§3), which §6 turns into an epoch-activity constraint.

L.3. Entries excluded from the released catalogue

The four classes of excluded entry. (i) Four target/bands (Barnard’s Star [B7], Wolf 359 [B7], SCR J1845-6357, Wolf 358) yielded no carrier result, each keeping a valid continuum non-detection. (ii) Giclas 9-38A and B [B3] are excluded entirely, the pair’s only matched MOUS pointing ~ 52 arcmin from either

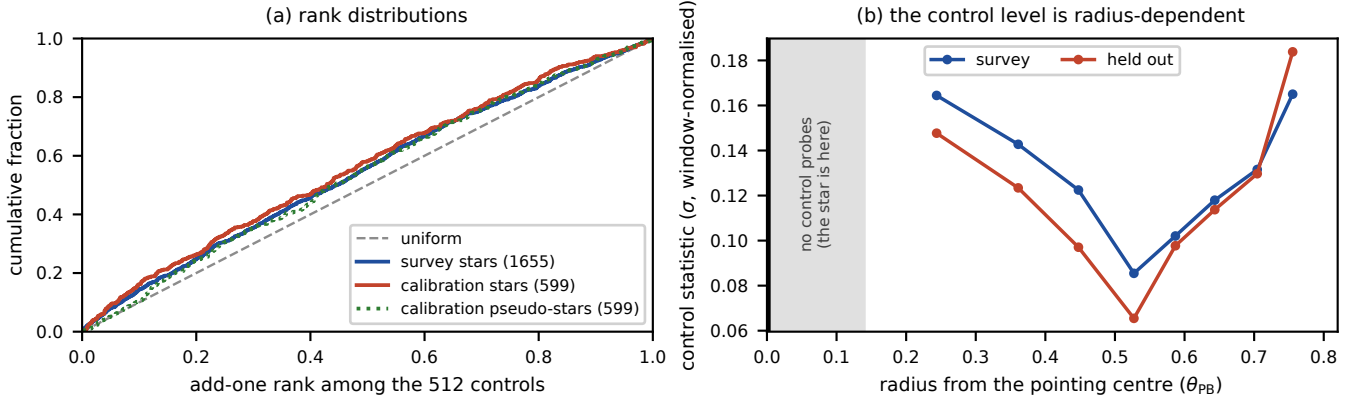


FIG. 14.— Why the spatial control ensemble is a calibrated screen and not an exchangeable rank test. (a) add-one rank of the star among its 512 controls, for the survey and for the calibration windows carrying no attributed line: both lie above the uniform diagonal. The dotted curve is the *pseudo-stars*, control positions at the inner edge of the annulus ranked against the other controls, which contain no star and are displaced just as far (§C.0.3), so the displacement is instrumental. (b) the control statistic itself against radius from the pointing centre, in units of each window's own noise scale. It is lowest at mid-annulus and rises towards both edges; the shaded strip inside $0.14\theta_{PB}$ holds no control probe at all, and that is where the star sits (median $0.0032\theta_{PB}$).

star, $35\times$ its primary-beam FWHM. (iii) 13 windows fail the noise test of §A.3, sitting 388–743 below the shallowest sibling of the same block and channel width. (iv) “ γ Lupi” resolves to HD 139664 at 17.40 pc, not the naked-eye star; all such entries are relabelled and no measured value changes.

The 12 Band 9/10 windows toward AU Mic and HD 61005 are included and none is a detection or spatial outlier ($T_\star \leq 4.91$). The quality audit is clean across all 1655 windows bar four χ^1 Ori Band 3 windows flagged as elevated, weakening their thresholds by $\lesssim 20$ per cent.

An external null from outside the sample. The archival sweep also searched 198 windows in 47 execution blocks toward 22 stars lying between 40.8 and 49.8 pc. They were queued by a beam-matching step that did not check distance, so they fall outside the sample this paper reports and are excluded from every number in it. They are not wasted. They went through the identical frozen pipeline, entered no threshold, mask or statistic decision, and no result of ours depends on them, which makes them an external null in the strict sense. Their stellar add-one ranks have median 0.437 against the 0.5 expected under exchangeability ($D = 0.11$, $p = 0.02$), and they contain 0 stage-1 flagged windows against 0.4 expected.

L.4. The three validation checks in full

Each of the three checks summarised in §C.0.2, §C.0.3 and §5.2.6 is given here with every number it rests on. The main text states the result; this is the working.

L.4.1. The hold-out reservation, block by block

The star-versus-control rank is used for candidate prioritisation only. It does not provide a detection p -value, because exact star/control exchangeability fails in the prospective hold-out sample. That is a result of this work rather than a caveat on it, and it is the reason every disposition in §5.2 rests on astrophysical evidence or on recurrence, and never on a rank alone.

The rest of this subsection establishes it. The statistic and the mask were settled on the 1655 windows above, so they cannot also calibrate themselves. Calibrating instead on whatever blocks happen to be processed last is a weak guarantee, because that set is defined by processing order rather than in advance. We therefore reserved a hold-out by a rule fixed and committed on 2026-09-14 (c75069040eab), *before* the archival search finished and before any reserved block was searched. The order of the three commitments is what makes the reservation worth anything, so we give it explicitly: the detection statistic of Eq. 3 was fixed on 2026-09-09 (0c465f2661bd), the promote/retire criteria on 2026 September 11 (119a68a6ab50), the hold-out rule 5 days after the statistic, and the first reserved block was searched 3 days after that. The detection statistic, the promote/retire criteria and the hold-out rule — the three choices that can manufacture a candidate — were all fixed before any reserved block was searched. Several *diagnostics* were built afterwards and are named here rather than left implicit: the radius-corrected normalisation (§C.0.3), the visibility-domain test (§5.2.1), the drift-resolved injection analysis (Appendix G) and the P_{90} uncertainty budget. None of them can promote a window — each either confirms a disposition reached

earlier or, in the case of the radius correction, removes flags — and the statistic they are compared against did not move. Local commit timestamps can be rewritten, so they are not by themselves evidence of anything: the dates above are corroborated by the GitHub server-side push times for the same commits, which the repository’s API exposes and which the authors cannot alter, and by the Zenodo deposit’s own minted date. A block is held out if and only if $\text{sha256}(\text{UID})_{[:8]} \bmod 5 = 0$, with two guards that depend only on identifiers and the already-published block list: blocks in the previous release stay in the survey, and no system may lose every block. The assignment is in `holdout_assignment_v381.json` and can be recomputed from the UIDs alone.

Of 484 blocks processed, 77 are reserved (15.9 per cent), giving 315 windows toward 36 stars. The headline sample loses no star and no system to the reservation, which is what guard two is for. Four things in this paper are calibrated empirically: the rank displacement, the false-alarm tail factor, the radial behaviour of the control ensemble, and the claim that stage-1 status alone carries no candidate significance. All four are measured on the reserved blocks and on nothing else.

The ranks are displaced out of sample, and by very nearly the amount measured in sample: median add-one rank 0.405 against 0.5 (block-clustered 95 per cent 0.341–0.458), $D = 0.109$ at $p = 0.0011$ over 315 windows. The false-alarm tail factor, the ratio of the measured pseudo-star first-rank rate to the rate ideal exchangeability would give, 1.95×10^{-3} , is 1.2 on 5040 trials, with a bootstrap clustered over the 77 reserved blocks giving 0.6–1.9. The reserved blocks contain 0 stage-1 spatial outliers against 0.61 expected. The radial profile of the standardised control level across the annulus runs $+0.12$, $+0.10$, $+0.12$, $+0.16$ from inner to outer edge, a mean per-window slope of $+0.067 \pm 0.033$: the displacement is reproduced out of sample but the steep inner-edge excess seen in sample is not, so we present the radial correction of §C.0.3 as one demonstrated mechanism for the displacement rather than the whole of it. A displacement that survives a hold-out chosen before the data is a property of the statistic, not of this sample.

The older, processing-order calibration set is retained below because it carries the repeat observations the hold-out does not. 60 windows cross the trigger at the star and 7 are stage-1 flagged windows. 4 of those are β Pictoris CO recovered at the stellar position, an empirical positive control where a real localised signal is present. The remaining 3 have no astrophysical attribution: 61 Vir, HD 14055 and HD 23484, at $T_\star = 6.16$, 6.03 and 5.30 against ring maxima 5.88, 5.96 and 5.25. *None recurs*. Re-searched at its own tuning in further blocks of the same star, each falls back into the body of its own control distribution: 4.83 with 119 controls above the star, 5.42 with 20, and 4.99 with 21.

Table 22 gives the CP–72 2713 repeat test in full and Table 25 sets all 4 beside it. All 4 are the same kind of event: a single cell a few per cent above the largest of 512 controls, in a window whose repeat observations put the stellar position back inside its own control distribution. All four are survey blocks, as `holdout_assignment_v381.json` records. What they show stands on the repeat blocks of §5.2: stage-1 status carries no candidate significance by itself, and recur-

TABLE 21

VISIBILITY-DOMAIN TEST FOR EMISSION AT THE STELLAR POSITION, FOR ALL 13 STAGE-1 WINDOWS. EVERY VISIBILITY IN THE EVENT’S OWN CHANNEL IS PHASE-ROTATED TO THE STAR AND AVERAGED: EMISSION CENTRED THERE GIVES A POSITIVE REAL PART AND AN IMAGINARY PART CONSISTENT WITH ZERO, WHILE EMISSION DISPLACED FROM THE STAR IS SUPPRESSED IN THE REAL PART AND IN GENERAL LEAVES A NON-ZERO IMAGINARY PART. N_{vis} IS THE NUMBER OF VISIBILITIES AVERAGED, σ THE STANDARD ERROR OVER THEM, AND ‘CTRL’ THE LARGEST REAL PART RETURNED BY EIGHT CONTROL POSITIONS ON THE ANNULUS OF §4.1. THE EVENTS ATTRIBUTED TO CARBON MONOXIDE ARE THE TEST’S CONTROLS: 7 ARE RECOVERED AT 10–59 σ , HD 48370 SHOWS THE DISPLACED-SOURCE SIGNATURE ITS FOREGROUND IDENTIFICATION PREDICTS, AND ONE β PICTORIS BLOCK COULD ONLY BE TESTED AT A SUBSTITUTED FREQUENCY. ALL 13 EXECUTION BLOCKS WERE RECALIBRATED FROM THE RAW ARCHIVE FOR THIS TEST.

Star	B	N_{vis}	Re/ σ	Im/ σ	χ^2/ν	ctrl	verdict
<i>Unattributed</i>							
61 Vir	7	12 375	−0.4	+0.7	1.00	+0.9	not above its own controls
HD14055	7	13 365	+2.5	+3.2	1.00	+1.1	not localised: Im $\neq$ 0
CP−72 2713	7	405 860	+2.3	−0.7	1.00	+2.1	nothing at the star
HD 23484	6	15 015	+2.3	+0.8	1.00	+0.3	nothing at the star
<i>Attributed to CO (positive controls)</i>							
β Pic	3	285 327	+23.7	+0.8	1.00	+0.9	emission at the stellar position
β Pic	3	288 435	+23.8	+2.5	1.00	+1.9	emission at the stellar position
HD 48370	6	9 832	+53.1	+7.9	1.00	+4.7	not localised: Im $\neq$ 0
β Pic	3	421 890	+0.5	+0.9	1.00	+0.5	not above its own controls
β Pic	3	274 700	+15.1	+1.1	1.00	+2.7	emission at the stellar position
β Pic	3	180 600	+13.9	−1.6	1.00	+0.7	emission at the stellar position
β Pic	6	347 613	+26.6	−0.2	1.00	+2.5	emission at the stellar position
β Pic	6	398 475	+10.0	−0.7	1.00	+1.1	emission at the stellar position
β Pic	6	405 900	+58.9	−0.7	1.00	+1.1	emission at the stellar position

TABLE 22

THE CP−72 2713 REPEAT TEST IN FULL, BOTH BLOCKS SIDE BY SIDE, SO THE READER CAN TELL “NOT DETECTED AGAIN” FROM “A PERSISTENT SIGNAL OF THE ORIGINAL STRENGTH WOULD HAVE BEEN RECOVERED”. THE REPEAT BLOCK IS THE DEEPER OF THE TWO, MATCHES THE FIRST IN TUNING, CHANNELISATION AND INTEGRATION, AND ITS FREQUENCY AXIS REGISTERS TO A FIFTH OF A CHANNEL. A SOURCE AT THE FIRST BLOCK’S FLUX WOULD HAVE RETURNED $T_\star = 6.21$ THERE; THE MEASURED VALUE IS 4.70. INTERMITTENT EMISSION IS NOT EXCLUDED BY THIS TEST, AND NO STATEMENT HERE ASSUMES OTHERWISE.

	first block	repeat block
Execution block	Xff0235_X4a6d	Xff0235_X502d
Start-to-start separation (h)	2.05	
On-source time (s)	2994	2994
Channel width (kHz)	488.28	488.28
Channel rms (mJy)	2.097	1.964
Primary-beam FWHM (arcsec)	18.2	18.2
Frequency axes registered to	0.18 channel	
Test frequency (GHz)	344.2697	
Test drift (kHz s $^{-1}$)	−2.06	
Drift track sweeps (channels)	15.8	
Flux at that cell (mJy)	11.3 $\pm$ 2.0	−1.0 $\pm$ 1.8
$T_\star$ in this window	5.81	4.70
Largest control statistic	5.68	5.93
$T_\star$ expected if persistent	—	6.21
Pair inconsistent at	4.6 σ	
First flux taken as exact	6.8 σ	

rence rather than rank is what settles a marginal event. Independently, the measured tail rate in the calibration sample predicts 3.7 such events over 1322 windows and 3 are seen there.

The rank distribution of the remaining 578 windows is the campaign’s most consequential result, and it is adverse to the method. The star’s add-one rank is not uniform. Its median is 0.405 where 0.5 is expected, 14 per cent of the ranks fall below 0.1 against 10 per cent expected and 6 per cent above 0.9 against the same, so the whole distribution is displaced towards the star; it does not simply grow a tail. Windows inside a block share a calibration and blocks of a unit set a target, so the window-level figure ($D = 0.08$, $p < 0.01$) treats as independent what is not; clustering removes neither the shift nor its significance, 99 of the 149 blocks having a mean rank below 0.5 (sign test $p < 0.001$) and a star-

TABLE 23

THE INJECTION CAMPAIGN RESOLVED BY CONFIGURATION. P_{50} AND P_{90} ARE THE INJECTED AMPLITUDES, IN UNITS OF THE LOCAL NOISE, AT WHICH HALF AND NINE-TENTHS OF TRIALS ARE RECOVERED. NO AXIS SHOWS A DEFICIT LARGE ENOUGH TO MOVE THE QUOTED THRESHOLD OUTSIDE THE WINDOW-TO-WINDOW BRACKET OF $\times 0.48$ – $\times 1.35$; THE SUB-CHANNEL-PHASE ROWS ARE MEAN RESPONSE FACTORS, NOT RECOVERY AMPLITUDES, AND ARE THE REASON THE THRESHOLD CARRIES A RESPONSE CORRECTION.

Axis	Level	Trials	P_{50} (σ)	P_{90} (σ)
ALMA band	B3	432	4.8	6.7
	B6	1872	4.7	6.1
	B7	1152	4.4	6.0
	B8	432	4.4	6.0
	B10	144	4.5	5.9
Channel width (kHz)	15.3	144	4.2	5.8
	61.0	288	3.0	4.6
	122.1	432	4.2	5.8
	244.1	864	4.7	6.5
	488.3	1152	4.7	6.2
Array	976.6	1152	5.0	6.0
	12 m	3600	4.6	6.0
	ACA	432	4.8	6.7
Drift rate / ceiling	<1/3	1341	4.6	6.1
	>2/3	1410	4.5	5.9
	1/3–2/3	1281	4.6	6.5
Integrations	<150	720	4.6	6.4
	>350	1872	4.5	5.9
	150–350	1440	4.6	6.4
Sub-channel phase	0.000	80	2.0	—
	0.125	80	2.2	—
	0.250	80	2.3	—
	0.375	80	2.5	—
	0.500	80	2.7	—

clustered bootstrap over these 578 windows placing the median at 0.440 (95 per cent interval 0.42–0.46); the pre-registered hold-out of §C.0.2 gives 0.405 on its own 315 windows, with its own interval 0.341–0.458. It is not peculiar to the campaign either. The survey’s own 1655 windows lean the same way without resolving it, median 0.45 against a star-clustered 95 per cent interval of 0.42–0.47, and the two distributions are statistically indistinguishable ($D = 0.03$, $p = 0.78$), so the in-sample checks above were underpowered against a shift of this size rather than contradicted by it. The control ensemble is therefore an empirically calibrated screen for extreme

TABLE 24

DATA PARTITIONS USED IN THIS STUDY. “SEEN BEFORE THE STATISTIC FROZE” RECORDS WHETHER THE DATA WERE AVAILABLE TO THE CHOICES THAT DEFINE THE DETECTION STATISTIC AND THE MASKS; THE HOLD-OUT WAS RESERVED BY THE RULE OF §C.0.2, COMMITTED BEFORE ANY OF ITS BLOCKS WAS SEARCHED. ONLY THE SCIENCE SAMPLE CONTRIBUTES TO THE HEADLINE SEARCH RESULT.

Partition	Blocks	Windows	Stars	Purpose	Seen before the statistic froze?	In the headline result?
Science sample	404	1655	90	the search re-	yes	yes
Pre-specified hold-out	77	315	36	ported here calibrates rank, tail and radial behaviour	no	no
Out-of-sample, beyond 40 pc	47	198	22	external null	no	no
No surviving window	3	–	–	failed the quality cut	n/a	no

TABLE 25

THE EXTERNAL CALIBRATION SAMPLE AS SEARCHED AT 2026 SEPTEMBER 17. (a) WHAT THE FROZEN PIPELINE RETURNED ON 1322 WINDOWS IT HAD NEVER SEEN. (b) THE 4 UNATTRIBUTED OUTLIERS, ALL OF THEM FROM THE SURVEY SAMPLE — AN EARLIER VERSION OF THIS TABLE ASSIGNED THREE OF THEM TO THE CALIBRATION SAMPLE, WHICH **HOLDOUT_ASSIGNMENT_v381.JSON** CONTRADICTS — EACH A SINGLE CELL A FEW PER CENT ABOVE THE LARGEST OF 512 CONTROLS, AND NONE RECURRING AT ITS OWN TUNING IN A FURTHER BLOCK. THE REPEAT ROW GIVES THE LARGEST STATISTIC THE FURTHER BLOCK RETURNS ANYWHERE ON ITS OWN GRID; THE *pinned* VALUE AT THE FIRST BLOCK’S OWN FREQUENCY AND DRIFT, WHICH IS THE STRONGER TEST, IS 3.60 FOR CP–72 2713 (§A.3.2).

(a) *Calibration sample, as searched*

Execution blocks / stars	326 / 56
Spectral windows	1322
Median add-one rank of the star	0.419
against uniform	$D = 0.095$
Trigger crossings at the star	60
Stage-1 outliers	7
β Pictoris CO at the star	4
no astrophysical attribution	3
Expected unattributed at the measured tail rate	3.7

(b) *The four unattributed single-epoch outliers*

	CP–72 2713	61 Vir	HD 14055	HD 23484
Sample	survey	survey	survey	survey
T_*	5.81	6.16	6.03	5.30
Ring maximum	5.68	5.88	5.96	5.25
Largest T_* in a	4.70	4.83	5.42	4.99
further block at the				
same tuning				
controls above the	many	119	20	21
star there				
Rekurs?	no	no	no	no

values and not an exactly exchangeable rank test, and the rest of this subsection is about which of those two statements the null result needs.

L.4.2. The radial gradient, measured on the probes

The cause is the noise scale. Images are corrected for the primary-beam response, so the true noise rises with distance from the pointing centre; the search, however, divides every position by one scale $\sigma(t, \nu)$ estimated from the probe ensemble as a whole, with no radial dependence (§4.1). Positions at large radius are therefore divided by a scale that is too small and stand too high, and the star, which sits inside the inner edge of the control annulus at $0.14\theta_{\text{PB}}$, is divided by a scale that is too large and stands too low. That predicts exactly the sign observed: the star’s rank is displaced *downward*, and the standardised control level rises outward across the annulus. Extended emission on short baselines, which is present in many of these fields, works the same way and adds to it.

The repair belongs at the measurement stage, and we have made it there. Correcting a statistic whose noise normalisation was wrong at construction, by debiting the

star afterwards, is weaker than normalising correctly in the first place. The control probes come from one fixed seed, so every stored control value can be paired with its radius: standardising each probe within its own radius bin, and the star on the innermost bin, removes the gradient by construction and introduces no free parameter. There is no stellar debit in that calculation.

We have run the corrected statistic over *all* 1614 windows, not a spot sample, and it does change the list. 9 windows are stage-1 flagged windows under both statistics. 4 lose the flag (CP–72 2713, HD14055, HD 23484, HD 48370) and 1 gains it (HD 207129), so the count falls from 13 to 10. The four that fall out are the marginal cases, every one of them already dispositioned on other grounds; the one that enters, HD 207129, is a debris-disc host whose crossing lies -150 km s^{-1} from CO(2 $\rightarrow$ 1) in the stellar frame, the same disposition the paper reaches for the other disc hosts.

We do not adopt the radius-corrected statistic as the primary one, and the reason is now empirical rather than procedural. Applied prospectively to 778 windows it played no part in building, it leaves the stellar rank inconsistent with uniformity and moves the median further from the exchangeable value (§5.2.3). A normalisation that does not deliver a calibrated statistic cannot be preferred merely because its mechanism is understood. The frozen statistic therefore defines every count reported here, and the corrected one is a robustness check. On the corrected statistic the survey would have 10 stage-1 flagged windows, 8 attributed to circumstellar or foreground CO and 2 unattributed — different windows, the same conclusion. The released catalogue carries both statistics per window, so either list can be reconstructed. Nothing in the paper’s conclusion moves: the corrected statistic removes more unexplained events than it adds, and the visibility-domain test of §5.2.1 finds no emission at the star in any of them. The survey median add-one rank does *not* move toward the exchangeable value of 0.5: it goes from 0.445 to 0.432, slightly further away. The correction removes the radial component of the displacement by construction, and what is left is larger than what it removed, so the residual is not radial and the screen remains a prioritisation rather than a test. We report this rather than the reverse because it is the direction the measurement gives.

The stellar debit $m(0)$ is no longer needed and is not used in the primary analysis. A successor should fold the primary-beam response into the per-position noise model from the start.

The same mechanism touches the completeness side, which the injections alone would not reveal, because they

are placed at the stellar position only. Fitting the out-of-sample radial profile and extrapolating to the stellar radius gives an offset of $+0.08\sigma$, so the scale applied at the star is too *large* by that amount and an injected signal is divided by too much. The quoted recovery fractions are therefore pessimistic by of order 2 per cent of the threshold, not optimistic. The effect is small and runs the safe way. It is not zero, so we do not treat the recovery curve as exact.

The control probes are not redrawn per window. The pipeline draws them from one fixed seed as $r = \sqrt{U(r_{\text{in}}^2, r_{\text{out}}^2)}$ with $r_{\text{in}} = 0.14\theta_{\text{PB}}$ and $r_{\text{out}} = 0.78\theta_{\text{PB}}$, so the fractional radii are identical in every window of the survey and each stored control statistic can be paired with the radius at which it was measured. The star is not at a comparable radius: its median offset from the pointing centre is $0.0032\theta_{\text{PB}}$, an order of magnitude inside the annulus's inner edge. Three tests follow, all from the released products and none needing a new search.

First, the control level itself depends on radius. In units of each window's own robust noise scale, the pooled control statistic runs $+0.16$ at $u = 0.24$, falls to $+0.09$ near mid-annulus, and returns to $+0.17$ at the outer edge. The calibration sample gives $+0.15$, $+0.07$ and $+0.18$. Positions near the pointing centre therefore sit high before any star is involved.

Second, matching on radius removes most of the departure. Rank the star against only the 54 innermost probes, those with $u \leq 0.30$, and the survey median moves from 0.444 to 0.455. The Kolmogorov–Smirnov statistic moves with it, from $D = 0.081$ ($p = 8 \times 10^{-8}$) to 0.066 ($p = 3 \times 10^{-5}$). The same effect appears in the geometry: the 31 windows whose star happens to fall inside the annulus have median rank 0.495, against 0.460 for the 727 that do not.

Third, a position with no star in it behaves the same way. Treat an innermost probe as a *pseudo-star* and rank it against the other 512–1 probes. Out of sample the pseudo-stars are displaced as far as the real ones, median 0.443 against 0.429 ($p = 0.003$); within the survey they are not (0.463, $p = 0.02$).

The displacement is therefore a property of *position in the field* and not of stars, which is what the same displacement would have meant had the pseudo-stars come out uniform.

This is a result about the method, and we state it as one. *Spatial control positions are a sound way to prioritise candidates in targeted interferometric archival data, and an unsound way to establish statistical significance in them.* The reason is structural rather than particular to this survey: the targets are objects someone chose to observe, so the field around them is not a blank sky, the primary-beam response and the imaging weights vary across the annulus the controls occupy, and the result is that the star's rank among its controls is displaced whether or not a star is there. Any future archival search that ranks a source against positions in its own field inherits this, and should measure the displacement out of sample before interpreting a rank as a probability. The remedy is not statistical but physical: a test that asks whether a point source sits at the stellar position, which the visibilities can answer directly (§6.4).

L.4.3. The line mask, species by species

The exclusion regions are defined before any crossing is classified, and they are *excluded from the experiment* rather than vetoed within it: the experiment declines to classify those frequencies rather than declaring anything found in them astrophysical. The searches as executed matched crossings to catalogued rest frequencies within one spectral channel at zero drift. The β Pictoris windows showed that inadequate: both fell outside that tolerance, by 10.6 and 21.9 MHz, while lying within 3.6 km s^{-1} of the stellar systemic of a resolved, mapped disc. Every crossing was therefore reclassified with a velocity-space mask, $\pm 50\text{ km s}^{-1}$ in the stellar frame around each catalogued transition, whose kinematic justification and the counter-argument that a communicator might choose these natural reference frequencies are in Appendix I. It is a post-hoc classification criterion and never a detection threshold, and its half-width was adopted after the β Pictoris crossings had been seen. Reclassifying all 75 crossings at ± 20 to $\pm 100\text{ km s}^{-1}$ costs 2.1 to 10 per cent of gross bandwidth and admits no new unattributed outlier. That test is one-sided by construction: widening the tube can only mask more, so it cannot show that the attributions survive a *narrower* tube. They do not rest on the half-width. Each attributed event is identified by a velocity agreement in the stellar frame computed through the full frame chain (Table 18), by an independently published detection of the same line in the same source, or by recurrence across execution blocks — not by falling inside a $\pm 50\text{ km s}^{-1}$ window. A reader who prefers a narrower mask should re-read the dispositions from that evidence, which is why it is tabulated separately. Masked regions are reported throughout as *excluded from the experiment*, not as searched-and-empty. What the mask cost in candidates, as opposed to bandwidth, is nothing: 17 of the 75 crossings fall inside a tube, 9 of them are the stage-1 flagged windows already dispositioned as CO in Table 9, and the remaining 8 do not exceed their own control ensembles, so no masked crossing would have entered candidate generation had the mask been absent.

The mask applied holds 17 transitions of 11 species at full JPL/CDMS precision (Pickett et al. 1998; Müller et al. 2005), applied in the LSR frame as well as the stellar frame and re-applied to the frozen crossing list as a post-hoc filter. The species criterion is the one objective rule we could state in advance: the transitions ALMA's own band definitions are built around, namely the CO isotopologues, HCN, HCO^+ , CS, SiO, SO, CN, H_2CO and [C I], which dominate the circumstellar and foreground emission of disc-selected fields. The choice is consequential rather than incidental: a full Splatalogue selection would mask 75.1 per cent of the union against the 4.1 per cent the adopted list masks, and would place 19 of the 20 crossings whose peak frequency the release records inside a tube (Appendix I), so “coincident with a known molecular line” is not a binary property but a statement about a catalogue. Since no masked crossing exceeds its own control ensemble (above), no plausible widening or narrowing of the species list changes a disposition. The cost of masking is measured on one definition, every JPL catalogued transition of the 9 frozen species displaced by each system's radial velocity, given

a $\pm 50 \text{ km s}^{-1}$ stellar-frame half-width, intersected with each window and merged: 4.1 per cent of the unique-frequency union and 4.7 per cent of the window-summed gross bandwidth, with CN and CO dominating the loss. Total exposure is $6.8 \times 10^6 \text{ s GHz}$, or 1900 star-hour-GHz, with 30 per cent of it in the five best-populated 2-GHz bins, which any frequency-prior calculation must weight (Appendix H). The mask governs dispositions and leaves the noise-derived thresholds alone. Table 15 gives the accounting band by band rather than as one survey total, because the loss is not uniform: it reaches 6.5 per cent in Band 9 and stays below 1.5 per cent in Bands 3–5, so a per-band frequency prior cannot use the global figure.

Masking is a design selection effect, and the distinction that matters is between *discovery* and *classification*. For discovery, line frequencies need not be removed at all: a carrier there is as detectable as anywhere else. For classification they are decisive, because the astrophysical prior at a catalogued transition is overwhelming and this pipeline cannot separate the two hypotheses on morphology alone. Removing them from the searched domain is the conservative choice and the one a non-detection can be quoted against; the better design, which we recommend for a successor, searches every frequency and attaches a *lower prior* to a line-coincident event rather than deleting the channels. A signal coincident with a molecular line is not thereby “false”, and these are exactly the frequencies a communicator wanting to be found near a natural reference might choose. *We place no constraint at all on artificial emission lying inside the masked regions,*

at any signal strength, and the reader should read every limit in this paper as conditional on that exclusion. The cost is 4.1 per cent of the unique-frequency union overall, but it is not shared equally between the two experiments: 0.45 GHz of the 47.7 GHz Class A union (0.9 per cent) against 0.23 GHz of 90.5 GHz in Class B (0.3 per cent). The fine-channel experiment therefore pays three times the fractional price, because its coverage is concentrated in the line-tuned windows a molecular mask is designed to remove. The release keeps those channels for the second of the two searches this dataset supports: a clean-frequency search, in which line coincidence is a veto, and a line-coincident search, in which a crossing is admitted only on independent evidence of spatial localisation, line morphology, polarisation, recurrence or non-thermal width. This paper reports the first; the β Pictoris crossings are what the second would have to handle. The exposure is not confined to the flagged windows either: 131 of 1655 windows contain an in-band catalogued transition of the masked species, and in none of them does a crossing fall within one native channel of a rest frequency.

L.5. Checks summarised rather than narrated

The following checks support the conclusions without changing them, and are summarised here rather than argued at length; the numbers behind each are in the released products, and the regeneration script reproduces them.

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TABLE 26

ROBUSTNESS CHECKS WHOSE OUTCOME IS REPORTED WITHOUT A NARRATIVE. EACH WAS RUN TO COMPLETION; NONE ALTERS A DISPOSITION.

Check	Outcome
R_σ residual scale at the four unattributed positions	Consistent with the control median to better than 1 per cent; the stellar positions carry no noise peculiar to themselves
Fully symmetric region-maximum statistic	Flags a strict subset of the frozen screen’s windows; no window is flagged by the symmetric form alone
Region-maximum audit, all windows	Both statistics released per window; the dependence of the candidate list on the choice is reproducible from the catalogue
Polarisation states of the searched blocks	198 of 200 resolvable blocks deliver both parallel hands; the search uses Stokes I only, so a per-hand search remains available at no observing cost
CP–72 2713 ancillary diagnostics and second-epoch detail	The recurrence test is reported in §5.2; the event does not recur, and no diagnostic changes that
β Pictoris recurrence block by block	The disc CO recurs in every block whose tuning covers it, which is what makes it the survey’s positive control

Zhang Z.-S., et al., 2020, ApJ, 891, 174